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NetSuite

# Manufacturing



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# Manufacturing

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# Manufacturing Overview

Manufacturing processes help organizations create finished goods from raw or semi-finished materials using some combination of labor and machinery. Finished goods are then sold, at a profit, to other manufacturers, wholesalers, or retailers who then sell them to consumers.

NetSuite integrates your manufacturing workflow from the sales order and process planning to building goods, tracking work orders, and releasing finished goods for shipping.

For information about how to work with NetSuite Manufacturing, see the following help topics:

- [Assembly Item Records:](#)

Define the members of an assembly and then track both the raw materials and the assembled items separately.

- [Assembly Work Orders:](#)

Track the production of assembly items needed for stock or to fill orders.

- [Advanced Bill of Materials:](#)

List the quantities of raw materials, assemblies, sub-components, and parts needed to manufacture a product at one or multiple facilities.



**Important:** The Advanced BOM record became available in NetSuite 2017.2. After your administrator enables the Advanced Bill of Materials feature, the Advanced BOM replaces the Assembly/Bill of Materials record.

- [Bill of Materials Member Control for Assembly Items:](#)

Ensure that the right components are included in assembly builds at the right time. Use BOM controls to plan to use and purchase components which are effective or obsolete within specified times.

- [Manufacturing Work In Process \(WIP\):](#)

Track work orders through the production process, from gathering materials, through shop floor assembly, to stocking finished goods.

- [Manufacturing Routing:](#)

Schedule and record manufacturing operational activities against a complex work order that requires multiple teams of employees, or work centers.

- [Manufacturing Preferences Overview:](#)

You can access manufacturing preferences at Setup > Manufacturing > Manufacturing Preferences.

- [Outsourced Manufacturing:](#)

Outsourced manufacturing helps customers manage their subcontracted manufacturing processes by purchasing outsourced assembly production from vendors.

- [SuiteAnalytics Manufacturing Workbook:](#)

NetSuite provides a **Manufacturing Transaction [Beta]** data source, which introduces the analytics transactions concept for manufacturing-specific transactions.

- [Advanced Manufacturing SuiteApp:](#)

Extend your NetSuite manufacturing routing into the Advanced Manufacturing Work Bench.

This connection enables manufacturers to define work instructions, associate material usage, compare resource supply with demand, and establish planned start and end times. To set up Advanced Manufacturing, you must install the Advanced Manufacturing SuiteApp.

- [Manufacturing Mobile:](#)

The NetSuite Manufacturing Mobile SuiteApp enables operators with little ERP knowledge to use mobile scanners to report manufacturing shop floor data. This mobile solution compliments the existing Advanced Manufacturing scanner to provide improved performance, scalability, customization support, and streamlined production activities.

- [Engineering Change Order:](#)

Generate Engineering Change Order (ECO) records to document changes to your Bills of Materials (BOMs) and authorize the implementation of those changes.

# Assembly Items

An assembly item is an inventory item made up of several components, but identified as a single item. Assemblies are manufactured by combining raw materials that you stock.

 **Note:** For details about distinctions between Groups, Kits, and Assemblies, see the help topic [Groups, Assemblies, and Kit/Packages](#).

After you create assembly item records that define the members of an assembly, you can track the raw materials and the assembled items separately.

For example, Wolfe Manufacturing sells the LogLeaper mountain bike that they assemble in-house. The LogLeaper is assembled from the following inventory components:

- one aluminum bicycle frame
- one set of handlebars
- one saddle
- one gearing assembly
- two wheel assemblies
- two sets of brakes
- two pedals

NetSuite tracks the stock of the LogLeaper and each component item separately. This enables Wolfe to track the stock levels of LogLeaper mountain bikes in inventory and available to ship to customers. Wolfe can also track the quantity of materials available to assemble more bicycles.

To use assembly items, you must complete the following tasks:

1. Enable the Assembly Items feature. For more information, see [Enabling Assembly Items](#).
2. To create assembly item records that define the assembly components, select the parts that make up the assembly. For more information, see [Assembly Item Records](#).
  - To create an assembly item record, go to Lists > Accounting > Items > New
  - On the **New Item** page, click **Assembly**. For more information, see the help topic [Creating Item Records](#).
3. Record an assembly build:
  - After you create an assembly item record, enter an assembly build to record assembly production. Physically manufacturing assemblies in a production run increases your stock of assembled items.
  - To record inventory level changes, go to Transactions > Inventory > Build Assemblies to enter an assembly build for each production run. For more information, see [Building Assembly Items](#).
  - After you create your assembly item, build the assembly in NetSuite to replenish stock. To record an assembly build, go to Transactions > Inventory > Build Assemblies.

NetSuite tracks assembly item and member component records separately. It also tracks the assembly and member item stock status individually. For each assembly build you record:

- the assembly item stock level increases
- the member items' individual stock levels decrease

## Enabling Assembly Items

Before you can create and use assembly items, an administrator must enable the Assembly Items and Inventory features.

### To enable the Assembly Items and Inventory features:

1. Go to Setup > Company > Setup Tasks > Enable Features (Administrator).
2. Click the **Items & Inventory** subtab.
3. Check the **Assembly Items** box.
4. Check the **Inventory** box.
5. Click **Save**.

## Assembly Item Records

You can create an assembly item record to track each assembly and its component items. The assembly record details member items and the quantity of each member required for each assembly. For more information, see [Building Assembly Items](#).



**Important:** With the release of NetSuite 2023.1 Bills of Materials, where components are embedded to an Assembly Item, will no longer be supported. Only business critical issues will be fixed. To continue working with this functionality you should transition to the [Advanced Bill of Materials](#) feature, which is free of charge. After you enable this feature, NetSuite will automatically migrate all of your Bill of Materials to the new structure.

## Available Member Items

The following table displays the available member item types:

| Regular Inventory Assembly | Serial or Lot Numbered Inventory |
|----------------------------|----------------------------------|
| Inventory                  | Inventory                        |
| Non-Inventory              | Serialized Inventory             |
| Other Charge               | Lot Numbered Inventory           |
| Service                    | Non-Inventory                    |
| Regular Assemblies         | Other Charge                     |
| Kit Items                  | Service                          |
|                            | Regular Assemblies               |
|                            | Serialized Assemblies            |
|                            | Lot Numbered Assemblies          |

## Assembly Items in Item Lists

To make an inventory item available in your assembly build but not for sale, don't choose an income account on the item record.

By not setting an income account on an inventory item's record, the item doesn't appear in the sales transaction's items list. The item is sold as part of the finished goods assembly the item belongs to, which has its own income account.

If you don't set an expense account on a non-inventory item for resale, the item doesn't appear in the purchase transaction's item list.

## Assemblies and Serial/Lot Numbered Members

Serialized and lot numbered items cannot be included as member items in a regular assembly item.

- A non-serialized assembly cannot include a serialized or lot numbered member item.
- A non-lot numbered assembly cannot include a serialized or lot numbered member item.

Serialized or lot numbered inventory items can be members of an assembly only if the assembly is serialized or lot numbered.

- A serialized assembly can include a serialized or lot numbered member item.
- A lot numbered assembly can include a serialized or lot numbered member item.

## Assemblies on Purchase Transactions, Web Sites, and Work Orders

NetSuite lets you make assembly items available in other product areas.

### Purchase Transactions

To make assemblies available to purchase transactions, go to Setup > Accounting > Preferences > Accounting Preferences. On the Order Management subtab, check the **Allow the Purchase of Assembly Items** box on the Order Management subtab, and then save. For information about preferences, see the help topic [Items/Transactions Accounting Preferences](#).

### Web Sites

You can offer an assembly in your web site using the **Store** and **Specials** subtabs. Edit the assembly item record to add the item to your web site.

### Work Orders

The Work Orders feature enables you to track the production of assembly items for stock or to fill orders. Work orders track the quantities of assemblies to be built, and the required quantities of components, or member items. Use **Special Order Work Orders** for a particular sale. Use **Production Work Orders** to increase stock. For more information, see [Assembly Work Orders](#).

## Matrix Assemblies

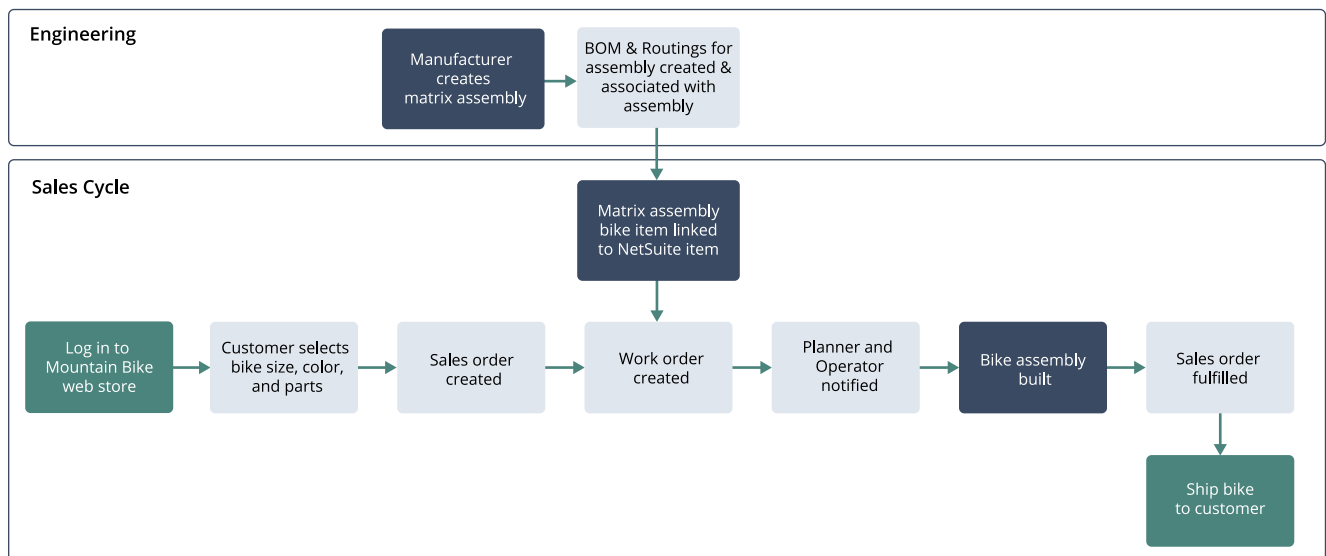
To use Matrix Assemblies, an administrator must enable the [Matrix Items](#) feature.

With the Assembly Items and Matrix Items features, you can create assembly item records that contain matrix options. These options help improve assembly item planning and production tracking. Matrix options make it possible to use Bills of Materials (BOM) and Manufacturing Routings to build items with numerous product styles and variations.

During production, you can create matrix assemblies using work orders and assembly builds.

For example, Wolfe Manufacturing produces mountain bikes and sells them only on their web site, not through retailers.

1. A customer logs on to the Wolfe web store to order a bicycle.
2. While completing the sales order, the customer selects the size of the bike, its color, brakes, wheels, and other components.  
The matrix assembly record links the custom bike created and sold in the web store to the item in NetSuite.
3. NetSuite automatically converts the completed sale order into a work order.  
Each bike configuration must be associated with a BOM and routing. These describe how to build the bike, components needed, manufacturing costs, and subsequent selling price.
4. The bicycle matrix assembly record is linked with the item in NetSuite.  
This informs planners and production operators which components are required for the BOM and which manufacturing routing will be used to assemble it.
5. After the bike is assembled, the sales order can be fulfilled.
6. The bike is shipped to the customer.



Matrix assembly records enable you to manage a BOM at the parent matrix assembly level and then implement BOM changes to specific configurations. BOM options can also be maintained on an individual item record basis.

- You can track demand for matrix assemblies and then make them available in the web store. You can also build them using work orders or assembly builds.
- For web store matrix assembly items, web store administrators can display only a parent matrix assembly item. Then make configuration options available in a list.

After matrix assembly sub-items are created, you may need to update the following matrix items:

- The BOM and its attributes such as the Cost Of Goods Sold (COGS) account or Unit of Measure.
- The parent assembly item and update some or all sub-items at one time.

For more information, see [Manufacturing Routing](#).

## Creating Matrix Assembly Records

You can create a matrix assembly item record in the same way you create other matrix items.

**Note:** After you create a matrix assembly, the Effective Date and Obsolete Date fields don't appear on the item record Component subtab.

To use BOM Control or set effective and obsolete dates, set them on the parent matrix assembly item record. Revision control must be set on the matrix assembly child item records. For more information, see [Revision Control BOM Management](#).

### To create matrix assembly records:

1. Select one of the following options:
  - **The Matrix Assistant:**
    1. Go to Lists > Accounting > Items > New.
    2. Click **Matrix Item Assistant**.

For more information, see the help topic [Using the Matrix Item Assistant](#).
  - **Automatically in NetSuite:**
    1. Manually create a parent matrix assembly item.
    2. Define options to create the new assembly item.
    3. Go to Lists > Accounting > Items > New.
    4. Click **Create Matrix Items**.

For more information, see the help topic [Creating a Matrix Item Manually](#).
2. Click **Save**.

## Updating Matrix Assembly Items

To learn how to update a matrix assembly item, see the help topic [Editing Matrix Items](#).

When you edit a matrix assembly item parent record, you can update child item BOMs.

### To update child item BOMs:

1. Click **Update Matrix**.
  - Clear the **Update BOM of Matrix Sub-items** box to not apply parent record changes to the child items.
  - Check the **Update BOM of Matrix Sub-items** box to apply parent record changes to the child items. The child items will be updated with the parent item BOM.

**Note:** Previous BOMs aren't updated with the changes entered.

2. Click **Save**.

## Matrix Assemblies on Transactions

Work order and assembly build transactions permit you to select only child matrix assemblies. The parent matrix assemblies aren't displayed in the list.



## Phantom Assemblies

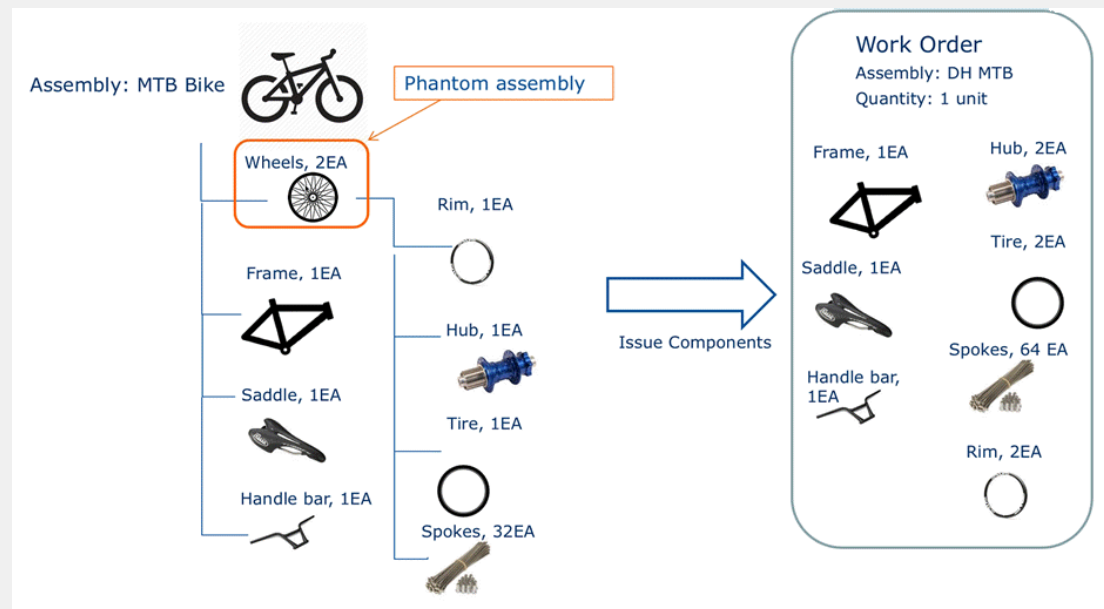
A phantom assembly is typically a non-stocked assembly that groups the components needed to produce a subassembly. For example, compare a phantom assembly to folders on a computer. The folder represents the phantom and the individual files are the components. They are called phantoms because they aren't counted as inventory items, but are created to fulfill the requirements of a higher-level assembly.

### Phantom Assemblies

Phantom assemblies provide the following advantages:

- No need to create multiple work orders. The phantom assembly is added to the higher-level work order.
- They work as build-to-order instructions. Components don't have to be issued from stock beforehand.
- They simplify Bill of Materials (BOM) management. If a component is changed on the Phantom BOM, it is automatically reflected in all BOMs that use the phantom as a sub-assembly.
- You can use them as configuration options for manufactured products when it would not be cost effective to keep the assembly in stock.

For example, Wolfe Manufacturing can use a phantom wheel assembly to build the wheels as part of the build assembly. With this phantom wheel assembly, Wolfe doesn't keep certain wheel types in stock. In this example, the phantom could include the following:



After you add a wheel assembly to the work order, you issue the components. Then, the system issues the components that make up that assembly and then adds them to the work order. For more information, see [Marking Assemblies to Create Work Orders](#).

While phantom assemblies are typically used as components in a larger assembly build, they can also be used to create stock items. For example, as part of a warranty claim, a batch of replacement wheels could be manufactured using a phantom assembly.

## Creating a Phantom Assembly

Use the following instructions to create a phantom assembly.

## To create a phantom assembly:

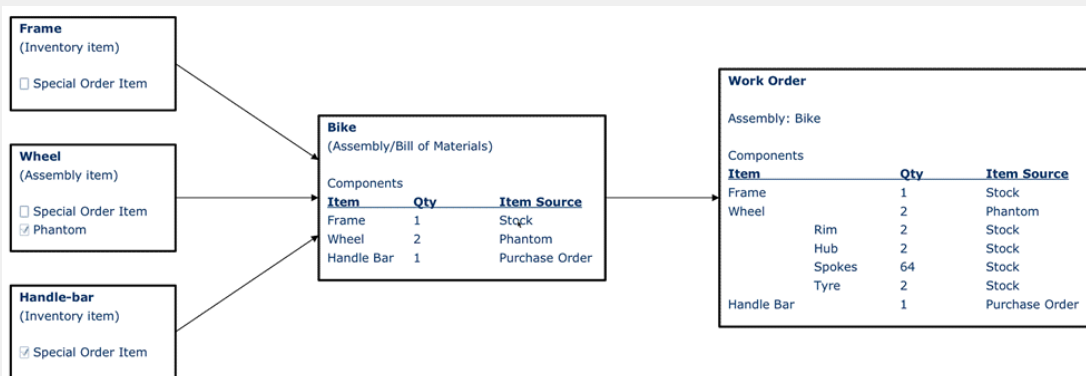
1. Go to Lists > Accounting > Items > New
2. Select **Assembly/Bill of Materials** from the **Item Type** column.
3. Complete the fields. Required fields display a red asterisk (\*). For more information, see the help topic [Creating Item Records](#).
4. On the **Purchasing/Inventory** tab Inventory Management section, check the **Phantom** box.
5. On the **Components** subtab, add the required components for the assembly and then update the **Item Source** field as required. For more information, see [Item Source Field](#).
6. If the phantom assembly contains sub-assemblies, check the **Mark Sub-Assemblies Phantom** box to mark them as phantoms.
7. Click **Save**.

## Item Source Field

The Item Source column appears on Bill of Materials (BOM), BOM Revision Record, and Work Order transactions. It is used to specify the preferred method of supply for a particular item. Source options include: Stock, Phantom, Purchase Order, and Work Order. The Assembly and BOM Revision Record item source values are then used as work order and assembly build defaults. This is then reflected in the Assembly/Bill of Material and Work Order records.

On assembly builds, NetSuite uses the item source values (stock or phantom) from the assembly and BOM revision record. However, the column field item source doesn't appear. The system expands the list of components on the assembly build when the list of components contains a sub-assembly with a phantom item source. You can override these default values on either the BOM record or the work order record.

The following diagram illustrates how NetSuite determines default item source values.



- The **Frame** is an inventory item. When the Special Order Item box is clear, the Item Source is marked as Stock.
- The **Wheel** assembly item is marked Phantom. Because the mark Sub-Assemblies Phantom box was clear, the wheel assembly work order is marked Phantom, but its sub components are taken from stock. If the Special-Order Item box is checked, the item source is marked Work Order.
- The **Handle bar** is normally an inventory item. When the Special Order Item box is checked, the Item Source is marked Purchase Order.

## Item Source Set to Phantom

When the Item Source field is set to Phantom, the sub-assembly components appear indented under the Phantom on the Bill of Materials. This cannot be edited.

In the following screenshot, Assembly Build for 0810 AssyB has two components. Component 0810 AssyA is an assembly and the 0810 AssyB BOM has Item Source set to Phantom, which is why the text is indented.

**Assembly Build** 6

Save Cancel Actions

**Primary Information**

Reference # 6

Quantity to Build 1

Inventory Detail

Date 25.5.2024

Posting Period May 2024

Memo

**Classification**

Department

Location East Coast : Georgia

Profit center

Class

Components Communication System Information Custom E-Document

Customize

| Component     | Quantity | Quantity On Hand | Units | Inventory Detail |
|---------------|----------|------------------|-------|------------------|
| 0810 InvItemB | 1        | 12               |       |                  |
| 0810 AssyA    | 1        |                  |       |                  |
| 0810 InvItemA | 1        |                  |       |                  |

Save Cancel Actions

## Inventory Detail Field

When the Item Source field is set to Phantom, the component sub-assembly is built as a part of the higher-level assembly.

For example, Wolfe Manufacturing sells road bikes in multiple configurations and have created a BOM for each configuration. The company doesn't stock carbon-fiber wheels, but has the wheel components in inventory and only assembles them when ordered. The carbon-fiber wheel is a sub-assembly of the road bike assembly and the Road Bike BOM. The item source for the carbon-fiber wheel is phantom.

Assembly and Inventory Items can be assigned a serial or lot number when you produce or receive them. After the item is consumed, in the **Work Order Item** subtab **Inventory Detail** field, select the serial or lot number being used.

## Phantom Bill of Materials on Assembly and Work Orders

To use the Phantom Bill of Materials functionality, the Assembly Items and Work Orders features must be enabled. These features are located on the Items & Inventory subtab on the Enable Features page.

When these features are enabled, phantom assembly options appear on the assembly and work order records.

The following table describes the phantom-related options:

| Option                             | Visible on                                                          | Action Performed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Phantom</b>                     | Assembly/Bill of Materials record                                   | Check this box to mark the Item Source for this item on any bill of materials as Phantom by default. This also applies to the item source value on any work order where the item is used.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Mark Sub-Assemblies Phantom</b> | Assembly/Bill of Materials and Work Order records                   | <p>Check this box to mark all sub-assemblies on the work order as Phantom. The sub-assemblies will be built as part of the higher-level work order rather than drawn from stock.</p> <p>Clearing this box doesn't refresh the items in the Item subtab. It enables in-line editing of individual components.</p> <p>Clearing the Mark Sub-Assemblies Phantom box doesn't refresh or remove sub-assembly components on the Items subtab. This is due to support of in-line editing of individual components. To reload the BOM for a top level assembly, select a different assembly in the Assembly field. To reload a BOM for a phantom sub-assembly, change the item source for the sub-assembly back to Phantom.</p> |
| <b>Item Source</b>                 | Components subtab Assembly/Bill of Materials and Work Order records | Adds Phantom as an option on the Item Source list. Enables sub-assembly components to be treated as phantoms on one BOM, and regular stock items on another.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

## Phantoms with Manufacturing Routing and Demand Planning

When you use manufacturing routing, define the routing steps associated with the sub-assembly build on the manufacturing routing of the higher-level assembly. For more information, see [Creating a Manufacturing Routing](#).

An assembly demand plan with a phantom sub-assembly in its bill of materials calculates dependent demand for components of the phantom item. However, it doesn't calculate for the phantom itself. For more information, see the help topic [Demand Planning](#).

## Assemblies and Units of Measure

To identify units of measure on assembly item records, the Multiple Units of Measure and the Assembly Items features must be enabled.

After you select a basic unit type on an assembly record you can define a default unit for the item on a particular transaction. On item records, designate a unit to default as a purchase unit, stock unit, or sales unit.

**Note:** You can set purchase and sale units when you create the item record. After you save the record, these units cannot be changed. You can make a selection for the stock units when you create the item record. You can change the stock units after you save the record.

On Assembly Builds, Assembly Unbuilds, and Work Orders for Assembly Items with Units of Measure, the Units field defaults to base units. You cannot change the default.

On transactions, units of measure are used as follows:

| Transaction                        | Editable Units                               | Display Units Only  |
|------------------------------------|----------------------------------------------|---------------------|
| Receive Order                      | Yes (in units that match the purchase order) |                     |
| Fulfill Orders                     | Yes (in units that match the sales order)    |                     |
| Adjust Inventory Worksheet         | Yes (in stock units)                         |                     |
| Replenish Location                 |                                              |                     |
| Build Assemblies                   |                                              | Yes (in base units) |
| Unbuild Assemblies                 |                                              |                     |
| Enter Work Orders                  |                                              |                     |
| Create Opportunities               | Yes                                          |                     |
| Prepare Estimates                  |                                              |                     |
| Enter Sales Orders                 |                                              |                     |
| Create Invoices                    |                                              |                     |
| Enter Purchase Orders              |                                              |                     |
| Bill Purchase Orders               |                                              |                     |
| Enter Vendor Return Authorizations |                                              |                     |
| Adjust Inventory                   |                                              |                     |
| Transfer Inventory                 |                                              |                     |
| Issue Credit Memos                 |                                              |                     |
| Write Checks                       |                                              |                     |
| Use Credit Card                    |                                              |                     |

- Purchase orders show the item in purchase units by default.
- Invoices show the item in sales units by default.
- Inventory adjustments show the item in stock units by default.

Generated reports display units of measure based on the units used in transactions.

## Serialized Assemblies

When working with serial-numbered assembly items:

- The base unit must be the lowest unit of measure when selecting a **Units Type**.
- Units cannot have a decimal unit of measure conversion with the base units when selecting **Stock Units**, **Purchase Units**, and **Sale Units**.
- The quantity of serial numbers entered must equal the quantity in base unit when a transaction is entered in a **non-base units**.

For example, you sell a serialized assembly that uses a base unit of Each and sale unit of Dozen. When you enter a sales order for 2 of the assembly, enter 24 serial numbers on the order to represent 24 each of the item.

## Lot Numbered Assemblies

After setting up a lot-numbered assembly item record, you can edit the purchase and sale units. You cannot, however, change the units type or stock units.

## Assembly Builds and Unbuilds

Assembly Builds and Unbuilds must be performed in the base units of measure. For more information, see the help topic [Using Item Records](#).

## Assemblies and Work Orders

If the assembly selected on a work order uses Units of Measure, they appear in the Units field.

Work orders you view that were previously generated by NetSuite display the units for reference only and cannot be changed.

If the work order was created from another source (such as a sales order), you can change the quantity by changing the source transaction. For more information, see [Assembly Work Orders](#).

## Building Assembly Items

Each time you physically manufacture assemblies in a production run, you increase your stock of the assembled items. Record each production run and update stock levels by entering an assembly build record.

### Building Assembly Items

For each assembly build you record:

- the assembly item stock levels increase
- the member items' individual stock levels decrease

Entering an assembly build for each production run updates your inventory levels.

 **Note:** If you use work orders and have entered work orders for assemblies, see [Building Work Orders](#) to complete builds for those assemblies.

### To enter an assembly build:

1. Go to Transactions > Manufacturing > Build Assemblies.
2. Under the **Primary Information** section, complete the following fields:
  - a. The **Reference #** field displays a system generated number.  
You can enter a different reference number to track this transaction.
  - b. In the **Assembly** list, select the assembly item you want to build.  
You can enter an assembly build only for assembly items on record. For more information, see the help topic [Creating Item Records](#).
  - c. The **BOM Revision** is defaulted based on the transaction date.  
The revision record effective date determines when this item is included as a member for an assembly.

When you select an assembly item, the maximum number that you can build appears in the **Buildable Quantity** field.

If you use locations, the quantity for the selected location is displayed in the Buildable Quantity field.

- d. In the **Quantity to Build** field, enter the number of assembly items you want to build.
- e. The **Projected Value** of your new assemblies is displayed.  
Projected value is the sum of the value of the member items times the quantity entered.
- f. To add assemblies to other bins, click the **Inventory Detail** icon.
- g. If you use serialized inventory, enter serial numbers for the assemblies you are building.
- h. In the **Date** field, enter a transaction date.
- i. Select the account **Posting Period** you want to post this transaction to.

You cannot post to closed periods.

- j. If you use bin management, the assembly item record preferred bin number is displayed in the **Bin Numbers** field. By default, all assemblies are added to the preferred bin. To add some assemblies to other bins, click the Bins icon or the Inventory Detail icon.

When you build this assembly, the bin quantity on hand for the assembly increases and the bin quantity on hand for each member item decreases.

- k. If you are building a lot numbered assembly item, enter the lot **Expiration Date** field.
  - 1. To receive a warning that a lot is about to expire, go to Setup > Accounting > Preferences > Accounting Preferences.
  - 2. On the **Items/Transactions** subtab, in **Days Before Lot Expiration Warning**, enter the number of days before a lot item's expiration to display a warning.
  - 3. Click **Save**.

- l. Enter a **Memo**. You can search for this text later to find this entry.

- 3. In the **Classification** section, complete the following fields:
  - a. If you use NetSuite OneWorld, select a **Subsidiary**.
  - b. If you track departments, select a **Department** for this transaction.
  - c. If you track classes, select a **Class** for this transaction.
  - d. If you track locations, select a **Location** for this transaction.
- 4. On the **Components** subtab, accept the default quantity of each component needed to complete the assembly. This number is taken from the assembly Item record.

You can adjust component quantities on a build-by-build basis to allow for fluctuations in material usage. For example, a member item record shows a component quantity of 2. You can create a build that has 3 of the component to fill a particular order. As the quantity changes, your projected value is updated.

**Note:** You can change the quantity of members on a serialized or lot numbered assembly. Use the assembly serial or lot numbers to track assembly items created with special member quantities. You cannot track non-serial or non-lot assemblies that could have special quantities of member items. Therefore, you may want to build non-serial or non-lot assemblies with special quantities. You can build when your available quantity of that assembly is zero and you are building assemblies with special quantities for a customer order.

- 5. Click the **Communication** subtab.
  - a. Enter events on the **Events** subtab. For example, maintenance, repair, or setup.

- i. Enter the event name or **Title**. For example, Assembly Work Center Setup.
- ii. Enter the event **Location**. For example, West Coast Assembly.
- iii. Accept today's **Date** or use the calendar to enter a different date.
- iv. If this is an all day event, check the **All Day** box.
- v. Enter event **Start** and **End Times**.
- vi. Click **Add**.

To add more events, repeat this procedure.

For example, Setup could start at 7:00 am and end at 7:30 am. Then, Assembly could start at 7:30 am and end at 3:30 pm. Finally, Breakdown could start at 3:30 pm and end at 4:30 pm.

- b. On the **Tasks** subtab, view or enter CRM tasks records. For information about tasks, see the help topic [Working with CRM Tasks](#).
- c. The **Phone Calls** subtab enables you to view or enter phone calls.
- d. The **Files** subtab enables you to attach files from the NetSuite File Cabinet, your computer, or the internet.
  - To add a file from the File Cabinet, select one from the **Attach File** list.  
After the file loads, NetSuite fills the **Folder**, **Size**, **Last Modified**, and **File Type** fields.
  - To add a new file, in the **Attach File** list, select **New**.  
Complete the List window.
- e. On the **User Notes** subtab, add and track notations.

- 6. Click **Save**.

After an assembly item is built, it is treated like an inventory item for inventory costing purposes. The built assembly item asset/costing value is the total value of the assembly's member items. These values act like the assembly item's purchase price for inventory costing calculations.

 **Note:** If you use NetSuite OneWorld, you can select the purchase currency on non-inventory items for purchase and resale.

Inventory costing is tracked for the assembly item based on the inventory costing method chosen at Setup > Accounting > Preferences > Accounting Preferences. For information about inventory costing, see the help topic [Using Item Records](#).

## Copy an Assembly Item

An assembly item is an inventory item made of several components, but identified as a single item. Assemblies are manufactured by combining raw materials you stock. Some assembly items have similar components or an assembly may be updated with some new slightly different components.

Instead of building a new assembly item, NetSuite enables you to copy an existing assembly item that you can then edit to reflect the differences in your new assembly.

For example, Your organization's flagship road bicycle is the RD Tour series. This year the RD Tour 5000 line is being built with an improved chain and gear components. Instead of building an new assembly for two new components, copy the previous RD Tour 5000 assembly item and update the component list.

### To copy an assembly item:



1. Go to Lists > Accounting > Items.
2. Beside the assembly item you want to copy, click **View**.
3. Click **Actions > Make Copy**.
4. Rename your new assembly item.
5. Complete the new assembly inventory item page.
6. Click **Save**.

## Unbuild Assembly Items

After you complete an inventory build, you may need to unbuild some assemblies. For example, Wolfe Manufacturing receives an order for one hundred mountain bikes. While building the bicycles, the customer cancels the order. Wolfe unbuids the assembly items that aren't sold, and maintains the stock as raw materials.

Unbuilding an assembly updates inventory levels on records for the finished assembly item and for each member component individually. For each assembly you unbuild:

- the assembly item stock level decreases
- the member item's individual stock levels increase

There are two methods to unbuild an assembly:

- You can unbuild an assembly from the Assembly Build transaction.
- You can generate a new unbuild transaction.

To learn more, see [Unbuilding an Assembly From the Assembly Build Transaction](#).

## Unbuilding an Assembly From the Assembly Build Transaction

NetSuite lets you unbuild an assembly from the original assembly build transaction. When you unbuild from this transaction, NetSuite generates a new unbuild transaction with all of the details from the original transaction.

### To unbuild an assembly from the assembly build transaction:

1. Go to Transactions > Manufacturing > Build Assemblies > List.
2. Click **View** next to the assembly you want to unbuild.
3. On the Assembly Build page, click **Unbuild**.

An unbuild transaction opens with the build information populated.

## Unbuilding an Assembly From a New Unbuild Transaction

If you don't want to unbuild an assembly from the original assembly build transaction, you can unbuild from a new unbuild transaction.

## To unbuild an assembly from a new unbuild transaction:

1. Go to Transactions > Manufacturing > Unbuild Assemblies
2. Under Primary Information, complete the following fields:
  - a. The **Reference #** field displays a system generated number. You can enter a different reference number to track this transaction.
  - b. In the **Assembly** list, select the assembly item you want to unbuild.  
Required fields display a red asterisk (\*).  
  
When you select an assembly item, the maximum number that you can unbuild appears in the **Quantity Built** field.  
  
If you use locations, the quantity that appears in the **Quantity Built** field is for the selected location.

 **Note:** If you use Multi-Location Inventory, when you select an assembly item and location, the maximum number you can unbuild appears in **Quantity Built**. If you don't select a location, the maximum number that you can unbuild doesn't appear in this field.

  - c. If you use serialized inventory, enter the **Serial Numbers** for the assemblies you plan to unbuild.
  - d. If you use bin management, select the **Assembly Item Record Bin Number** from **Bin Numbers** list. By default, all assemblies are removed from the preferred bin. To remove some items from other associated bins, click the Bins icon or Inventory Detail icon.  
  
When you unbuild this assembly, the bin quantity on hand decreases, and the bin quantity on hand for each member item increases.
  - e. If you enabled Advanced Bill of Materials, select a **Bill of Materials**.
  - f. If you enabled Advanced Bill of Materials, select a **Bill of Materials Revision**.
  - g. In the **Quantity to Unbuild** field, enter the number of assembly items you want to unbuild.  
You cannot enter a quantity that exceeds the number in the **Quantity Built** field.
  - h. The projected value of your disassembled items appears in the **Projected Value** field.  
Projected Value is the sum of the value of the member items times the quantity entered.
  - i. In the **Date** field, enter the transaction date.
  - j. If you use accounting periods, select a transaction **Posting Period**. You cannot post to a closed period.
  - k. Enter a **Memo**. You can search for this text later to find this entry.
3. In the Classification section, complete the following fields:
  - a. If you use NetSuite OneWorld, select a **Subsidiary**.
  - b. If you track departments, select a **Department** for this transaction.
  - c. If you track classes, select a **Class** for this transaction.
  - d. If you track locations, select a **Location** for this transaction.
4. The **Components** subtab provides the inventory detail for each component.
5. Click the **Communication** subtab.
  - a. Enter events on the **Events** subtab. For example, maintenance, repair, or setup.
    - i. Enter the event name or **Title**. For example, Assembly Work Center Setup.
    - ii. Enter the event **Location**. For example, West Coast Assembly.

- iii. Accept today's **Date** or use the calendar to enter a different date.
- iv. If this is an all day event, check the **All Day** box.
- v. Enter event **Start** and **End Times**.
- vi. Click **Add**.

To add more events, repeat this procedure.

For example, Setup could start at 7:00 am and end at 7:30 am. Then, Assembly could start at 7:30 am and end at 3:30 pm. Finally, Breakdown could start at 3:30 pm and end at 4:30 pm.

- b. On the **Tasks** subtab, view or enter CRM tasks records. For information about tasks, see the help topic [Working with CRM Tasks](#).
  - c. The **Phone Calls** subtab enables you to view or enter phone calls.
  - d. The **Files** subtab enables you to attach files from the NetSuite File Cabinet, your computer, or the internet.
    - To add a file from the File Cabinet, select one from the **Attach File** list.  
After the file loads, NetSuite fills the **Folder**, **Size**, **Last Modified**, and **File Type** fields.
    - To add a new file, in the **Attach File** list, select **New**.  
Complete the List window.
  - e. On the **User Notes** subtab, add and track notations.
6. Click **Save**.

After you save the transaction, your inventory count of the assembly item decreases and the inventory count of the member items increases accordingly.

## Assembly Unbuild Variance Posting

Assembly item unbuids could generate a difference between the item cost at assembly time and the item cost at unbuild time. In the assembly item record **Unbuild Variance** field, select an account for posting variance amounts.

For example, when you unbuild an assembly, each component is restocked and the item cost for each component is calculated. Any variance between the assembly cost and the unbuild cost posts to the selected Unbuild Variance account on the item record for that assembly.

If no variance account is selected for an assembly item, variance amounts for that item post to the Cost of Goods Sold (COGS) account.

### To set the variance account for an item:

1. To open the assembly item record, go to Lists > Accounting > Item.
2. In the **Unbuild Variance Account** field, select the account where you want to post variance amounts.
3. Click **Save**.

## Assembly Unbuilds and the Adjust Inventory Worksheet

When you enter an Adjust Inventory Worksheet, it deletes the previous assembly unbuild cost history. To maintain costing history for an assembly, enter an inventory adjustment, not a worksheet.

## Unbuilding Assemblies and Purchased Assembly Costing

If you use the **Allow Purchase of Assembly Items** preference, the following paragraph explains how NetSuite handles costing for assemblies you unbuild. For information about manufacturing preferences, see the help topic [Manufacturing Preferences Overview](#).

When unbuilding an assembly item, NetSuite uses historical costing to determine member item values. For example, a member item's historical transactions display a cost of \$25. NetSuite uses this amount to calculate cost for the member item after the assembly unbuild.

If the assembly being unbuild was purchased from a vendor, a member item has no previous transactions. The historical cost to be considered is \$0 and member items in this case would have a value of \$0.

If you want to adjust your inventory prior to an unbuild or rather than unbuilding an assembly, refer to the following procedures:

- [Entering an Inventory Adjustment Before the Unbuild](#)
- [Entering an Inventory Adjustment Rather Than an Assembly Unbuild](#)

### Entering an Inventory Adjustment Before the Unbuild

You can enter an inventory adjustment before the unbuild to establish item cost. Add a quantity of 1 and save the form, edit the form, and then enter a new line to remove a quantity of 1.

The following example describes the transaction sequence for assembly item ABC.

#### To enter an inventory adjustment prior to unbuild:

1. Go to Transactions > Inventory > Adjust Inventory.
2. In the **Adjustments** subtab, in the **Item** field, select an item. For example, Member item ABC.  
Required fields display a red asterisk (\*).
3. Enter a value in the **Adjust Qty. By** field. For example, enter 1.
4. Enter a **Current Value**. For example, \$25.00.
5. Click **Save**.
6. Click **Edit** on the adjustment.
7. In the **Item** field on a new line, select the item you edited. In this example, click Member item ABC.
8. Enter a quantity of **-1**.
9. Verify a value of **\$25.00**.
10. Click **Add** and then click **Save**.

When the assembly is unbuild, member item ABC is valued at \$25.

### Entering an Inventory Adjustment Rather Than an Assembly Unbuild

In the following procedure, you enter an inventory adjustment rather than an assembly unbuild. This removes the assembly from stock and adds the member items.

#### To enter an inventory adjustment instead of an assembly unbuild:

1. Go to Transactions > Inventory > Adjust Inventory.

2. In the **Item** field, select **Assembly Item XYZ**.  
Required fields display a red asterisk (\*).
3. Enter a quantity of **-1**.
4. Enter a value of **\$100.00** and then click **Add**.
5. In the **Item** field, select **Member item 1**.
6. Enter a quantity of **1**.
7. Enter a value of **\$75.00** and then click **Add**.
8. In the **Item** field, select **Member item 2**.
9. Enter a quantity of **2**.
10. Enter a value of **\$25.00**.
11. Click **Add** and then click **Save**.

The assembly is removed from inventory and the appropriate number of member items is added to inventory.

## Marking Work Orders Built

NetSuite enables you to show work order items as assembled without completing all the steps.

When you mark an order as built, the required items are marked built and added to inventory.

 **Note:** Associated variances aren't created when you mark an order built.

### To mark a work order built:

1. Go to Transactions > Manufacturing > Mark Work Orders Built.
2. Select a **Location** to filter the list of orders.
3. Select an **Item** to filter the list of orders.
4. Select a **Customer** to filter the list of orders.
5. Check the **Mark Built** box next to each order.
6. Click **Save**.

## Marking Work Orders Closed

For some work orders you may want to show the items as being assembled without completing all the steps. In such a case you can skip the steps and mark the order as closed.

When you mark an order as closed, the required items are marked closed and added to inventory.

 **Note:** Associated variances are created when you mark an order built. For information about variances, see the help topic [Using Item Records](#).

### To mark a work order closed:

1. Go to Transactions > Manufacturing > Close Work Orders.
2. In Primary Options and Criteria, complete the following:

- a. Select the **Posting Period** you want to post this transaction to. You cannot post to a closed period.
  - b. Enter a transaction **Date**.
  - c. Select a transaction **Location**.
  - d. Select an assembly **Item** to filter the list and show only work orders for the item.
  - e. Select a **Customer** to filter the list for transactions associated with the customer. Select All to show all transactions.
  - f. Check the **Include In-Process Work Orders** box to include work orders with In Process and Built status.
  - g. In the **Under-Produced Variance Tolerance (%)** field, enter a percentage to close only orders that produced less than planned. This helps you determine whether to keep some orders open because they haven't produced enough finished product.  
 Order variance calculation:  $(\text{quantity ordered} - \text{quantity built}) / \text{quantity ordered} = \text{variance percentage}$   
 Entering a percentage in this field filters the list of orders to close to show only orders with a variance lower than the percentage entered.
  - h. In the **Production Variance Tolerance (%)** field, enter a percentage to filter orders with a specific value variance.  
 Order variance calculation:  $\text{absolute value (remaining WIP value} / \text{WIP of assembly)} = \text{variance percentage}$   
 Entering a percentage in this field filters the list of orders to close to show only orders with a variance lower than the percentage entered.
3. In Date Range Criteria, choose from the following:
    - To filter by an order date range, enter a beginning date in the **Order Date From** field. Enter an end date in the **Order Date To** field.
    - To filter by a production start date range, enter a beginning date in the **Production Start Date From** field. Enter an end date in the **Production Start Date To** field.
    - To filter by a production end date range, enter a beginning date in the **Production End Date From** field. Enter an end date in the **Production End Date To** field.
  4. In Orders, complete the following:
    - a. Check the **Close** box next to each order to be closed.
    - b. Verify the quantities and variances displayed for each order.
    - c. Click **Submit**.

## Printing an Item Bill of Materials

A Bill of Materials (BOM) lists all the components of your assembly item, the assembly quantity, and the total quantity for each.

If one of the components of your assembly item is an assembly item, each item subcomponent appears. Each subcomponent displays the quantity needed for each subcomponent to complete the assembly.

For example, your assembly item contains four components—Item A, Item B, Item C, Item D. Item B is an assembly item made up of Widget 1 and Widget 2. Two of each widget are necessary for assembly of Item B. And two of Item B are needed to complete the parent assembly. The quantity needed to complete assembly Item B must be doubled to complete the parent assembly.

The following table displays the assembly bill of materials:

**Parent Assembly**

| Name     | Assembly Quantity | Total Quantity |
|----------|-------------------|----------------|
| Item A   | 1                 | 1              |
| Item B   | 2                 | 2              |
| Widget 1 | 2                 | 4              |
| Widget 2 | 2                 | 4              |
| Item C   | 5                 | 5              |
| Item D   | 1                 | 1              |

**To print a bill of materials for an assembly item:**

1. Go to Lists > Accounting > Items.
2. To open the **Filters** section, click the + icon.
3. In the **Type** field, select **Assembly** to filter your item list to show assembly items.
4. Click **View** next to the item you want to print the bill of materials (BOM) for.
5. To print the BOM, click the **Print** icon.
6. When the window containing your BOM opens, click the **Print**  icon.

You can also click **Export** in this window to open or save your BOM as a CSV file.

To learn another way to view BOM details, see [Bill of Materials Inquiry](#).

To learn how to print a bill of materials for work orders that you enter, see [Printing a Work Order Bill of Materials](#).

## Bill of Materials Inquiry

If you use Assembly Items, the Bill of Materials (BOM) Inquiry enables you to see the build requirements for an assembly item. The BOM Inquiry displays the member components of the assembly, and the number of each component needed for each assembly.

For example, you can run BOM Inquiry to identify the materials needed to assemble a Mountain Bike. The inquiry shows that you need two wheels, one frame, one seat, and one handle bar. The inquiry also shows the sub-assembly (phantom assembly) components for each wheel: one rim, one hub, one tube, one tire, and spokes.



**Note:** Custom roles must specify access to view this inquiry. For more information, see the help topic [Customizing or Creating NetSuite Roles](#).

**To run a Bill of Materials Inquiry:**

1. Go to Transactions > Manufacturing > Bill of Materials Inquiry.
2. Select the assembly you want to show a BOM for.
3. If you use the multi-location inventory feature, select a **Location** to view data for that location.  
Required fields display a red asterisk (\*).  
If you use Advanced BOM, see [Running a BOM Inquiry](#).

4. Choose a level of detail:
  - Check the **Top Level Only** box to show only the top level member items details (sub-assembly information isn't shown).
  - Clear the **Top Level Only** box to show details about all levels of member items.
5. In the **BOM Display Control** list, select **By Date** or **By Revision** to decide what to include in an assembly.



**Note:** This field is available only when an assembly item uses Revision Control for its Effective BOM Control value.

When assembly items use Effective Date as Effective BOM Control value, the **Date** field determines the date when querying components of the assembly item.

Components are displayed based on the following criteria:

- Selected date is greater than or equal to the effective date.
- Selected date is less than or equal to the obsolete date.

If an assembly item uses Revision Control, you can query the components either By Date or By Revision.

- If you select **By Date**, then follow the process described in step 5 above.
- If you select **By Revision**, in the **Revision** field, select a revision.

The active components for the revision are displayed and the date field displays the revision effective date.

The inquiry displays all components used in a multi-level bill of materials structure, using a nested tree view.

The Bill of Materials Inquiry displays the following items:

| Column Label                           | Explanation                                                                                                                                                                                                                                                     |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Component Name</b>                  | The name of the Component as defined in the Item Name field                                                                                                                                                                                                     |
| <b>Level</b>                           | Where the component appears on the BOM tree structure                                                                                                                                                                                                           |
| <b>Component Yield</b>                 | Shows how much of this component is available for final assembly, after accounting for loss/scrap in the production process<br><br>A yield factor of 0.9 means that 90% of the usage quantity of the component on a bill becomes part of the finished assembly. |
| <b>BOM Quantity per Assembly</b>       | The quantity required for this assembly according to the BOM                                                                                                                                                                                                    |
| <b>Quantity per Assembly</b>           | The quantity required when component yield is taken into consideration                                                                                                                                                                                          |
| <b>Quantity per Top Level Assembly</b> | The total quantity of this component required to make the top-level assembly<br><br>Top level items are typically finished products. For example, a Barbecue grill set.                                                                                         |
| <b>On Hand</b>                         | Number of items physically held at the specified location                                                                                                                                                                                                       |
| <b>Available</b>                       | Uncommitted stock of item                                                                                                                                                                                                                                       |
| <b>Back Ordered</b>                    | The quantity of any unfulfilled order or existing commitment for this component                                                                                                                                                                                 |
| <b>On Order</b>                        | The total quantity ordered of this component across all current work orders                                                                                                                                                                                     |

To display information stored in custom item fields for each component, click **Customize**.



## Export or Print a Bill of Materials Inquiry

For more flexibility to work with data outside of NetSuite, you can print or export the results of a Bill of Materials Inquiry.

Click the printer icon at the top of the inquiry page to print or export (Excel or CSV) the inquiry results.

**Note:** The data in the inquiry header (such as location and date) isn't exported. Only the table data resulting from the inquiry is exported. Only the columns shown on the query export page can be exported or printed.

For more information, see [Printing an Item Bill of Materials](#) for an alternate way to view BOM details.

## Costed Bill of Materials Inquiry

Costed BOM Inquiry enables you to examine the standard cost breakdown of an assembly. The Costed BOM Inquiry report displays individual and aggregated assembly costs. This report shows how different cost components roll up to an assembly item based on the Bill of Materials (BOM). This includes material and conversion costs (labor and machine costs).

For example, the Costed BOM Inquiry helps a cost accountant in a manufacturing environment see the different cost components associated with building an assembly. It can also show how each component is calculated and rolled up to the finished goods.

This provides visibility into variances between inventory values at a certain time (displayed on the Inventory Valuation reports). It also provides visibility in the desegregated component costs of the assembly.

**Note:** To use the Costed BOM Inquiry, enable the Standard Costing and Assemblies features. This inquiry can be run only for a standard cost assembly item.

After you select the subsidiary, location, and assembly, the inquiry displays the cost breakdown of the assembly and its member components. This assembly cost is calculated based on the standard cost of components.

For assemblies that have a conversion cost, the assembly cost is calculated as follows:

- Component cost is based on the component cost displayed in the inventory revaluation transaction.
- Conversion cost is based on the assembly inventory revaluation transaction.

For assemblies that don't use routing, the assembly cost is based on the component cost displayed in the inventory revaluation transaction.

### To run a Costed Bill of Materials Inquiry:

1. Go to Transactions > Manufacturing > Costed Bill of Materials Inquiry.
2. If you use OneWorld, select a **Subsidiary**.
3. Select the **Assembly** you want to show a costed BOM for.
4. If you use the multi-location inventory feature, select the **Location** you want to view data for.
5. To show only the top level member items details check the **Top Level Only** box. Sub-assembly information isn't shown.  
Clear the Top Level Only box to show all member items.
6. To decide what to include in an assembly, select a **BOM Display Control** option:

- **By Date** – determines the date for querying assembly item components.

Components are displayed based on the following criteria:

- Selected date is greater than or equal to the effective date.
- Selected date is less than or equal to the obsolete date.

- **By Revision**

This field is available only when an assembly item uses Revision Control for its Effective BOM Control value.

When assembly items use Effective Date as the Effective BOM Control value, the date field determines the date querying components of the assembly item.

If an assembly item uses Revision Control, you can query the components either by Date or by Revision.

- If you select **By Date**, then follow the process described in step 5 above.
- If you select **By Revision**, then in the **Revision** field, select a revision.

The active components for the revision are shown and the **Date** field displays the revision **Effective Date**.



**Note:** If you have enabled the Advanced BOM feature, the Bill of Materials and Bill of Materials Revision lists will be displayed.

7. Select the **Bill of Materials** you want to run.

For more information, see [Advanced Bill of Materials](#).

The **Bill of Materials Revision** field is automatically populated.

8. To change the inquiry **Effective Date**, enter a date, or click the calendar icon to select a date.

- The **Top Level Material Cost** field displays the material cost for the top level assembly selected. This cost is summed for all components.
- The **Top Level Conversion Cost** field displays the routing cost for the top level assembly. This field is available only if you enable the Manufacturing Routing feature.
- The **Total Unit Cost** field displays the sum of Top Level Material Cost plus Top Level Conversion Cost.

Component costs are displayed only for standard cost components. Non-standard cost components are displayed with a cost of zero.

9. To display information stored in custom item fields for each component, click **Customize**.



**Note:** To display a value, sub-assemblies must be a standard cost type. If a sub assembly contains a component, the unit cost value is taken as a sum of all components costs.

## Export or Print a Costed Bill of Materials Inquiry

For more flexibility to work with data outside of NetSuite, you can print or export the results of a Costed Bill of Materials Inquiry.

Click the printer icon at the top of the inquiry page to print or export (Excel or CSV) the inquiry results.

The data in the inquiry header (location and date) doesn't get exported. Only the tabular data resulting from the inquiry is exported. Only the columns shown on the query export page can be exported or printed.

For more information, see [Printing an Item Bill of Materials](#) for an alternate way to view BOM details.

## Costed Bill of Materials

The Costed Bill of Materials SuiteApp calculates the total cost of an assembly based on a bill of materials revision, manufacturing routing, and cost templates used. The cost is calculated by adding up the costs of the components.

The SuiteApp also enables you to do the following:

- View the comprehensive cost breakdown of an assembly item, including bills of materials, operations, and cost types based on selected bill of materials revision.
- View and analyze the costs of each component needed for an assembly, including material and conversion costs for each component.
- Estimate the potential costs of manufacturing assembly items for bill of materials based on costing method, purchase price, and item default price.
- Access cost breakdowns for bills of materials as part of your product engineering and costing processes.

You can access the SuiteApp by navigating to Reports > Cost Accounting > Costed Bill of Materials.

 **Note:** The cost calculations are not based on planned standard cost, cost re-evaluation, or cost roll-up values for any standard cost assembly item.

Watch the following help video for an overview of the Costed Bill of Materials SuiteApp:

 [Costed Bill of Materials SuiteApp Overview](#)

## Installing Costed Bill of Materials

The Costed Bill of Materials SuiteApp requires a NetSuite OneWorld account.

Only users with Administrator role can install the SuiteApp. For more information about installing a SuiteApp, see the help topics [Installing from the SuiteApp Marketplace](#) and [SuiteApp Marketplace in NetSuite](#).

## Prerequisites for Installing Costed Bill of Materials

To use the Costed Bill of Materials SuiteApp, enable the features in the following table. For more information, see the help topic [Enabling Features](#).

To work with subsidiaries, you need access to a NetSuite One World account and have enabled Subsidiaries. To learn more, see the help topic [Working with Subsidiaries](#).

| Subtab            | Feature Name                     |
|-------------------|----------------------------------|
| Items & Inventory | Advanced Bill of Materials       |
|                   | Assembly Items                   |
| SuiteCloud        | Custom Records                   |
|                   | Server SuiteScript               |
|                   | Client SuiteScript               |
|                   | SuiteCloud Development Framework |



**Note:** Following are some general information for the Costed Bill of Materials SuiteApp:

- The currency of the cost displayed in the report is shown based on the default currency of the selected subsidiary.
- The subassemblies should have a master default bill of materials and the default routing for location to get the detailed estimated cost.
- The SuiteApp displays details in the **Materials Costs, Top Level Costs, and Fully Exploded Costs** subtabs irrespective of whether a work center, cost template, or routing is active or inactive.
- The SuiteApp supports up to 900 levels of assembly items and up to 400 sub-assemblies in each level.

## Creating or Customizing Roles and Permissions to Use Costed Bill of Materials

If you are an administrator, you can create new roles or customize roles for using the Costed Bill of Materials SuiteApp. For more information, see the help topics [Assigning Roles to an Employee](#) and [Customizing or Creating NetSuite Roles](#).

A custom role **Costed\_BOM\_Role** is created during the SuiteApp installation which will be used as a default value for **Execute As Role** field in the Script Deployment page. It is updated in the **Costed Bill of Materials** and **Costed BOM DataService** script.

Refer to the following table for minimum access levels of permissions required for SuiteApp deployments. These minimum permissions must be updated to the existing role or while creating a new role.

| Permissions                   | Minimum Access Level |
|-------------------------------|----------------------|
| Lists > Bill of Materials     | View                 |
| Lists > CRM Groups            | View                 |
| Lists > Custom Record Entries | Full                 |
| Lists > Documents and Files   | View                 |
| Lists > Items                 | View                 |
| Lists > Locations             | View                 |
| Lists > Perform Search        | View                 |
| Lists > Subsidiaries          | View                 |
| Lists > Units                 | View                 |
| Lists > Vendors               | View                 |
| Setup > Accounting Lists      | View                 |
| Setup > Custom Record Types   | Full                 |
| Setup > Enable Features       | Full                 |

| Permissions                                      | Minimum Access Level |
|--------------------------------------------------|----------------------|
| Setup > SuiteScript                              | View                 |
| Setup > SuiteScript Scheduling                   | Full                 |
| Setup > Set Up Company                           | Full                 |
| Custom Record > Costed BOM MR Assembly Instances | Full                 |

Refer to the following table for the minimum permissions required for logged-in user role.

| Permissions                                                                                                                                                                                             | Minimum Access Level |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| Lists > Bill of Materials                                                                                                                                                                               | View                 |
| Lists > Items                                                                                                                                                                                           | View                 |
| Lists > Subsidiaries                                                                                                                                                                                    | View                 |
| Lists > Locations                                                                                                                                                                                       | View                 |
| Lists > Documents and Files                                                                                                                                                                             | View                 |
| Lists > Manufacturing Routing                                                                                                                                                                           | View                 |
|  <b>Note:</b> This permission is required only if the <b>Manufacturing Routing and Work Center</b> feature is enabled. |                      |

 **Note:** The currency and unit symbols are displayed in the three subtabs (**Material Costs**, **Top Level Costs**, and **Fully Exploded Costs**). The symbols are displayed irrespective of the currency and unit permissions given to the logged-in user.

## Enabling Manufacturing Routing And Work Center Feature

When **Manufacturing Routing And Work Center** feature is enabled in the **Enable Features** page after the SuiteApp is installed, you must perform the following steps.

To enable Manufacturing Routing And Work Center feature:

1. Go to Setup > User/Roles > Manage Roles.
2. On the **Costed\_BOM\_Role** role row, click **Customize**. The Role page opens.
3. Create a custom role.
  - a. (Optional) To rename the custom role, enter the role name in the **Name** field.
  - b. In the **Permissions** subtab, click the **Lists** subtab.
  - c. In a new row, select **Manufacturing Cost Template**.
  - d. From the level list, select **View**.
  - e. Click **Add**.
  - f. From the list, select **Manufacturing Routing**.
  - g. From the level list, select **View**.
  - h. Click **Save**.
4. Add the custom role to the scripts:

- a. Go to Customization > Scripting > Scripts.
- b. On the **Costed Bill of Materials** script row, click **Deployments**.
- c. On the Script Deployments page, click **Edit**.
- d. In the **Execute As Role** field, select the custom role that you created in step 3.
- e. Click **Save**.
- f. Repeat steps 4.1 to 4.5 for **Costed BOM DataService** script.

## Enabling Multiple Currency Feature

When **Multiple Currencies** feature is enabled in the **Enable Features** page after the SuiteApp is installed, you must perform the following steps.

To enable Multiple Currency feature:

1. Go to Setup > User/Roles > Manage Roles.
2. On the **Costed\_BOM\_Role** role row, click **Customize**. The Role page opens.
3. Create a custom role.
  - a. (Optional) To rename the custom role, enter the role name in the **Name** field.
  - b. In the **Permissions** subtab, click the **Lists** subtab.
  - c. In a new row, select **Currency**.
  - d. From the level list, select **View**.
  - e. Click **Add**.
  - f. Click **Save**.
4. Add the custom role to the scripts:
  - a. Go to Customization > Scripting > Scripts.
  - b. On the **Costed Bill of Materials** script row, click **Deployments**.
  - c. On the Script Deployments page, click **Edit**.
  - d. In the **Execute As Role** field, select the custom role that you created in step 3.
  - e. Click **Save**.
  - f. Repeat steps 4.1 to 4.5 for **Costed BOM DataService** script.

## Supported Browsers for Costed Bill of Materials

The Costed Bill of Materials SuiteApp supports the following browsers and operating systems:

| Browser                         | Operating System      |
|---------------------------------|-----------------------|
| Google Chrome                   | Windows 10            |
|                                 | Mac OS 10.15 or newer |
| Microsoft Edge (Chromium-based) | Windows 10            |
| Mozilla Firefox                 | Windows 10            |
|                                 | Mac OS 10.15 or newer |

| Browser | Operating System      |
|---------|-----------------------|
| Safari  | Mac OS 10.15 or newer |

**Note:** In the **Item** field of the Costed Bill of Materials page, type three characters and click on the field for the items to be populated in the list, based on the search. This is applicable only for Safari browser.

## Creating Costed Bill of Materials

The Costed Bill of Materials SuiteApp displays the total cost breakdown of an assembly item using the bill of materials revision, routing steps, and costing method.

### To create a costed bill of materials:

1. Go to Reports > Cost Accounting > Costed Bill of Materials.
2. From the **Subsidiary** list, select a subsidiary for the bill of materials.
3. From the **Item** list, select the assembly item for which you want the bill of materials.
4. From the **Location** list, select a location to view data for that location.
5. From the **Bill of Materials** list, select the bill of materials to automatically populate the **Bill of Materials Revision** and **Effective Date** fields.
6. From the **Display Control** list, select either **By Date** or **By Revision** to decide what to include in an assembly. This field is available only when an assembly item uses revision control for its effective display control value.
7. The **Date/Effective Date** field displays the date used to decide when a component should be included in an assembly. This field is available only when an assembly item uses revision control for its effective **display control** value.

The field is editable only if the **display control** value is set to **By Date**. The **Date** field controls which date is used when querying the components of the assembly item. Then, components are shown based on the following criteria:

- The selected date is greater than or equal to the effective start date.
- The selected date is less than or equal to the effective end date.

8. From the **Bill of Materials Revision** field, choose the bill of materials revision from the list.
9. Check the **By Manufacturing Routing** box only if the assembly item has an associated routing. The box will be disabled if the item has no associated routing.

**Note:** You must check the **By Manufacturing Routing** box to see data in **Top Level Costs** and **Fully Exploded Costs** subtabs.

10. From the **Manufacturing Routing** list, select a manufacturing routing. This field is enabled when you check the **By Manufacturing Routing** box.
11. In the **Build Quantity** field, enter the build quantity of the item you want to see for the costed bill of materials. The default value is 1. However, if **Manufacturing Routing And Work Center** feature is enabled in **Enable Features** page, the default value will be the costing lot size mentioned in selected assembly item record.
12. Check the **Top Level Only** box to limit the details to the selected assembly item, its components, and operation only. Information about sub-assemblies is not displayed. Clear this box to display details at all levels of the fully exploded Bill of Materials. This field cannot be checked if the **By Manufacturing Routing** box is selected.

**Note:** If the **Top Level Only** field is checked, the costs for the selected assembly item are sourced directly from the assembly item.

If the **Top Level Only** field is not checked, the costs for the selected assembly item are sourced and aggregated from the component items in the exploded bill of materials.

13. From the **Material Cost** list, you can select the cost type to be used for all components. For more information, see [Cost Types for Material Cost Calculation](#).

14. Click **Submit**.

You can view the three subtabs (**Material Costs**, **Top Level Costs**, and **Fully Exploded Costs**) after the process is complete.

15. (Optional) You can download the report by clicking the **Download** button.

**Note:** Following points are applicable to the downloaded file in the SuiteApp.

- The downloaded file has the file name in the format **Assembly Name\_BOM name\_Revision name\_Routing name\_timestamp.xls** with a character limit of 199 characters. If **Assembly Name\_BOM name\_Revision name\_Routing name** is of 170 characters, the SuiteApp will automatically truncate the characters and allocate the remaining characters for **\_timestamp.xls**.
- The file name will add the display name as a part of the assembly name if the **Show Display Name with Item Codes** preference is enabled.
- Unsupported characters (\, /, :, \*, ?, ", <, >, |, %) will be converted to underscore (\_) in the file name.

16. (Optional) To email the report, click the **Email** button which goes to a new browser tab.
  - a. In the **Email Recipients** field, enter the email addresses. By default, the field contains the email address of the logged-in user.

**Note:** Following points are applicable to the email option in the SuiteApp.

- The email addresses in the **Email Recipients** field must be separated by commas.
- You can add a maximum of 10 email recipients in the **Email Recipients** field.
- You cannot email the Costed Bill of Materials report if the file size is more than 10 MB. So, you can download the report and email it instead.

- b. Click **Submit**.
17. (Optional) To simulate the report with modified unit cost (for non-assembly items) and build quantity values:
  - a. In the **Unit Cost** column of the **Material Costs** subtab, modify the value for any nonassembly item.
  - b. In the **Simulate Build Quantity** field of the **Material Costs** subtab, modify the value for which you want to generate a simulated report.



**Note:** The **Build Quantity** field value in the **Filters** section takes up the value provided in the **Simulate Build Quantity** field after simulation. However, the modified values will not be reflected in the original report.

- c. Click **Simulate** to display the simulated Costed Bill of Materials report.
- d. To download the simulated report, click the **Download** button.

This Costed Bill of Materials report shows the modified values in all subtabs based on the updated unit cost of the nonassembly items.

**Note:** Following points are applicable to the simulate option in the SuiteApp:

- The modified changes of the simulated report will be replaced with the original values after clicking the **Submit** button.
- The modified unit cost is updated in each bill of materials where the item is used if:
  - The unit cost is modified for any nonassembly item.
  - The item is used in more than one bill of materials within the exploded bill of materials for the assembly.
- When the values are modified in the **Unit Cost** column of the **Material Costs** subtab, the values are highlighted in blue color.
- The downloaded file has the file name in the format **Assembly Name\_BOM name\_Revision name\_Routing name\_simulated\_timestamp.xls** with a character limit of 199 characters. If **Assembly Name\_BOM name\_Revision name\_Routing name** exceeds 170 characters, the SuiteApp automatically truncates the excess characters and allocates the remaining characters for **simulated\_timestamp.xls**.

## Material Costs Subtab

The **Material Costs** subtab provides a breakdown of the cost of the components used in the assembly. You can view material costs based on item costing method, purchase price, or item default price on this subtab. Using these details, you analyze the cost of the materials used in the assembly.

The following table provides details about the fields in the **Material Costs** subtab.

| Field                  | Description                                                                                   |
|------------------------|-----------------------------------------------------------------------------------------------|
| Total Material Cost    | Total material cost calculates based on the build quantity and cost of the end assembly item. |
| Material Cost per Unit | Material cost of a single unit of the end assembly item.                                      |

The following table provides details about the columns in the **Material Costs** subtab.

| Column Name | Description          |
|-------------|----------------------|
| Item Name   | Component item name. |

| Column Name                     | Description                                                                                                                                                                                                             |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Level                           | Level represents the level in the hierarchy of the exploded bill of materials.                                                                                                                                          |
| Item Display Name               | Display name of the component item.                                                                                                                                                                                     |
| Quantity                        | Quantity of the component item as per the bill of materials required to produce one unit of the assembly at this level.                                                                                                 |
| Component Yield                 | Component yield is retrieved from the selected bill of materials revision of the assembly item.                                                                                                                         |
| Quantity Per Assembly           | Quantity of the component item required to produce one unit of the assembly at this level based on component yield.                                                                                                     |
| Quantity Per Top Level Assembly | Quantity of the component item required to produce one unit of the end assembly.                                                                                                                                        |
| Quantity for Build              | Quantity for Build calculates based on the product of build quantity and quantity per top level assembly.                                                                                                               |
| Unit Cost                       | Unit cost of the component as per the selected material cost source. This column can be edited to generate simulated reports. For more information, read step 17 in <a href="#">Creating Costed Bill of Materials</a> . |
| Units                           | Consumption unit of the component.                                                                                                                                                                                      |
| Total Cost                      | Total cost of the component.                                                                                                                                                                                            |

**Note:** The SuiteApp selects the bill of materials revision for a subassembly in the following precedence order:

1. If a BOM contains multiple revisions, the active revision is considered based on the current date falling within the effective start and end date range.
2. If a BOM contains multiple revisions with the effective date range as expired or future effective, the BOM revision with latest or last modified date is considered.

Inactive BOM revisions are not considered for estimations.

Following points show how the bill of materials is associated with an effective revision of a subassembly:

- A BOM associated with the **Default for Location** field is prioritized over the BOM mapped to the **Master Default** field.
- If a subassembly is not mapped to a BOM that is associated with **Default for Location** field, the BOM mapped to **Master Default** field is considered.
- Inactive BOM is not considered even if it is associated with **Default for Location** or **Master Default** fields.
- If there are multiple BOMs with multiple revisions, and revisions have expired or future effective dates, the BOM revision associated with **Default for Location** field is considered.
- If a subassembly is mapped to following BOMs, the **Master Default** BOM with active revision is considered.
  - BOM that contains multiple revisions with expired or future effective date and associated to **Default for Location** field.
  - BOM with active revision and mapped to the **Master Default** field.

# Top Level Costs Subtab

The **Top Level Costs** subtab provides a breakdown of the costs of the assembly by cost type. You can view and analyze cost breakdown for manufacturing operations on this subtab. The details can help you understand the cost of each manufacturing operation involved in producing the item.

The following table provides details about the fields in the **Top Level Costs** subtab.

| Field                | Description                                                                                                     |
|----------------------|-----------------------------------------------------------------------------------------------------------------|
| Total Assembled Cost | Total assembled cost calculates based on the sum of material cost and conversion cost of the end assembly item. |

| Field                   | Description                                               |
|-------------------------|-----------------------------------------------------------|
| Assembled Cost per Unit | Assembled cost of a single unit of the end assembly item. |

The following table provides details about the columns in the **Top Level Costs** subtab.

| Column Name            | Description                                                                               |
|------------------------|-------------------------------------------------------------------------------------------|
| Operation              | The sequence of the operation in the routing.                                             |
| Operation Name         | Name of the operation sequence.                                                           |
| Item Name              | Component item name.                                                                      |
| Material Costs         | Cost of materials as per the selected material cost source to produce the final assembly. |
| Labor Run              | Cost of labor run to produce the final assembly.                                          |
| Labor Run Overhead     | Cost of labor overhead to produce the final assembly.                                     |
| Labor Setup            | Cost of labor setup to produce the final assembly.                                        |
| Labor Setup Overhead   | Cost of labor setup overhead to produce the final assembly.                               |
| Machine Run            | Cost of machine run to produce the final assembly.                                        |
| Machine Run Overhead   | Cost of machine run overhead to produce the final assembly.                               |
| Machine Setup          | Cost of machine setup to produce the final assembly.                                      |
| Machine Setup Overhead | Cost of machine setup overhead to produce the final assembly.                             |
| Total Cost             | Total cost to produce the final assembly.                                                 |

For more information about the conversion cost calculations for top level costs, see [Conversion Cost Calculations for Top Level Costs and Fully Exploded Costs Subtabs](#).

**Note:** The following information is applicable to both **Top Level Costs** and **Fully Exploded Costs** subtabs:

- When a subassembly is not associated with a default routing, the conversion cost is populated as 0. This calculation is based on the unavailability of a default routing for the subassembly. The conversion cost will remain 0 even if there are other routing options for the subassembly, but are not marked as default.
- The assembly item is displayed with the conversion cost when the following two conditions are true:
  - The subassembly does not have any component in the revision.
  - Default routing is associated with the item.

The material cost will be 0 and on expanding the assembly item, no components will be displayed.
- The assembly item displays conversion cost as zero when the following two conditions are true:
  - The subassembly does not have an effective bill of materials or revision.
  - Default routing is associated with the item.
- If a **Operation** is not defined for a component in the **Component Per Operation** subtab of a routing record, the first operation will be considered for that component.

## Fully Exploded Costs Subtab

The **Fully Exploded Costs** subtab provides a detailed breakdown of material and operation costs by cost types on this subtab. You can analyze the costs associated with each material and operation used in the assembly.

The following table provides details about the fields in the **Fully Exploded Costs** subtab.

| Field                  | Description                                                                                                                            |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Total Exploded Cost    | Total exploded cost is calculated based on the sum of material cost and conversion cost of all subassemblies of the end assembly item. |
| Exploded Cost per Unit | Exploded cost of a single unit of the end assembly item.                                                                               |
| Material Cost %        | Percentage of the material cost from the total exploded cost percentage.                                                               |
| Conversion Cost %      | Percentage of the conversion cost from the total exploded cost percentage.                                                             |

The following table provides details about the columns in the **Fully Exploded Costs** subtab.

| Column Name | Description                                                                    |
|-------------|--------------------------------------------------------------------------------|
| Item Name   | Component item name.                                                           |
| Level       | Level represents the level in the hierarchy of the exploded bill of materials. |

| Column Name            | Description                                                                               |
|------------------------|-------------------------------------------------------------------------------------------|
| Material Costs         | Cost of materials as per the selected material cost source to produce the final assembly. |
| Conversion Costs       | Cost types associated with each assembly to produce the final assembly.                   |
| Labor Run              | Cost of labor run to produce the final assembly.                                          |
| Labor Run Overhead     | Cost of labor overhead to produce the final assembly.                                     |
| Labor Setup            | Cost of labor setup to produce the final assembly.                                        |
| Labor Setup Overhead   | Cost of labor setup overhead to produce the final assembly.                               |
| Machine Run            | Cost of machine run to produce the final assembly.                                        |
| Machine Run Overhead   | Cost of machine run overhead to produce the final assembly.                               |
| Machine Setup          | Cost of machine setup to produce the final assembly.                                      |
| Machine Setup Overhead | Cost of machine setup overhead to produce the final assembly.                             |
| Total Cost             | Total cost to produce the final assembly.                                                 |

For more information about the conversion cost calculations for fully exploded costs, read [Conversion Cost Calculations for Top Level Costs and Fully Exploded Costs Subtabs](#).

## Cost Types for Material Cost Calculation

You can select the cost type from the **Material Cost** field in the Costed Bill of Materials page.

Following are the cost types.

- [Per Costing Method](#)
- [Per Purchase Price](#)
- [Per Item Default](#)



**Note:** Following information is applicable to all cost types:

- Item purchase price values are converted to the currency of the selected subsidiary.
- Service, non-inventory, or charge items do not have any costing method associated with them. So, the value of these items is retrieved from item purchase price. For currency conversion, currency exchange rates values are used.
- If an assembly item is not associated with a bill of materials or bill of materials revision, then the costs for the assembly item will be sourced the way the inventory items are sourced.
- You will see blank conversion cost values if the values for conversion costs are not defined in the cost template. Some examples for conversion costs are labor setup, machine run, and so on.

Watch the following help video for cost calculation methods:

 Cost Calculation Methods

## Per Costing Method

Cost is calculated based on the costing method assigned to the item. For example: LIFO, FIFO, standard, average, and so on.

The costs are sourced from the items in the sequence outlined in the table. For example, refer to row 1 in the below table. An assembly item will source by item location average cost if the value is greater than 0. Else, it moves to the next source (item location last purchase price) and checks if the value is greater than 0. It then goes to the next source (Item purchase price) and checks if the value is greater than 0. If none of the source types provide a value greater than 0, then the price of the assembly item will be 0.

**Note:** If the **Top Level Only** field is not checked, the costs for the assembly item are sourced and aggregated from the component items in the exploded bill of materials.

| Item Type                       | Top Level | Costing Method                                                 | Cost Source                                                                                                                                                                                                                                                                                                            |
|---------------------------------|-----------|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Assembly item or inventory item | Yes       | Group Average<br>Average                                       | <ol style="list-style-type: none"> <li>1. Item location average cost</li> <li>2. Item location last purchase price</li> <li>3. Item purchase price</li> </ol>                                                                                                                                                          |
|                                 |           | Standard                                                       | <ol style="list-style-type: none"> <li>1. For assembly item, the cost is a summation of component item cost.<br/>For inventory item, the cost is based on item location standard cost.</li> <li>2. Item location average cost</li> <li>3. Item location last purchase price</li> <li>4. Item purchase price</li> </ol> |
|                                 |           | Lot Numbered<br>Specific<br>Actual<br>FIFO<br>LIFO, and so on. | <ol style="list-style-type: none"> <li>1. For assembly item, the cost is a summation of component item cost.<br/>For inventory item, the cost is based on item location default cost.</li> <li>2. Item location average cost</li> <li>3. Item location last purchase price</li> <li>4. Item purchase price</li> </ol>  |
| Inventory item                  | No        | Group Average<br>Average                                       | <ol style="list-style-type: none"> <li>1. Item location average cost</li> <li>2. Item location last purchase price</li> <li>3. Item purchase price</li> </ol>                                                                                                                                                          |
|                                 |           | Standard                                                       | <ol style="list-style-type: none"> <li>1. Item location default cost</li> <li>2. Item location average cost</li> <li>3. Item location last purchase price</li> <li>4. Item purchase price</li> </ol>                                                                                                                   |
|                                 |           | Lot Numbered<br>Specific<br>Actual<br>FIFO                     | <ol style="list-style-type: none"> <li>1. Item location default cost</li> <li>2. Item location average cost</li> <li>3. Item location last purchase price</li> <li>4. Item purchase price</li> </ol>                                                                                                                   |

| Item Type | Top Level | Costing Method   | Cost Source |
|-----------|-----------|------------------|-------------|
|           |           | LIFO, and so on. |             |

## Per Purchase Price

Cost is calculated as per the last purchase price of the item (assembly item or inventory item) in the location based on top level selection. If the last purchase price is not available, the report uses the default item purchase price.

## Per Item Default

Cost is calculated as per the standard cost of the item in the location. This cost type is not displayed in the **Material Cost** list if the **Standard Costing** feature is disabled.

## Conversion Cost Calculations for Top Level Costs and Fully Exploded Costs Subtabs

The following table contains formulas and criteria based on which the conversion cost for top level costs and fully exploded items costs subtabs are calculated:

| Cost Calculation                         | Criteria                                   | Formula                                                                             |
|------------------------------------------|--------------------------------------------|-------------------------------------------------------------------------------------|
| Labor Run and Labor Run Overhead         |                                            | Cost * Labor Resources * (Run Rate/60) * Quantity for build                         |
| Labor Setup and Labor setup Overhead     | if (Quantity for build < Costing Lot Size) | Cost * Labor Resources * (Setup Time/60)* (Quantity for Build / Costing Lot Size)   |
| Labor Setup and Labor setup Overhead     | if (Quantity for build > Costing Lot Size) | Cost * Labor Resources * (Setup Time/60)                                            |
| Machine Run and Machine Run Overhead     |                                            | Cost * Machine Resources * (Run Rate/60) * Quantity for build                       |
| Machine Setup and Machine Setup Overhead | if (Quantity for build < Costing Lot Size) | Cost * Machine Resources * (Setup Time/60)* (Quantity for build / Costing Lot Size) |
| Machine Setup and Machine Setup Overhead | if (Quantity for build > Costing Lot Size) | Cost * Machine Resources * (Setup Time/60)                                          |

**Note:** Quantity for build is calculated by considering the base unit (unit of measure) of the .

The following table contains details of where the formula values are retrieved from:

| Formula Component | Retrieved From                   |
|-------------------|----------------------------------|
| Cost              | Manufacturing Cost Template page |
| Labor Resources   | Manufacturing Routing page       |
| Machine Resources | Manufacturing Routing page       |
| Setup Time        | Manufacturing Routing page       |



| Formula Component  | Retrieved From                                                 |
|--------------------|----------------------------------------------------------------|
| Run Rate           | Manufacturing Routing page                                     |
| Costing Lot Size   | Item Location Configuration page                               |
| Quantity for build | Product of quantity per top level assembly and build quantity. |

The following calculations for conversion costs are applied for the phantom assemblies:

- Conversion costs are not calculated for phantom assemblies that are labelled as subassemblies under a main assembly.
- On the **Fully Exploded Costs** subtab, if the selected main assembly is phantom with associated manufacturing routing, the SuiteApp calculates the conversion cost for phantom assembly. However, if the phantom is a subassembly of the selected main assembly, the SuiteApp does not calculate the conversion cost.
- If phantom is a subassembly, the associated conversion cost is not included in the cost setup of the main assembly. The SuiteApp calculates and includes the conversion costs only when the phantom is labelled as main assembly.

For example:

The main assembly has a conversion cost of \$100, whereas the phantom assembly has a conversion cost of \$20 according to its manufacturing routing. In this case:

- When you run Costed BOM process for main assembly, the results show total conversion cost as \$100. In this case, phantom conversion cost is not calculated. This conversion cost value of \$100 is shown on the **Fully Exploded Costs** subtab.
- When you run Costed BOM process for phantom assembly, the result shows total conversion cost as \$20. In this case, phantom conversion cost is calculated, since the SuiteApp considers it as an end assembly. This conversion cost value of \$20 is shown on the **Fully Exploded Costs** subtab.

 **Note:** For an assembly that is labelled as a subassembly under the phantom subassembly breakdown, the conversion cost for that assembly is zero.

Watch the following help video for cost calculation methods:

 [Cost Calculation Methods](#)

## Cost Variance Analysis

The Cost Variance Analysis SuiteApp analyzes the variances that happen while you're running work orders. These variances are associated with quantity and cost between the planned values for the work order and the actual values recorded against the work order.

This SuiteApp also enables you to do the following:

- Measure how your work orders perform and help prevent cost overruns.
- Provide visibility to review work order cost variances at component and operation level. Review the work order including work order, cost type and cost category, with access to detailed information up to the transaction level.

You can access the SuiteApp from two places:

- Go to Reports > Cost Accounting > Cost Variance Analysis.
- Click on the **Cost Variance Analysis** button in a work order record.

You'll see the **Cost Variance Analysis** button in a work order record if:

- The work order status is not **Planned** or **Released**.
- The work order contains actual production start date.
- The work order is in view mode.
- The work order is not outsourced.

 **Note:** Cost analysis isn't supported for work orders that use standard cost items.

## Installing Cost Variance Analysis

The Cost Variance Analysis SuiteApp requires a NetSuite OneWorld account.

Only users with the Administrator role can install the SuiteApp. For more information about installing a SuiteApp, see the help topics [Installing from the SuiteApp Marketplace](#) and [SuiteApp Marketplace in NetSuite](#).

## Prerequisites for Installing Cost Variance Analysis

To use the Cost Variance Analysis SuiteApp, enable the features in the following table. For more information, see the help topic [Enabling Features](#).

| Subtab            | Feature Name       |
|-------------------|--------------------|
| Items & Inventory | Work Orders        |
| SuiteCloud        | Custom Records     |
|                   | Server SuiteScript |
|                   | Client SuiteScript |

## Creating or Customizing Roles and Permissions to Use Cost Variance Analysis

If you are an administrator, you can create new roles or customize roles for using the Cost Variance Analysis SuiteApp. For more information, see the help topics [Assigning Roles to an Employee](#) and [Customizing or Creating NetSuite Roles](#).

A custom role **Cost\_Variance\_Custom\_Role** is created during the SuiteApp installation. This custom role is used as a default value for **Execute As Role** field in the Script Deployment page. The role is also updated in the **Cost Variance Analysis SL** and **Cost Variance DataService** script.

Refer to the following table for minimum access levels of permissions required for SuiteApp deployments. These minimum permissions must be updated to the existing role or while creating a new role.

| Permissions                      | Minimum Access Level |
|----------------------------------|----------------------|
| Transactions > Build Assemblies  | View                 |
| Transactions > Build Work Orders | View                 |
| Transactions > Find Transaction  | View                 |

| Permissions                                | Minimum Access Level |
|--------------------------------------------|----------------------|
| Transactions > Work Order                  | View                 |
| Transactions > Work Order Close            | View                 |
| Transactions > Work Order Completion       | View                 |
| Transactions > Work Order Issue            | View                 |
| Lists > Address List in Search             | Full                 |
| Lists > Bill of Materials                  | View                 |
| Lists > Currency                           | View                 |
| Lists > Custom Record Entries              | Full                 |
| Lists > Documents and Files                | Edit                 |
| Lists > Email Template                     | View                 |
| Lists > Items                              | View                 |
| Lists > Locations                          | View                 |
| Lists > Manufacturing Routing              | View                 |
| Lists > Perform Search                     | View                 |
| Lists > Subsidiaries                       | View                 |
| Setup > Accounting Lists                   | View                 |
| Setup > Enable Features                    | Full                 |
| Setup > SuiteScript                        | View                 |
| Setup > SuiteScript Scheduling             | Full                 |
| Setup > Set Up Company                     | View                 |
| Custom Record > Cost Variance MR Instances | Full                 |

Refer to the following table for the minimum permissions required for logged-in user role.

| Permissions                     | Minimum Access Level |
|---------------------------------|----------------------|
| Lists > Items                   | View                 |
| Lists > Subsidiaries            | View                 |
| Lists > Locations               | View                 |
| Lists > Documents and Files     | View                 |
| Transactions > Find Transaction | View                 |
| Transactions > Work Order       | View                 |

## Supported Browsers for Cost Variance Analysis

The Cost Variance Analysis SuiteApp supports the following browsers and operating systems:

| Browser                         | Operating System                      |
|---------------------------------|---------------------------------------|
| Google Chrome                   | Windows 10                            |
|                                 | Mac OS 10.15 or newer                 |
| Microsoft Edge (Chromium-based) | Windows 10 or newer                   |
| Mozilla Firefox                 | Windows 10 or newer                   |
|                                 | Mac OS 10.15 or newer                 |
| Safari                          | Two most recent stable major versions |

## Creating Cost Variance Analysis

The Cost Variance Analysis SuiteApp calculates the cost variances of work orders based on the selection criteria in the **Filters** section. You can view the cost variances of each work order in the **Work Order Summary** section. This calculation is based on the difference between the planned values for the work order and the actual values recorded against the work order.

### To create a cost variance analysis:

1. Go to Reports > Cost Accounting > Cost Variance Analysis.
2. From the **Item** list, select the assembly item for which you want the cost variance analysis.
3. (Optional) From the **Location** list, select a location to view data for that location.
4. From the **From** list, select a date to define the beginning of the date range.
5. From the **To** list, select a date to define the end of the date range.



**Note:** Following points are applicable to the **To** field:

- The from and to date range filters the work orders based on the actual production start date.
- The period between the from and to date should not be more than a year.

6. From the **Work Order** list, select a work order for which you want to see the detailed cost variance analysis.



**Note:** Following points are applicable to the **Work Order** list:

- The field value in the **Work Order** list is displayed in WorkOrderNumber\_ActualProductionStartDate format.
- The work order must have an actual production start date.

7. Click **Submit**.  
You can view the **Work Orders Summary** section after the process is complete.
8. (Optional) To download the report, click **Download**.

 **Note:** Following points are applicable to the downloaded file in the SuiteApp:

- The downloaded file has the file name in the format  
ItemName\_LocationName\_FromDate\_ToDate\_timestamp.xls.
- Unsupported characters (\, /, :, \*, ?, ", <, >, |, %) will be converted to underscore (\_) in the file name.
- A scheduled script runs once a week to delete the generated cost variance report files from the file cabinet.

9. (Optional) To email the report:

1. Click the **Email** button.

A new browser tab opens.

2. In the **Email Recipients** field, enter the email addresses. By default, the field shows the email address of the user who's logged in.

 **Note:** Following points are applicable to the email option in the SuiteApp:

- The email addresses in the **Email Recipients** field must be separated by commas.
- You can add up to 10 email recipients in the **Email Recipients** field.
- You cannot email the Cost Variance Analysis report if the file size is more than 10 MB. So, you can download the report and email it instead.

3. Click **Submit**.

**Note:** Following points are applicable to the Cost Variance Analysis page in the SuiteApp.

- To generate a cost variance analysis report, you can either:
  - Enter a work order number in the **Work Order** field to see its detailed cost variance analysis.
  - Select an assembly item in the **Item** field to view the list of work orders.
- After you have selected the work order, the **Item** and **Location** fields will be grayed out. The item and location specified become defaulted in the work order.
- The work orders in the **Work Order Summary** section are sorted in descending order of the work order created date.

## Work Order Summary Section

Select an assembly item, from and to date in the Cost Variance Analysis page, and you'll see the relevant work orders in the **Work Order Summary** section.

The following table provides details about the columns in the **Work Order Summary** section.

| Column Name            | Description                                                                                                                                         |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Work Order             | Work order number. Click on the work order number to generate the cost variance analysis report.                                                    |
| Status                 | Work order status.                                                                                                                                  |
| Location               | Location of the assembly item in the work order.                                                                                                    |
| Actual Start Date      | Actual production start date of the work order.                                                                                                     |
| Actual End Date        | Actual production end date of the work order.                                                                                                       |
| Build Quantity         | Quantity to be built for the work order.                                                                                                            |
| Built Quantity         | Quantity built for the work order. If the built quantity is zero, the cost variance per unit will be empty.                                         |
| Planned Cost           | Planned cost for completing the work order. If the planned cost is zero and the actual cost is non-zero, the cost variance percentage will be 100%. |
| Actual Cost            | Actual cost when completing the work order. If the planned cost is zero and the actual cost is non-zero, the cost variance percentage will be 100%. |
| Cost Variance          | Cost difference between the actual cost and planned cost of the work order.                                                                         |
| Cost Variance (%)      | Percentage of cost variance per planned cost of the work order.                                                                                     |
| Cost Variance per Unit | The difference between actual cost and planned cost for each unit built.                                                                            |

## Summary Section

The **Summary** section of the Cost Variance Analysis page includes the work order information, **Cost Category** subtab, **Component** subtab, and **Conversion Costs** subtab. You can sort the column values in any subtab.

You'll be directed to the **Summary** section if you've picked a work order in the **Filters** section. Else, click the work order number in the **Work Order Summary** section to get there.



**Note:** The following points are applicable to the **Cost Category, Component, and Conversion Costs** subtabs:

- Only the variance and variance percentage with value other than zero are highlighted.
- Positive variance is highlighted in red.
- Negative variance is highlighted in green.

The table below explains the work order information fields.

| Field                      | Description                                                                                                                                    |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Subsidiary                 | Subsidiary name of the work order.                                                                                                             |
| Bill of Materials          | Bill of Materials of the work order.                                                                                                           |
| Bill of Materials Revision | Revision version of the bill of materials.                                                                                                     |
| Manufacturing Routing      | Manufacturing routing associated with the work order.                                                                                          |
| Actual Start Date          | Actual production start date of the work order.                                                                                                |
| Actual End Date            | Actual production end date of the work order.                                                                                                  |
| Planned Build Quantity     | Quantity to be built for the work order. If the planned build quantity is zero, the planned cost per unit is empty.                            |
| Actual Built Quantity      | Build quantity of the selected work order. If the actual built quantity is zero, the are empty.                                                |
| Build Quantity Variance    | Difference between the planned build quantity and the actual build quantity of the work order.                                                 |
| Work Order Status          | Status of the work order.                                                                                                                      |
| Planned Cost               | Planned cost for completing the work order. If the planned cost is zero and the actual cost is non-zero, the cost variance percentage is 100%. |
| Actual Cost                | Actual cost for completing the work order. If the planned cost is zero and the actual cost is non-zero, the cost variance percentage is 100%.  |
| Cost Variance              | Cost difference between actual cost and planned cost of the work order.                                                                        |
| Cost Variance (%)          | Percentage of cost variance per planned cost of the work order.                                                                                |
| Planned Cost per Unit      | Planned cost per unit for completing the work order.                                                                                           |
| Actual Cost per Unit       | Actual cost per unit for completing the work order                                                                                             |
| Cost Variance per Unit     | Cost difference between actual cost per unit and planned cost per unit.                                                                        |

- Note:** The following points are applicable to number of records that you can see in the cost variance analysis summary report:
- The SuiteApp supports a maximum of 5000 work order completions in a single work order.
  - If you have multiple manufacturing operations, the SuiteApp supports up to 1500 work order completions with 100 manufacturing routing operations in a single work order.
  - You can only display up to 150,000 records with the combination of work order completions and manufacturing routings in the cost variance analysis summary report. For example, if you have 1500 completions and 100 routing operations in one work order, you'll see 150,000 records (1500\*100).
  - The SuiteApp supports a maximum of 1000 different work orders in the work order summary report.

If the data exceeds the above-mentioned limits, the SuiteApp will show zero value in the **Conversion Costs** subtab for further routing operations. Also, the actual cost and actual cost per unit will be zero in the cost variance analysis **Summary** section.

## Cost Category Subtab

The **Cost Category** subtab displays the variances for a work order by cost category. This subtab calculates the sum of all the costs under each cost category. This subtab is available for both work in progress and non-work in progress work orders.

- Note:** If no cost category is associated with any of the component items, there will be no data in this subtab.

The following table provides details about the columns in the **Cost Category** subtab.

| Field             | Description                                                             |
|-------------------|-------------------------------------------------------------------------|
| Cost Category     | Cost category type.                                                     |
| Planned Cost      | Planned cost for completing the work order.                             |
| Actual Cost       | Actual cost for completing the work order.                              |
| Cost Variance     | Cost difference between actual cost and planned cost of the work order. |
| Cost Variance (%) | Percentage of cost variance per planned cost of the work order.         |

## Component Subtab

The **Component** subtab displays variances for a work order by component. You can see the component item name even if the quantity issued or used is zero.

Check the **Only Display Variance** box to show only rows with a variance.



The table below explains the columns in the **Component** subtab.

| Field                 | Description                                                                     |
|-----------------------|---------------------------------------------------------------------------------|
| Operation             | Operation sequence in the manufacturing routing associated with the work order. |
| Item                  | Item name of the component.                                                     |
| Display Name          | Another name for the item.                                                      |
| Planned Cost          | Planned cost for completing the work order.                                     |
| Actual Cost           | Actual cost for completing the work order.                                      |
| Cost Variance         | Cost difference between actual cost and planned cost of the work order.         |
| Cost Variance (%)     | Percentage of cost variance per planned cost of the work order.                 |
| Planned Quantity      | Quantity to be built for the work order.                                        |
| Actual Quantity       | Built quantity of the work order.                                               |
| Quantity Variance     | Quantity difference between planned cost and actual cost of the work order.     |
| Quantity Variance (%) | The difference between planned quantity and actual quantity for the work order. |

## Conversion Costs Subtab

The **Conversion Costs** subtab shows costs for operations like labor cost, machine cost, and so on, for the selected work order. Total cost is the sum of labor cost and machine cost.

This subtab won't show any data if the work order isn't linked to a manufacturing routing.

The table below explains the columns in this subtab.

| Field                   | Description                                                                                                                                                                                                                                                                                 |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Operation               | Operation sequence in the manufacturing routing associated with the work order.                                                                                                                                                                                                             |
| Name                    | Operation name of the operation sequence.                                                                                                                                                                                                                                                   |
| Planned Total Cost      | Sum of planned labor cost and planned machine cost. If the planned total cost is zero and the actual total cost is non-zero, the total cost variance percentage is 100%.                                                                                                                    |
| Actual Total Cost       | Sum of actual labor cost and actual machine cost. If the planned total cost is zero and the actual total cost is non-zero, the total cost variance percentage is 100%.                                                                                                                      |
| Total Cost Variance     | The difference between actual total cost and planned total cost.                                                                                                                                                                                                                            |
| Total Cost Variance (%) | Percentage of total cost variance per planned total cost.                                                                                                                                                                                                                                   |
| Planned Labor Cost      | Sum of all the planned labor costs associated with the operation. If the planned labor cost is zero and the actual labor cost is non-zero, the labor cost variance percentage is 100%.<br>For more information, see <a href="#">Conversion Cost Formulas for Planned and Actual Costs</a> . |
| Actual Labor Cost       | Sum of all the actual labor costs associated with the operation. If the planned labor cost is zero and the actual labor cost is non-zero, the labor cost variance percentage is 100%.<br>For more information, see <a href="#">Conversion Cost Formulas for Planned and Actual Costs</a> .  |
| Labor Cost Variance     | The difference between planned labor cost and actual labor cost for the work order.                                                                                                                                                                                                         |

| Field                     | Description                                                                                                                                                                                                                                                                                             |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Labor Cost Variance (%)   | Percentage of labor cost variance per planned labor cost of the work order.                                                                                                                                                                                                                             |
| Planned Machine Cost      | Sum of all the planned machine costs associated with the operation. If the planned machine cost is zero and the actual machine cost is non-zero, the machine cost variance percentage is 100%.<br><br>For more information, see <a href="#">Conversion Cost Formulas for Planned and Actual Costs</a> . |
| Actual Machine Cost       | Sum of all the actual machine costs associated with the operation. If the planned machine cost is zero and the actual machine cost is non-zero, the machine cost variance percentage is 100%.<br><br>For more information, see <a href="#">Conversion Cost Formulas for Planned and Actual Costs</a> .  |
| Machine Cost Variance     | The difference between planned machine cost and actual machine cost.                                                                                                                                                                                                                                    |
| Machine Cost Variance (%) | Percentage of machine cost variance per planned machine cost.                                                                                                                                                                                                                                           |

## Conversion Cost Formulas for Planned and Actual Costs

The following table contains conversion cost formula for planned and actual costs.

| Cost Calculation                         | Planned Cost Formula                                                                    | Actual Cost Formula                                                        |
|------------------------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Labor Run and Labor Run Overhead         | $\text{Cost} * \text{Labor Resources} * (\text{Run Rate}/60) * \text{Build Quantity}$   | $\text{Cost} * \text{Labor Resources} * (\text{Labor Run Time}/60)$        |
| Labor Setup and Labor Setup Overhead     | $\text{Cost} * \text{Labor Resources} * (\text{Setup Time}/60)$                         | $\text{Cost} * \text{Labor Resources} * (\text{Labor Setup Time} /60)$     |
| Machine Run and Machine Run Overhead     | $\text{Cost} * \text{Machine Resources} * (\text{Run Rate}/60) * \text{Build Quantity}$ | $\text{Cost} * \text{Machine Resources} * (\text{Machine Run Time} /60)$   |
| Machine Setup and Machine setup Overhead | $\text{Cost} * \text{Machine Resources} * (\text{Setup Time}/60)$                       | $\text{Cost} * \text{Machine Resources} * (\text{Machine Setup Time} /60)$ |



**Note:** The following points are applicable for conversion costs calculation:

- The parameters related to planned cost formula are taken from the work orders and their respective manufacturing operation tasks.
- The parameters related to actual cost formula are taken from the work order completions.

## Printing Assembly Item Materials on Transactions

You can print all the members of an assembly item, including display names, quantities, descriptions, and rates on an invoice or other transaction form. Alternatively, you can print only the description and amount of the assembly item on the form.



**Note:** If you have enabled Advanced Bills of Materials, this feature isn't available.

**To print raw materials on transactions:**

1. Go to Lists > Accounting > Items.
2. In the **Type** field, select **Assembly** to filter your item list to show assembly items.
3. Click **Edit** next to the assembly item you want to change.
4. Check the **Display Components on Transaction** box.
5. Click **Save**.

 **Note:** You must enter a display name on the member item records for an item name to appear.

Your administrator can customize the layout of your printed forms using Advanced PDF/HTML templates. For more information, see the help topic [Advanced PDF/HTML Templates](#).

## Printing Assembly Labels

The following procedure describes how to print labels for assembly member items and the assembly item.

### To print labels for each member item and the assembly item:

1. Go to Transactions > Management > Print Checks and Forms.
2. Click **Item Labels**.
3. On the **Print Item Labels** page, click **Customize**.
4. On the **Customize Sublist** page, click the **Additional Filters** sublist.
5. Check the **Component Of** box.
6. Click **Save**.
7. On the **Print Item Labels** page, select the assembly from the **Component Of** list.
8. Click **Mark All**.
9. Click **Print**.

## Running the Component Where Used Inquiry

The Component Where Used Inquiry enables you to view where components are used in assembly items, including parent assembly and sub-assembly items.

For example, item#24567 (a table leg) is a component in items that you assemble. Run the Component Where Used Inquiry to learn where that component is used:

- item#44555, small square table, requires 4 table legs per assembly
- item#55666, medium console table, requires 3 table legs per assembly
- item#66777, large rectangle table, requires 6 table legs per assembly

Data is also returned for sub-assembly component requirements:

- item#77888, 3-piece table set, includes one small square table, one medium console table, and one large rectangle table.
- To produce one of item #77888 requires 13 table legs per assembly.

## To run the Component Where Used Inquiry:

1. Go to Transactions > Manufacturing > Component Where Used Inquiry.
2. Enter criteria in the following fields to filter the results returned:
  1. Select a **Component** that is required for an assembly item or sub-assembly.  
Components represent level zero in the structure.
  2. If you use the Multi-Location Inventory feature, select a **Location**.
  3. Check the **Single Level Only** box to limit the display to only one level below the component.  
Clear this box to display all levels for the assembly.
  4. Enter an active **Date** for revision control and obsolete dating.  
This represents the date an assembly must be active to be included in the results list.  
If specified on the item record, assembly active dates start on the Effective Date and end on the Obsolete Date,  
For example, an assembly item Effective Date is January 1, 2017 and its Obsolete Date is December 31, 2019.
    - If you enter an active date of January 1, 2016, the assembly item doesn't appear in the results.
    - If you enter an active date of January 1, 2018, the assembly item appears in the results.



**Note:** If a parent assembly item isn't active for the specified date, then child item doesn't appear in the list.

3. Based on the criteria selected, you could see the following fields:
  - **Assembly** – The name of the assembly item that uses the component selected in the header.
  - **Level** – The level in the Bill of Materials (BOM) structure where the component resides.  
For example, a level 1 item is a sub-member of the parent (selected) component item. A level 2 item is a sub-member of the level 1 item.
  - **Quantity per Assembly** – The amount of the component required to build the assembly.
  - **On Hand / Available / Back Ordered / On Order** – Quantities are displayed for the selected location.
  - **Units** – The units for the component.
  - **Effective Date** – The date the assembly becomes effective.
  - **Obsolete Date** – The date the assembly is no longer effective.

The inquiry in the following example is run to determine where the Table Leg item is used as a component.

The following is selected in the inquiry header:

- **Component:** Table Leg
- **Single-Level only:** No

| Item                   | Level | Qty Required per Assembly | Qty Required per Top-Level Assembly |
|------------------------|-------|---------------------------|-------------------------------------|
| Table: European        | 1     | 4                         | 4                                   |
| European Table Set Box | 2     | 1                         | 4                                   |

| Item                   | Level | Qty Required per Assembly | Qty Required per Top-Level Assembly |
|------------------------|-------|---------------------------|-------------------------------------|
| Table: American        | 1     | 6                         | 6                                   |
| American Table Set Box | 2     | 1                         | 6                                   |
| Patio Side Table       | 1     | 4                         | 4                                   |
| Patio Table Pair Set   | 2     | 2                         | 8                                   |
| Patio Table Box Set    | 3     | 1                         | 8                                   |

The following is the same inquiry, with Single-Level only set to Yes.

| Item             | Level | Qty Required per Assembly | Qty Required per Top-Level Assembly |
|------------------|-------|---------------------------|-------------------------------------|
| Table: European  | 1     | 4                         | 4                                   |
| Table: American  | 1     | 6                         | 6                                   |
| Patio Side Table | 1     | 4                         | 4                                   |

# Assembly Work Orders

When you enable the Work Orders feature, you can use assembly work orders for your assembly items.

Assembly work orders track the production of assembly items needed for stock, or to fill orders. Work orders track the quantities of assemblies that need to be built, and the quantities of components, or member items, needed to do so.

For example, if you stock and sell mountain bikes, you can enter a work order to do the following:

- Track the quantity of mountain bikes that need to be assembled
- Commit member items available in stock to the work order
- Track when the mountain bikes are assembled and the work order is completed to ensure mountain bikes can be stocked or sold

## Sales Orders Linked to Work Orders

If a work order is created from a sales order, the two transactions are linked. Please note the following about making changes on sales orders or work orders that are linked:

- If you close a line on a sales order that links to an assembly, the link to the work order remains. The work order remains open and the finished assembly from the build is added to general inventory.
- If you change the quantity on a work order line, note the following. If the quantity exceeds the amount on the corresponding sales order line, the link to the sales order remains. When the build is completed, the excess assemblies are added to general inventory.
- Sales orders that are cancelled are no longer linked to work orders.

## Assemblies and Advanced Inventory Management Calculations

If you use Advanced Inventory Management and auto-calculation for inventory items that are components of assemblies, note the following. NetSuite bases demand on work orders as well as sales using the following calculation:

Qty of assembly \* qty per assembly

Orders for finished assemblies are included in calculations of demand, reorder points, and preferred stock levels for member inventory items.

You can choose to base demand on sales instead of sales orders if you use the **Transactions to Consider** preference. This preference determines if work orders and builds are included in demand calculations for assembly components. For more information, see the help topic [Setting Up Advanced Inventory Management](#).



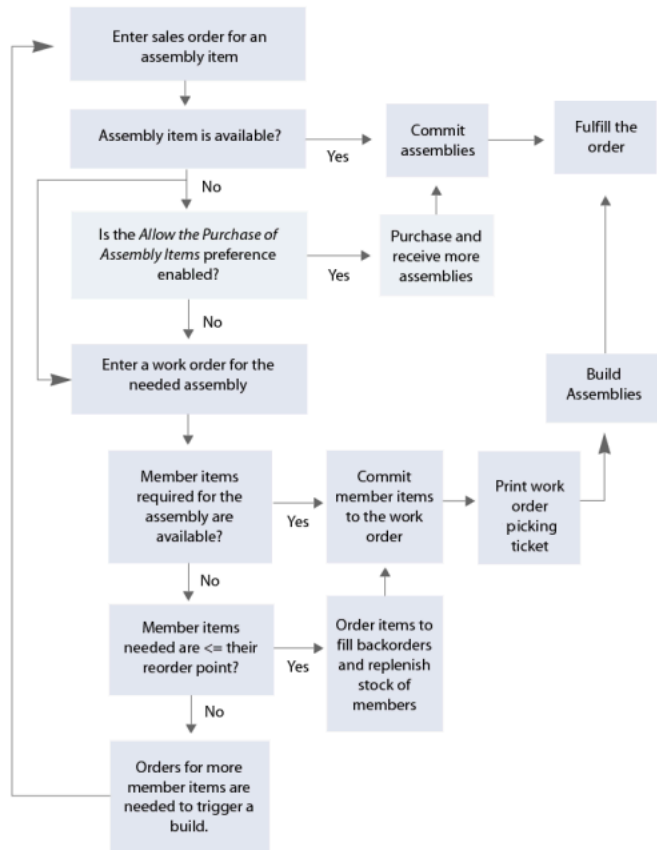
**Note:** For transaction customization purposes, assembly work order forms are classified as sales forms. If you create a custom transaction field and apply it to sales transactions, it shows on work order forms.

### Assemblies and Units of Measure

If you use Multiple Units of Measure, see [Assemblies and Units of Measure](#).

You can make changes on work orders after you create them. For more information, see [Editing a Work Order](#).

## Assembly Work Orders Workflow Chart



## Work Order Statuses

Work orders can have the following statuses:

- **Planned** – No components are committed regardless of commit option settings.
- **Released** – No transaction has posted and no activities have been recorded. Components can be committed based on commit option settings.
- **In Process** – A transaction has been posted.
- **Built** – The quantity built is equal to the quantity planned.
- **Closed**

## Two Types of Assembly Work Orders

There are two types of assembly work orders you can use:

### ■ Special Order Work Orders

Special order work orders track assemblies to be built for a particular sale. The work order can be linked to the originating sale.

Sold assembly items where member items are in stock but not finished goods, NetSuite adds a work order to the work order creation queue. Based on the amount of finished product needed for the order, member items are committed to the work order to build the finished assemblies.

### ■ Production Work Orders

Production work orders track assemblies that need to be built to increase stock and aren't intended for a particular sale. Production work orders aren't linked to a sales order and have no customer assignment.

Production work orders are generated when the back ordered quantity of an assembly reaches its assigned build point. After the build point is reached, NetSuite adds a work order in the Mass Create Work Orders queue.

For each work order, a bill of materials (BOM) is generated to facilitate picking member items for the build. When this work order completes, the regular stock level of the assembly increases and the finished goods are committed to open sales orders.

Both production and special order work orders use the same Work Order transaction record in NetSuite. A work order can be entered individually, or be automatically added to the work order creation queue based on inventory settings, as described below:

#### ■ Entering Individual Work Orders

You can enter work orders for assemblies one at a time. They can be special order or production work orders. For more information, see [Entering an Individual Work Order](#).

#### ■ Mass Creating Work Orders

After an assembly item reaches its assigned build point, NetSuite adds a work order in the Mass Create Work Orders queue. The work order's suggested quantity restores the item's preferred stock level. For more information, see [Mass Creating Work Orders](#).



**Warning:** Automatically creating seven hundred plus special work orders can impact system processing.

The following is true for both types of work order forms. If you check the Mark Sub-Assemblies Phantom box, member items that are assemblies themselves are also built to complete the work order.

If you use the Work Orders and Demand Planning features, see [Work Orders and Sub-Assemblies](#).

You can also create a work order for an assembly item when you add it to a sales order. When you select an assembly on a sales order, check the **Create WO** box. When the order is saved or approved, NetSuite adds a work order to the Mass Create Work Order queue. For more information, see [Marking Assemblies to Create Work Orders](#). After work orders are entered, completing an assembly build for the work order closes the order.

## Enabling the Work Orders Feature

An administrator can use the following procedure to enable the Work Orders feature.

### To enable the Work Orders feature:

1. Go to Setup > Company > Enable Features.



2. Click the **Items and Inventory** subtab.
3. In the Inventory section, check the **Work Orders** box.
4. Click **Save**.

After the feature is enabled, you can set the Build Based On Commitment accounting preference. For more information, see the help topic [Order Management Accounting Preferences](#).

## Entering an Individual Work Order

Enter work orders to track the production of assembly items needed for stock or to fill open orders. A work order is considered a non-posting transaction.

Special Order work orders track assemblies for a particular sale. Production work orders track assemblies to increase stock. Both use the same work order form, but production work orders don't link to a sale transaction.

Work orders you enter list the members, or components, of the assembly item to be built.

 [Manufacturing Assembly Work Orders](#)

For information about work order statuses, see [Work Order Statuses](#)

### To enter individual work orders:

Click one of the following links for instructions on entering individual work orders:

- [Entering a Work Order](#)
- [Adding Custom Fields to Work Orders](#)
- [Actual Production Start or End Dates](#)

 **Note:** When the Run Rate of previous operation is set to 0, the lag of current operation is also set to 0. This results in no production from the previous operation.

## Entering a Work Order

Complete the following procedure to enter either a custom or standard work order. The order and availability of these fields may differ depending on the form you select.

### To enter the work order primary information section:

1. Go to Transactions > Manufacturing > Enter Work Orders.  
If a **Custom Form** is defined for this work order, select a form.
2. The **Order#** is generated automatically.
  - a. To enter another order number, go to Setup > Company > Setup Tasks > Auto-Generated Numbers (Administrator).
  - b. On the **Transactions** subtab, next to **Work Order**, check the **Allow Override** box.
  - c. Click **Save**.

You can enter another number, however the next order number will revert to the standard pattern.

3. Select a **Customer**.

4. Select the **Assembly** you need to build.

After you select the item, the assembly components appear on the **Items** subtab.

After you associate an assembly build with this work order, you cannot change this field.

5. If you use Advanced Bills of Materials, the related **Bill of Materials** is automatically selected.

If you use Advanced Bills of Materials, the related **Bill of Materials Revision** is automatically selected based on revision effective dates and Production Start Date (the date when the backward scheduling method selected).

6. To build member assembly items to complete the work order, check the **Mark Sub-Assemblies Phantom** box.

Clearing this box doesn't refresh or remove sub-assembly components on the Items subtab. To reload the BOM for a top level assembly, select a different assembly in the Assembly field.

To reload a BOM for a phantom sub-assembly, change the item source for the sub-assembly to Phantom.



**Note:** After you associate an assembly build with this work order, you cannot change this field.

For information about using the work order and demand planning, see [Work Orders and Sub-Assemblies](#).

7. To designate the work order to use WIP instead of a standard assembly build, check the **WIP** box.

This option is only available when the order status is Released.

8. Select the related **Manufacturing Routing**.

This list is only available when the WIP box is checked.

9. To calculate lag times for operation tasks, check the **Auto-Calculate Lag** box.

For more information, see [Operations Overlap](#)

10. Enter the **Quantity** of assembly items you want to create. This can be a fractional number.

The item's available quantity is calculated based on availability across all locations. Changing the header location doesn't affect the item availability used on the line.



**Important:** After an assembly build is associated with this work order, the quantity can be changed only by closing the work order and then making a copy or creating a new work order. Closing the work order sets the back order amount to zero for the component items. However, it has no financial impact on the created assembly Item.

11. Click the calendar icon to select a new **Date**, or accept the default date.

The default **Status** is Released.

12. To firm the order, check the **Firmed** box.

This box is checked by default for Released orders.

13. Optionally, enter a **Memo** for this work order.

You can search for the memo text to find this work order.

This field defaults to show the transaction date plus lead time.

14. Routing and demand planning generate supply work orders that enable you to select one of the following **Scheduling Methods** to calculate production requirements:

- **Forward Scheduling** – set the production start date. NetSuite calculates the time, materials, and resources required to complete all necessary operations to finish the task. The production end date is determined based on these calculations.
- **Backward Scheduling** – set the production end date. This is the date items need to be completed. NetSuite calculates the time, materials, and resources required to complete all necessary operations. The production start date is determined based on these calculations.

To learn more, see [Production Scheduling Methods Overview](#).

15. In the **Supply Required by Date** field, enter the date the item must be shipped to the customer.
16. The **Production Start Date** automatically defaults to the current date. Depending on your settings, the current date will reflect your location or subsidiary time zone.

Production start date automatically defaults to the current day in your location time zone, if it is set. Otherwise, NetSuite will default to the subsidiary time zone if it is set. If neither location or subsidiary are set as default time zones, NetSuite defaults to your company time zone.

17. The **Production End Date** defaults to the production start date plus lead time.

When you use Routing you can set the Scheduling Method to Forward (production end date is computed automatically) or Backward (production start date is computed automatically based on the selected production end date).

To learn more, see [Production Scheduling Methods Overview](#).

18. The **Actual Production Start Date** field is automatically populated based on the earliest date of the related Issue Components, Completion, or Assembly Build transactions

To override an actual production date, check **Enter Manually**. Select a new date from the calendar and then click Save.

19. The **Actual Production End Date** field is automatically populated based on the latest date of the related Completion (after the Close transaction entry) or Assembly Build transactions.
20. If you use the Outsourced Manufacturing feature, click the **Outsourcing** subtab and then complete the fields, as required. For more information, see [Outsourced Manufacturing](#).
21. Click the **Relationships** subtab, and then follow the steps below, as required.
  - a. Check the **Update Customer** box to update the sales team on the customer's record with changes made.
  - b. Select a **Partner** and partner role, if necessary.
  - c. Check the box in the **Primary** column if this partner is the lead.
  - d. In the **Contribution %** column, enter the contribution percentage for each team member.
  - e. Click **Add**.
22. If you use Team Selling, click the **Sales Team** subtab, and then complete the following steps, as required.
  - a. Select the sales team responsible for this sale.  
The members of the sales team appear below. You can edit each team member's sales role and contribution for this transaction or add team members, if needed.
  - b. Select the sales team responsible for this sale.  
The members of the sales team appear below. You can edit each team member's sales role and contribution for this transaction or add team members, if needed.
  - c. In the **Choose Team** field, select a sales team to associate with this transaction. To create a sales team, go to Lists > Relationships > New > Select Sales Team Members.
  - d. Check the **Update Customer** box if you want to update the sales team on the customer's record with changes you make here.

- e. Select an **Employee** and **Sales Role** if necessary.
  - f. Check the box in the **Primary** column if this employee is the lead.
  - g. In the **Contribution %** column, enter the contribution percentage for each team member.
  - h. Click **Add**.
23. Click the **Communication** subtab, and then complete the following steps, as required.
- a. On the **Events** subtab, enter events.
  - b. On the **Tasks** subtab, view or enter CRM tasks records.  
For information about tasks, see the help topic [Working with CRM Tasks](#).
  - c. On the **Phone Calls** subtab, view or enter new phone calls.
  - d. On the **Files** subtab, you can select and add files from the File Cabinet that are associated with this contact.  
Select **-New-** to upload a new file to the File Cabinet.
  - e. On the **User Notes** subtab, add and track notations.
24. Click **Save**.

### To complete the work order classification section:

1. Select a **Subsidiary**.
2. Select a **Department** or **Class** if you track them. The selected department or class appears on the assembly build.
3. Select a **Location** to display only work orders for that location. This location is also displayed in the assembly build.  
If you use the Multi-Location Inventory, component inventory items are committed from the selected location.  
To learn more, see [Mass Creating Work Orders](#).

 **Note:** Department, Class, and Location fields are pre-selected. However, after an Assembly is selected, they are overridden using values defined for the related Assembly.

### To complete the work order items subtab:

1. The **Items** subtab displays the following information:

| Field                       | Description                                                                                                                                 |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Items</b>                | The components needed to make the assembly item.                                                                                            |
| <b>Days Late</b>            | The number of days the delivery is late.                                                                                                    |
| <b>Component Yield</b>      | The anticipated yield loss during the manufacturing process.                                                                                |
| <b>BOM Quantity</b>         | The quantity of the component used to build an item assuming no loss.                                                                       |
| <b>Quantity</b>             | The amount required for the assembly when one unit of the assembly is being built. This value is automatically calculated from the entries. |
| <b>Units</b>                | The base units of the component used in the parent assembly.                                                                                |
| <b>Inventory Detail</b>     | Set the bin, inventory status, or or serial detail.                                                                                         |
| <b>Expected Supply Date</b> | The estimated date for the item to be provided.                                                                                             |

| Field                        | Description                                                                                                                         |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reallocate Order Item</b> | The Reallocate Order item generates intelligent supply re-allocation recommendations with reasonably little effect on other orders. |
| <b>Description</b>           | The item description as recorded on the item record.                                                                                |
| <b>Allocation Strategy</b>   | The allocation strategy used to define the rules to allocate selected inventory items.                                              |
| <b>Commitment Confirmed</b>  | When the <b>Commitment Confirmed</b> box is checked, you cannot reallocate the committed quantities to a different order            |
| <b>Order Priority</b>        | To change the order priority of added items, specify a value in the <b>Order Priority</b> column.                                   |
| <b>Options</b>               | Any custom options associated with the item.                                                                                        |
| <b>Item Source</b>           | Specifies the preferred method of supply for a particular item: Stock, Phantom, Purchase Order, and Work Order.                     |

After the work order is recorded, you can enter an assembly build against the order to close it.

## Actual Production Start or End Dates

NetSuite automatically calculates Actual Production Dates by default based on related work order transactions. This includes Assembly Builds for Standard Work Orders and Issue Components and Work Order Completions for WIP Work Orders. This functionality enables you to track actual production dates to help assess production plans, identify problems, and optimize production.

NetSuite starts calculating production when the first build transaction is initiated or, if WIP is enabled, the date components have been issued. The system end date is recorded when the work order is closed. The end date must occur on the same day or some time after the start date. You cannot record an end date if you haven't entered a start date.

For example, you start to build mountain bikes for an assembly build. The work order displays both the actual production start and end dates as January 16. The work order instructions require that 10 bicycles be built. On January 15, two bicycles were built in your facility, so the system records the Actual Production Start Date as January 15. However the other eight bicycles are assembled earlier, on January 10. NetSuite automatically adjusts the work order Actual Production Start Date as January 10 and the Actual Production End Date to January 15.

Automation logic always uses transaction dates. In the preceding example, the system used the dates from the assembly builds for these non-WIP work orders. However, NetSuite lets you manually change actual start and end dates.

## Overriding Actual Production Dates

The following procedure describes how to manually change actual production dates.

### To override a actual production date:

1. On the work order, below the **Actual Production Start** or **End Date**, click **Enter Manually**.
2. Select a new date from the calendar.  
For example, you can change the date from Dec. 2 to Dec. 3.
3. Click **Save**.

## Disabling Actual Production Dates

If you prefer to control production dates manually, and don't want NetSuite to calculate production dates, you can disable this functionality. However, you must enter production dates manually on each work order.

### To disable production dates:

1. Go to Setup > Manufacturing > Manufacturing Preferences.
2. In the **Production Execution** section, clear the **Automatically Fill Actual Production Start and End Dates** box.
3. Click **Save**.

When you create a work order, the **Production Start Date** is editable.

When you enter an **Production Start Date**, the **Production End Date** is editable.

## Adding Custom Fields to Work Orders

The following procedure provides details to adding custom fields to the Work Order page.

### To add a custom field to the Work Order page:

1. Go to Transactions > Manufacturing > Enter Work Orders.
2. Click **Customize** on the top right of the page:
  - For a custom body field, click **New Body Field**.
  - For a custom column field, click **New Column Field**.
3. On the Transaction Column Field page, enter a **Label** for the new field:
  - For body fields, this label is displayed next to the field on transactions.
  - For column fields, this name appears as a column heading on transactions.
4. On the **Applies To** subtab, check the box next to transactions you want the field to appear on:
  - For body fields, check the **Work Order / Assembly Build** box.
  - For column fields, check the **Work Order** box.
5. Click **Save**.

The custom field appears on your work order form.

## Mass Creating Work Orders

You enter work orders to track the production of assembly items needed either for stock or to fill orders. The work order lists the members, or components, of the assembly item to be built. A work order is a non-posting transaction.

Some work orders in the queue aren't intended for a particular sale. Production work orders are generated when the back ordered quantity of an assembly reaches its assigned build point. After the build point is reached, a work order is added in the Mass Create Work Orders queue.

For each work order, a bill of materials (BOM) is generated to facilitate picking member items for the build. When this work order completes, the regular stock level of the assembly is increased and the finished goods are committed to open sales orders.

Special Order work orders track assemblies for a particular sale. Production work orders track assemblies to increase stock. Both use the same work order form, but production work orders don't link to a sale transaction.

After work orders are entered, completing an assembly build for the work order closes the order.

For information about entering individual work orders, see [Entering an Individual Work Order](#).

### To mass create work orders:

1. Go to Transactions > Manufacturing > Mass Create Work Orders.
2. Select a **Location** to show only work orders for that location. Select **All** to show work orders for all locations.

If you use Multi-Location Inventory, the location selected is the one that component inventory items are committed from.

 **Note:** All items on one work order must be committed from the same location. Items can commit only from the location specified. This is true even if there are no available items at the specified location, and there are items available at another location.

3. Select a **Department** or **Class** if you track them. The selected department or class appears on the assembly build.
4. Select a **Parent Item** to show only child items for that parent.
5. Enter a **Minimum Quantity** to filter the list by the minimum set on the item record.
6. Complete the fields on the **Time Phased Items** subtab.

The **Time Phased Items** subtab displays a list of items that need to be ordered based on time-phased replenishment.

 **Note:** To use these enhancements for time-phased planned items, the Demand Planning feature must be enabled.

- a. Check the box in the **Order** column next to each item you want to create a work order for.
- b. Select an **Order Date**.
- c. Select a **Production End Date**.
- d. Check the box next to each item to order.  
Click the **Mark All** button to check all boxes or click the **Unmark All** button to clear all boxes.
- e. Accept the suggested **Quantity** or enter a new amount.

 **Note:** When you use Demand Planning, see the help topic [Demand Planning on Item Records](#) for information about suggested quantities.

When you use Multiple Units of Measure, the quantity for members of an assembly item is always defined in base units on work orders.

- f. Check the **Mark Sub-Assemblies Phantom** box to treat the sub-assemblies within the selected assembly as phantoms. The sub-assembly components are included in the assembly work order that is generated.

For information about work orders and demand planning, see, [Work Orders and Sub-Assemblies](#).

7. Click the **Reorder Point Items** subtab.

The **Reorder Point Items** subtab displays a list of items that need to be ordered based on designated reorder point. These items have a quantity available that is less than the reorder point indicated on the item record.

 **Note:** To use these enhancements for time-phased planned items, the Demand Planning feature must be enabled.

- a. Check the box in the **Order** column next to items you want to order.  
Click the **Mark All** button to check all boxes or click the **Unmark All** button to clear all boxes.
- b. Accept the suggested amount to order in the **Quantity** column, or enter a new quantity.  
The suggested NetSuite order calculation is: (preferred stock level + quantity needed) less (quantity available + quantity on order).
- c. If you use Make Departments required and Allow Per-Line Departments, select a **Department**. Departments are used for the corresponding line items on generated purchase orders. For more information, see the help topic [Using Per-Line Classifications](#).
- d. Check the **Mark Sub-Assemblies Phantom** box to mark an individual sub-assembly as a phantom assembly. A phantom assembly is typically a non-stocked assembly that groups together material needed to produce a subassembly. The Phantom BOM option lets you define the item source for the subassembly on a line-by-line basis.  
  
If you use the Work Orders and Demand Planning features, see [Work Orders and Sub-Assemblies](#).

8. Click **Submit**.

Work Orders are generated for the items you have indicated. Work orders generated for assembly items that use the Reorder Point replenishment method use Forward Scheduling. This is true regardless of the default scheduling method set in the account preferences. For more information, see [Production Scheduling Methods Overview](#).

## Marking Assemblies to Create Work Orders

When you use the Work Orders feature, you can create work orders for assembly items from sales orders. You can mark items to create work orders in two ways. You can mark the item when you create the sales order, or you can mark the item record. For more information, see [Creating Work Orders From a Sales Orders](#) and [Creating Work Orders From Item Records](#).

## Creating Work Orders From a Sales Orders

You can mark items to create work orders when you enter a sales transaction by checking the box in the **Create WO** column. Then, when you save or approve the order, NetSuite creates a work order that is linked to the sale.

### To create a work order from the sales order:

1. Go to Transactions > Sales > Enter Sales Orders.
2. Enter information in the transaction header as needed.  
Required fields display a red asterisk (\*).
3. In the **Item** field, select an assembly item.



4. Check the box in the **Create WO** column.
5. Enter additional information as needed for this line item.
6. Click **Add**.
7. Click **Save**.

## Creating Work Orders From Item Records

You can set an assembly item to default to create a work order by tagging the item record. Then, when the item is selected on a sales order, the box in the Create WO column is automatically checked.

### To set an assembly item to default to create a work order:

1. Go to Lists > Accounting > Items.
2. In the **Type** field, select **Assembly** to filter your item list to show assembly items.
3. Click **Edit** next to the item.  
Only assembly item records can be set up to default to create work orders.
4. On the **Purchasing/Inventory** subtab of the assembly item record, check the **Special Work Order Item** box.
5. Click **Save**.

## Planned Work Orders

You can enter work orders that have a status of Planned. A planned work order operates as follows:

- Includes component information
- Includes resource information
- doesn't commit items until the work order is released

Planned work orders can be either Open or Firmed.

- **Open** – Open planned work orders are deleted before supply planning runs.
- **Firmed** – Firmed planned work orders aren't deleted before supply planning runs. For more information, see [Marking Work Orders Firmed](#).

You can manually create planned work orders, or you can have the NetSuite supply planning process generated the planned work orders. For more information, see [Manually Entering Planned Work Orders](#) and [Automatically Generating Planned Work Orders](#).

## Manually Entering Planned Work Orders

The following procedure explains how to manually enter planned work orders.

### To manually enter planned work orders:

1. Go to Transactions > Manufacturing > Enter Work Orders.
2. In the **Status** field, select **Planned**.

3. Complete additional fields on the form as necessary.
4. Click **Save**.

## Automatically Generating Planned Work Orders

Set preferences for the NetSuite planning process to generate planned work orders. When you let NetSuite generate planned work orders, you can see the resources and materials required immediately after a planning process run.

### To set up preferences:

1. Go to Setup > Manufacturing > Manufacturing Preferences.
2. To define the default for supply planning work orders, select a **Create Work Orders in Supply Planning** option:
  - **don't Generate**
  - **Generate in Firm Planned Status**
  - **Generate in Open Planned Status**
  - **Generate in Released Status**

Your selection defines the default status of new work orders generated by a planning process run.



**Note:** If you **generate orders** and also use the Manufacturing Routing and Demand Planning features, you can define production scheduling methods on work orders. For more information, see [Production Scheduling Methods Overview](#) and [Supply Planning and Routing](#).

3. In the **Default Work Order Status** field, choose one of the following:
  - **Firm Planned**
  - **Open Planned**
  - **Released**

Your selection defines the default status of new work orders you manually create.

4. Click **Save**.

## Marking Work Orders Firmed

You must change the status of Open to Firmed for the order to be saved and processed when the supply planning process runs. You can open an individual order and set it to Firm. You can also use the steps below to update the status of many orders at one time.

### To mark work orders firmed:

1. Go to Transactions > Manufacturing > Mark Work Orders Firmed.
2. In the **Item** field, select an item to filter the list and show only orders that include that item.
3. In the **Customer** field, select a customer to filter the list and show only orders for that customer.
4. Check the box in the **Mark Firmed** column next to each order you want to firm.
5. Click **Submit**.

## Marking Work Orders Released

Any work order in a Firm Planned state must be set to the Released status to be able to commit items to it. You can open an individual order and set it to released. You can also use the steps below to update the status of many orders at one time.

### To mark work orders released:

1. Go to Transactions > Manufacturing > Mark Work Orders Released.
2. In the **Item** field, select an item to filter the list and show only orders that include that item.
3. In the **Customer** field, select a customer to filter the list and show only orders for that customer.
4. Check the box in the **Mark Released** column next to each order you want to firm.
5. Click **Submit**.

## Component Yield Preferences

Assembly Component preferences can help you with component requirement calculations. To use these component preferences, enable the Work Orders feature and adjust your item records settings that are available on the following:

- **Use Component Yield** – Account for material component yield loss during the ordering and planning process
- **Round Up Quantity as Component** – Round the component quantity up in the units used on work order

## Setting the Use Component Yield Preference

The Use Component Yield preference lets you automatically calculate quantity allowances for expected raw material loss during processing. You no longer need to manually adjust raw material purchase quantities to account for issues. For example, quality of material/defects or machine processing/scrap.

Accounting for material variances lets you adjust your planned usable quantity by ordering more units than the target build quantity. Work orders can reflect item order quantities based on accurate expected component yield assumptions, not the quantities listed on the bill of materials.

For example, you sell a coffee cup assembly that is made up of 3 components: 1 cup, 1 lid and 1 sleeve. Every 100 assemblies produces 5 defective cups. Therefore, when ordering the cup assembly you must order 105 units to produce 100 cups.

### To set the Use Component Yield preference:

1. Create a new BOM record.  
For more information, see [Creating a New BOM Record](#).
2. To enable NetSuite to calculate the necessary quantity to build or order based on component yield settings, in the Bill of Materials page, check the **Use Component Yield** box.  
You can find the Use Component Yield in the Bill of Material form. To learn more, see [Creating a New BOM Record](#).

NetSuite applies this option to the assembly and any sub-assembly components required for the top level assembly. For more information, see [Example Three: Sub-assemblies and Component Yield](#).



**Important:** Review your customization scripts prior to checking this box.

To prevent NetSuite from calculating the necessary quantity to build/order, clear the **Use Component Yield** box.

3. If you check the **Use Component Yield** box, you can set the following on the item record:

- The **Quantity** field on the **Components** subtab is disabled because this quantity is calculated by NetSuite.

This field displays the amount required for the assembly when one unit of the assembly is being built. This value is automatically calculated from the entries in the **Component Yield** and **BOM Quantity** fields.

- The **BOM Quantity** column displays the quantity of the component used to build an item assuming no loss.
- The **Component Yield** column displays the anticipated yield due to loss during the manufacturing process.

For example, a Sleeve is a component of the Coffee Cup assembly. Each Coffee Cup assembly requires a sleeve. The sleeve has an anticipated loss amount of 50%. Enter the following on the Coffee Cup assembly item record:

- Check the **Use Component Yield** box
- Sleeve Component Yield is 50%
- Widget 1 BOM Quantity is 1

When a work order is created for the Coffee Cup item, NetSuite calculates a requirement of 2 sleeves. NetSuite displays 2 in the **Quantity** field on the **Components** subtab.

## Setting the Round Up Quantity as Component Preference

Use the Round Up Quantity as Component Preference on a work order component quantity to round up in the units used on work order. Depending on your settings, the component yield calculation could result in a fractional quantity.

For example, a mountain bike assembly requires 2 units of the brake component. The component yield is 99%. To build 5 of these assemblies requires 10.1 units of the brake component. Because you can consume components only in whole numbers, you cannot consume 10.1 units. You need to round up to the next highest unit. For example, 11 brake component units.

On a work order for an assembly that uses component yield, NetSuite indicates the Component Quantity, BOM Quantity and Component yield for rounded-up components.

### To set the Round Up Quantity as Component preference:

1. Create or edit an item record for an Inventory item or an Assembly item that is used as a sub-assembly component.
2. In the **Purchasing/Inventory** subtab **Inventory Management** section, do one of the following:
  - To enable NetSuite to round up the quantity consumed for this item, check the **Round Up Quantity as Component** box.

- To prevent NetSuite from rounding up the quantity consumed for this item, clear this box.

When you check this option, the **Quantity** field on the **Components** subtab is disabled because this quantity is calculated by NetSuite.



**Important:** Review your customization scripts before checking this box.

## Component Yield Examples

The following table displays the Table Assembly item and its components:

| Level 1   | Level 2          | Level 3    |
|-----------|------------------|------------|
| Table (1) |                  |            |
|           | Table Top (1)    |            |
|           | Leg Assembly (2) | Rod (1)    |
|           |                  | Filler (1) |

### Example One: Assembly without Component Yield

Each table assembly you manufacture requires 1 table top and 4 legs. For each leg, you require 1 rod and 1 filler. The Build Subassembly box is checked on the Table Top Assembly item record. When a work order for a table top assembly is created, 4 rods and 4 fillers are required.

### Example Two: Assembly with Component Yield

The item Table Assembly item record Use Component Yield box is checked.

Ten percent of the leg subassemblies are processed incorrectly and are waste. When a work order is created for 100 table assemblies, NetSuite calculates that the order requires 110 rods and 110 fillers.

### Example Three: Sub-assemblies and Component Yield

When a work order is created for an assembly, the top level assembly component yield option applies to all sub-level components.

Example 3a:

- Table assembly doesn't use component yield
- Leg subassembly uses component yield

The item Table Assembly and the item record Use Component Yield box isn't checked.

After a work order is created for 100 Table Assemblies, NetSuite follows the component yield option for the top level assembly and sub-level components.

Because component yield isn't used for the top level, NetSuite doesn't use component yield for the subassembly. The work order displays a requirement of 100 rods and 100 fillers.

Example 3b:

- Table assembly uses component yield

- Leg sub-assembly doesn't use component yield

After the Build Subassembly box is checked, NetSuite performs a yield calculation for all subassembly components.

|                        |                               |                       |
|------------------------|-------------------------------|-----------------------|
| Item A                 |                               |                       |
| ■ Uses component yield |                               |                       |
|                        | Item B                        |                       |
|                        | ■ doesn't use component yield |                       |
|                        | ■ is a component of A         |                       |
|                        |                               | Item C                |
|                        |                               | ■ is a component of B |
|                        |                               | Item D                |
|                        |                               | ■ is a component of B |

A work order for Item A is created and the Build Subassembly box is checked. NetSuite uses component yield for all subassemblies (C and D) because it respects the setting of the top level assembly.

## Editing a Work Order

You can add and edit components on any work order which is in Planned, In Process or Released state. You can also remove components from a Work Order if they haven't already been used in the build. In such cases, amend the associated Work Order Issue or Work Order Completion, before attempting to remove the item from the work order. The quantity of components cannot be changed to a number less than the quantity used in a build.

### To edit individual work orders:

1. Go to Transactions > Manufacturing > Enter Work Orders > List.
2. Beside a work order that is in **In Process** or **Released** status, click **Edit**.
3. In the **Item** subtab, select the item you want to edit.
  - Click **Cancel** to delete any changes you made and return to the previous item configuration.
  - To reuse the existing item details to create a new item, click **Make Copy**.
  - Click **Insert** to add another item.
  - Click **Remove** to delete the item information.
4. Click **Save**.



**Note:** If you use Advanced Manufacturing, you can track updates to your work order. For more information, see the help topic [Scheduling Work Orders](#).

## Printing a Work Order Bill of Materials

For work orders you have entered, you can print a Bill of Materials (BOM). The BOM shows the types and quantities of items you need to complete the work order.

Following are the two ways to print a BOM.

### Print from the work order:

1. Go to Transactions > Manufacturing > Enter Work Orders > List.
2. Click **View** next to the work order.
3. On the work order, click **Print BOM**.

Only information from the header of the work order appears in the BOM. If you want to print assembly details of the work order, you must open the assembly item record and click **Print**.

### Print from the print queue:

1. Go to Transactions > Management > Print Checks and Forms.
2. Click **Bill of Materials**.
3. In the **Filter By** field, choose one of the following to filter the work orders shown:
  - **Some Items Committed** – The list shows orders that have one or more items committed to be built.
  - **All Items Committed** – The list shows orders that have all items committed to be built.
  - **Ignore Item Availability** – The list shows all open orders regardless of the availability.
4. Select a location to filter the list for orders for that location.
5. Select a form to use for this print run.

This field defaults to the preferred form, but you can choose a form you have previously customized.

If you are printing packing slips and use the Advanced Shipping feature, you can also use a custom invoice form when printing packing slips. For example, you can customize an invoice form to show the item rate and amount, and the order total. Then, when you print the packing slip using the custom form, the packing slip shows the additional information.

To customize a form, go to Customization > Forms > Transaction Forms. Click **Customize** next to the appropriate form.

6. The **Documents in Queue** field shows the number of forms you have selected to print. This field updates as you check bills of materials to print.
  7. Check the **Allow Reprinting** box to reprint previously printed transactions.
- When you check this box, all documents appear at the bottom of the page in segments. Clear this box to allow documents to be printed only one time.



**Note:** The work order tracks whether a bill of materials has been printed. It resets this flag whenever components are committed so that new top-level assemblies may be built.

8. Click the **Select Order Number** field to enter or scan in transaction bar codes.
9. Check the box in the **Print** column next to each order you want to print a bill of materials for.
10. Click **Print**.

The bill of materials prints in two sections:

#### ■ Section One: Bill of Materials

This section is a complete list of all items needed to complete the build for the work order. It shows the specific items and the quantity needed for each, including components of assembly members.

#### ■ Section Two: Assembly Hierarchy

This section shows how many of each component are needed for each unit.

**Bill Of Materials**

**DB96 Doc All Access Test Account - 563214**

**Date** 7.8.2018  
**Order #** 7

**Bill of Materials Revi...**  
**Bill of Materials**  
**Customer** Anonymous Shopper  
**Units**  
**Assembly** Assembled widget  
**Qty. Required** 100

| Item                          | Qty. Requi... | Units | Options                                                           | Inventory Detail | Description             |
|-------------------------------|---------------|-------|-------------------------------------------------------------------|------------------|-------------------------|
| \$100 shake                   | 100           |       | Gift Wrap: No                                                     |                  |                         |
| Bandage-Sterile-25-ig-blu-cir | 100           |       | Color: Blue<br>Size: Large<br>Shape: Circular                     |                  |                         |
| Bandage-Sterile-25-ig-blu-ov  | 100           |       | Gift Wrap: No<br>Color: Blue<br>Size: Large<br>Shape: Oval        |                  |                         |
| Bandage-Sterile-25-ig-blu-rec | 100           |       | Gift Wrap: No<br>Color: Blue<br>Size: Large<br>Shape: Rectangular |                  |                         |
| Bandage-Sterile-25-ig-blu-sq  | 100           |       | Gift Wrap: No<br>Color: Blue<br>Size: Large<br>Shape: Square      |                  |                         |
| 10 Person-First-Aid Kit       | 100           |       | Gift Wrap: No                                                     |                  | 10 Person-First-Aid Kit |

| Item                             | Assembly Required | Quantity |
|----------------------------------|-------------------|----------|
| Assembled widget                 |                   | 0        |
| \$100 shake                      |                   | 1        |
| Assorted Bandages - Large - Blue | Yes               | 1        |
| Bandage-Sterile-25-ig-blu-cir    |                   | 1        |
| Bandage-Sterile-25-ig-blu-ov     |                   | 1        |
| Bandage-Sterile-25-ig-blu-rec    |                   | 1        |
| Bandage-Sterile-25-ig-blu-sq     |                   | 1        |
| 10 Person-First-Aid Kit          |                   | 1        |

## Print Instructions with the Bill of Materials

If you print your BOM in PDF format, you can append the PDF file to print additional materials with the BOM. For example, you can print a diagram or instructions about the assembly process for the top level assembly item. For more information, see [Appending a PDF File to Print with the Bill of Materials](#).

An alternative way to view and print the BOM is by using the [Bill of Materials Inquiry](#).

## Appending a PDF File to Print with the Bill of Materials

If you print your bills of materials (BOM) in PDF format, you can append the PDF file to print additional materials with the BOM. These additional materials are for use when picking and assembling the item. For example, you can print a diagram or instructions about the assembly process for the top level assembly item.

### To append a file to a bill of materials:

1. Go to Lists > Accounting > Items.



2. In the **Type** field, select **Assembly** to filter your item list to show assembly items.
3. Click **Edit** next to the assembly item.
4. Click the **Communication** subtab of the item record.
5. In the **Attach File** field, attach the PDF file that contains the diagram, instructions, or other information you want to print with the BOM.
6. Check the **Print with BOM** box.  
The printed file is in PDF format and the contents of the PDF file are appended to printed BOMs for this assembly item.
7. Click **Add**.
8. Click **Save**.

## Building Work Orders

To close a work order, you must complete a build that assembles the necessary items.

### To build work orders:

1. Go to Transactions > Manufacturing > Build Work Orders.
2. Select an assembly **Item** to filter the list and show only work orders for the item.
3. Select a **Customer** to filter the list for work orders associated with that customer.  
Select **All** to display all work orders.
4. Select the **Posting Period** you want to post this transaction to. If a period is closed, you cannot post to that period.
5. By default, NetSuite inserts today's **Date** as the transaction date. You can enter another date, or click the calendar icon to select another date.
6. Select the **Bulk Build From Location** use to build the orders.
7. Check the **Filter** box to filter the list to show only orders associated with the location selected in **Bulk Fulfill From Location**.  
Clear this box to build orders associated with any location or associated with no location.
8. In the **Filter By** field, choose one of the following work order filters:
  - **Some Items Committed** – Display orders with one or more items committed to be built
  - **All Items Committed** – Displays orders with all items committed to be built
  - **Ignore Item Availability** – Displays all open orders
9. Click the **Select Order Number** field to enter or scan in transaction bar codes.
10. Check **Build** box beside each order you want to build.
11. Click **Submit**.

An assembly build is recorded for each of the work orders.

## Work Orders and Demand Planning

When you use the Work Orders and Demand Planning features, you can create work orders to replenish stock, based on demand for assembly items. These work orders use information from item records to calculate lead times for orders.

## Lead Time for Supply Planning

When you use the Work Orders and Demand Planning features, assembly item records show the **Work Order Lead Time** field.

In the **Work Order Lead Time** field, enter the lead time (in days) to build one assembly in the base unit. Then, NetSuite calculates the lead time for a work order using the following:

Lead time for a work order=

Work Order Lead Time on Item Record \* Quantity in base unit of measure

## Work Order Start and End Dates

When you use Demand Planning, if either the start date or end date is left blank, it can be calculated in the following way:

- When an **End Date** is entered but the **Start Date** is blank, the start date is calculated as follows:

Start Date = End Date - (Work Order Lead Time from the item record \* Quantity in base unit of measure)

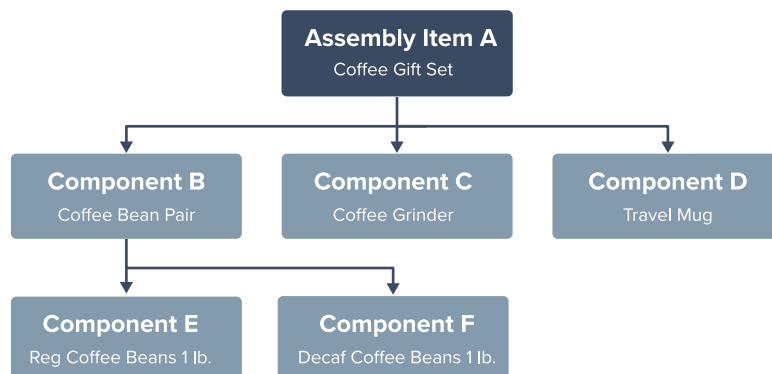
- When a **Start Date** is entered but the **End Date** is blank, the end date is calculated as follows:

End Date = Start Date + (Work Order Lead Time from the Item Record \* Quantity in base unit of measure)

**Note:** The natural rounding method is used to determine the start and end date of work orders.

## Work Orders and Sub-Assemblies

When you generate a new supply plan for an assembly item, it calculates material requirements based on the lowest level of component items needed. This is true for assembly items that have the **Mark Sub-Assemblies Phantom** box checked on the item record. For example, the following diagram depicts the Coffee Gift Set assembly item and has the following component structure.



- Assembly Item A: Coffee Gift Set
  - Component B: Coffee Bean Pair

- Component E: Bag of Regular Beans, 1 lb.
- Component F: Bag of Decaf Beans, 1 lb.
- Component C: Coffee Grinder
- Component D: Travel Mug

The **Mark Sub-assemblies Phantom** box is checked on the item record for Item A: Coffee Gift Set. When a planned work order is generated for a Coffee Gift Set, the component assembly requirements are for items E, F, C and D. And, the purchase order created shows these items. Notice that Item B: Coffee Bean Pair isn't a requirement itself, only its member components are required.

Also, the Mass Create Work Order page displays the lines from the supply plan with the Mark Sub-assemblies Phantom box disabled.

By contrast, the **Mark Sub-assemblies Phantom** box is clear on the item record for Item A: Coffee Gift Set. The work order is created only for items B, C, and D. If B isn't available, NetSuite doesn't create a work order.

# Supply 360

The Supply 360 SuiteApp analyzes component shortages across work orders based on the selected location and date filters. You can also view planned supplies of any shortage component.

You can access the SuiteApp by navigating to Reports > Supply 360 > Supply 360. The SuiteApp opens in the **Overview** subtab of the Supply 360 page.

To view information in the **Overview**, **Shortages by Work Order**, **Shortages by Component**, and **All Components Requirements** subtabs:

1. Select a location from the **Location** filter.
2. Change the suggested dates by selecting an appropriate from and to date in the **Date Range** field.

By default, the SuiteApp sets a seven-day date range starting from today. For example, if the current date is 5th of July, the **From** date will be 5th of July and the **To** date will be 11th of July.

Click the **Refresh** button to see updated data for your chosen location, date range, and subtab.



**Important:** Following points are applicable to the Supply 360 SuiteApp:

- Purchase Order, Transfer Order, Inter Company Transfer Order, Work Order, and Inbound Shipment are eligible supply types if their supply receipt date is greater than or equal to the current date, based on user preferences.
- Only work orders are considered as demand transactions. The SuiteApp displays work orders whose start date is between the selected date range.
- Past dated work orders (start date < current date) with back ordered quantity are considered under insufficient supplies, as the SuiteApp does not assign any planned supplies to past dated work orders.
- For work orders with **Planned** status, the supply source will be blank.
- Supplies waiting for approval aren't counted as potential supplies.
- Supply is considered for demand only if the supply receipt date (expect receipt date, completion date or expected delivery date) is on or after the start date.
- The Supply 360 SuiteApp always checks the available item quantity before looking at supplies for that demand.

## Installing Supply 360 SuiteApp

The Supply 360 SuiteApp requires a NetSuite OneWorld account. Only users with Administrator role can install the SuiteApp. For more information about installing a SuiteApp, see the help topics [Installing from the SuiteApp Marketplace](#) and [SuiteApp Marketplace in NetSuite](#).

## Prerequisites for Installing Supply 360

To use the Supply 360 SuiteApp, enable the features in the following table. For more information, see the help topic [Enabling Features](#).

| Subtab            | Feature Name |
|-------------------|--------------|
| Items & Inventory | Work Orders  |
| Company           | Locations    |

## Creating or Customizing Roles and Permissions to Use Supply 360

If you're an administrator, you can create new roles or customize existing ones to use the Supply 360 SuiteApp. For more information, see the help topics [Assigning Roles to an Employee](#) and [Customizing or Creating NetSuite Roles](#).

Refer to the following table for minimum access levels of permissions required for SuiteApp deployments. Ensure to update these minimum permissions for existing roles or when you create a new role.

| Subtab       | Permission       | Minimum Access Level | Purpose                                                                                               |
|--------------|------------------|----------------------|-------------------------------------------------------------------------------------------------------|
| Transactions | Work Order       | View                 | Required for Supply 360 SuiteApp.                                                                     |
|              | Find Transaction | View                 | Required for Supply 360 SuiteApp.                                                                     |
| Lists        | Locations        | View                 | Required for Supply 360 SuiteApp.                                                                     |
| Lists        | Items            | View                 | Required for Supply 360 SuiteApp.                                                                     |
| Lists        | Vendors          | View                 | Required for Supply 360 SuiteApp.                                                                     |
| Lists        | Inbound Shipment | View                 | Required for Supply 360 SuiteApp (When Inbound Shipment feature is enabled for the account).          |
| Lists        | Units            | View                 | Required for Supply 360 SuiteApp (When Multiple Units of Measure feature is enabled for the account). |
| Setup        | SuiteScript      | Edit                 | Required for Supply 360 SuiteApp.                                                                     |

## Supported Browsers for Supply 360

The Supply 360 SuiteApp supports the following browsers and operating systems:

| Browser                         | Operating System      |
|---------------------------------|-----------------------|
| Google Chrome                   | Windows 10            |
|                                 | Mac OS 10.15 or newer |
| Microsoft Edge (Chromium-based) | Windows 10 or newer   |
| Mozilla Firefox                 | Windows 10 or newer   |

| Browser | Operating System                      |
|---------|---------------------------------------|
|         | Mac OS 10.15 or newer                 |
| Safari  | Two most recent stable major versions |

## Supply 360 Overview Subtab

The **Overview** subtab is the default page the SuiteApp takes you to. This subtab shows the total number of work orders, components, and assemblies with insufficient inventory, along with related alerts.

### Alerts

Alerts show you the total number of issues.

The following alerts list is displayed on the **Overview** subtab:

- Number of work orders scheduled today have components with insufficient inventory on hand.
- Number of components have insufficient planned supply.

### Tiles

The three titles on the **Overview** subtab are as below:

- The **Work Orders** tile displays the total count of work orders with components that have insufficient inventory on hand (at least one component is in shortage). Click **View Details** to go to the **Shortages by Work Order** subtab. Here, rows in red color indicate the work orders with insufficient planned supplies.
- The **Components** tile displays the total count of components with insufficient inventory on hand where shortage quantity > 0 in at least one work order. Click **View Details** to go to the **Shortages by Component** subtab. Here, rows in red color indicate the components are completely in shortage.
- The **Assemblies** tile displays the total count of assemblies with insufficient inventory on hand on respective work orders. Click **View Details** to go to the **Shortages by Work Order** subtab. Here, rows in red color indicate the work orders with components in shortage.

## Shortages by Work Order Subtab

The **Shortages by Work Order** subtab shows a list of work orders and their related assemblies that has potential shortages. The results are based on the dates chosen in the **Date Range** field. You can modify the date range based on your business requirement. The SuiteApp prioritizes and displays the work order results based on work order production start dates or transaction dates (if Demand Planning feature is enabled). You can filter the work order results based on the Start Date, Sales Order, Work Order, Assembly, Component, Status, and Insufficient Supply.

You can use the **Choose Columns** option to select which columns you want to see on the **Shortages by Work Order** subtab grid.

### To Select Columns for the Subtab Grid:

1. Go to Reports > Supply 360 > Supply 360 > Shortages by Work Order subtab.

2. On the **Shortages by Work Order** subtab, click **Choose Columns**.
3. On the **Choose Columns** window, check the boxes for the required columns to show on the grid.
4. Click **Save**.

To retain the default columns available on the grid, click **Reset**.

The following table provides details about the columns available in the **Shortages by Work Order** subtab.

| Column Name       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Start Date        | If Demand Planning is on, you'll use the work order's production start date. Otherwise, the transaction date of the work order is considered. For more info, see <a href="#">Enabling Features</a> . For more information, see the help topic <a href="#">Enabling Features</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Sales Order       | Sales order associated to the work order.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Work Order        | Click the link to go to the work order transaction.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Assembly          | Assembly item associated to work order.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Component         | Name of the component.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Status            | Status of the work order.<br><br><div>  <b>Note:</b> Work orders with <b>Planned</b> status are not considered when calculating the short quantity. </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Short Quantity    | The back ordered quantity for the component (at line level) in your selected date range.<br><br><div>  <b>Note:</b> The Short Quantity column displays in color coded format for the following conditions: <ul style="list-style-type: none"> <li>■ If any of the following conditions is true, the <b>Short Quantity</b> column appears in red color for that line: <ul style="list-style-type: none"> <li>□ When supply quantity is less than the short quantity.</li> <li>□ If Work Orders start date is less than the system date or current date of the system.</li> <li>□ If there are no planned supplies for the components.</li> </ul> </li> <li>■ In the following condition, the <b>Short Quantity</b> column appears in orange color for that line: <p>If the supply is fully available for a work order and one of the supply component receipt date is same as the production start date.</p> </li> <li>■ In the following condition, the Short Quantity column is blank without any color: <ul style="list-style-type: none"> <li>□ If the supply is fully available for a work order and the supplies arrive before the production start date.</li> <li>□ For planned work orders.</li> </ul> </li> </ul> </div> |
| Units             | Component Unit of Measure (Line level).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Required Quantity | Component Ordered Quantity.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

| Column Name            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Supply Source          | <p>Supply transactions displayed with a hyperlink. Any one of the following supply category is displayed:</p> <ul style="list-style-type: none"> <li>Work order</li> <li>Purchase order</li> <li>Transfer order</li> <li>Inter company transfer order</li> <li>Inbound shipment</li> </ul> <p>In case of multiple supplies, the <b>Multiple</b> hyperlink opens a popup which displays all eligible supply transactions within a selected date range.</p>   |
| First Receipt Quantity | First receipt quantity from the first source (if the source is multiple) whose expected receipt date, production end date, or expected delivery date which is at the earliest. If it is a single source, the corresponding first receipt quantity is displayed associated to a supply type.                                                                                                                                                                 |
| Operation Sequence     | The component's operation sequence number for the work order.                                                                                                                                                                                                                                                                                                                                                                                               |
| Operation Name         | Name of the operation related to the component.                                                                                                                                                                                                                                                                                                                                                                                                             |
| First Receipt Date     | <p>It shows the earliest supply date. The column value is derived from one of the following transactions:</p> <ul style="list-style-type: none"> <li>Expected receipt date for purchase order, transfer order, or inter company transfer order.</li> <li>Production end date of the work order. (if Demand Planning feature is disabled, transaction date of the work order is considered)</li> <li>Expected delivery date for inbound shipment.</li> </ul> |
| Confidence             | <p>The vendor's predicted risk confidence.</p> <p>In case of multiple supplies, if the earliest supply is a purchase order, confidence percentage associated to vendor is displayed on results. If none of the supplies contain purchase order, the confidence will be blank.</p>                                                                                                                                                                           |

To download data from the **Shortages by Work Order** subtab, click **Export** button. This creates an .xls file with the data from the subtab.

## Shortages by Component Subtab

The **Shortages by Component** subtab shows a list of components used in multiple work orders that might have shortages. The results are based on the dates you enter in the **Date Range** filter. Components with back ordered quantity greater than zero in work orders are listed here. You can filter the results based on the first required date, component, and insufficient supply.

You can use **Choose Columns** option to select the required columns to display on the **Shortages by Component** subtab grid.

### To Select Columns for the Subtab Grid:

1. Go to Reports > Supply 360 > Supply 360 > Shortages by Component subtab.
2. On the **Shortages by Component** subtab, click **Choose Columns**.
3. On the **Choose Columns** window, check the boxes for the columns you want to see on the grid.
4. Click **Save**.



To retain the default columns available on the grid, click **Reset**.

The following table provides details about the columns in the **Shortages by Component** subtab.

| Column Name           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| First Required Date   | If Demand Planning is on, you'll use the work order's production start date. Otherwise, the transaction date of the work order is considered.                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Component             | Name of the component.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Total Short Quantity  | <p>Sum of back ordered quantities for a component across all the work orders within the selected date range.</p> <div>  <b>Note:</b> In any of the below conditions, the <b>Total Short Quantity</b> column appears in red color for that line: <ul style="list-style-type: none"> <li>■ If any one of the work order has supply quantity lesser than the short quantity.</li> <li>■ If there is a historical dated work order in the selected time frame.</li> </ul> </div> |
| Units                 | Base unit of the component item.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Supply Source         | <p>Supply transactions displayed with a hyperlink. (Work order, purchase order, transfer order, inter company transfer order, or inbound shipment)</p> <p>In case of multiple supplies, the <b>Multiple</b> hyperlink opens a popup which displays all eligible supply transactions within a selected date range.</p>                                                                                                                                                                                                                                         |
| Supply Quantity       | Sum of the supply quantity from all supply types across multiple work orders. If there are no planned supplies, it will be zero.                                                                                                                                                                                                                                                                                                                                                                                                                              |
| First Supply Quantity | <p>Supply quantity is taken from the first source (if source is multiple) whose expected receipt date, production end date, or expected delivery date which is at the earliest.</p> <p>If it is a single source, the corresponding supply quantity is displayed associated to a supply type.</p>                                                                                                                                                                                                                                                              |
| Confidence            | <p>The vendor's predicted risk confidence.</p> <p>If there are multiple supplies and the earliest one's a purchase order, you'll see the vendor's confidence percentage. If none of the supplies contain purchase order, the confidence will be blank.</p>                                                                                                                                                                                                                                                                                                    |
| First Receipt Date    | <p>It shows the earliest supply date. The column value is derived from one of the following transactions:</p> <ul style="list-style-type: none"> <li>■ Expected receipt date for purchase order, transfer order, or inter company transfer order.</li> <li>■ Production end date of the work order. (if Demand Planning feature is disabled, transaction date of the work order is considered)</li> <li>■ Expected delivery date for inbound shipment.</li> </ul>                                                                                             |

To download data from the **Shortages by Component** subtab, click **Export** button. This creates an .xls file with the data from the subtab.

## All Components Requirements Subtab

The **All Components Requirements** subtab displays the list of all components used in work orders. The results are displayed based on the dates you provide in the **Date Range** field. These results can also be filtered based on the component.

The following table provides details about the columns in the **All Components Requirements** subtab.

| Column Name             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Component               | Name of the component.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Total Required Quantity | Total ordered quantities across all the work orders.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Units                   | Base unit of the item.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| On Hand                 | Displays on hand quantity from the item record.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Committed               | Displays committed quantity from the item record.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Available               | Displays available quantity from the item record.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Total Planned Supplies  | <p>Total of all planned supply quantities of a component across all the work orders. The SuiteApp shows total planned supplies based on the following date range.</p> <ul style="list-style-type: none"> <li>Current system date to selected future date.</li> </ul> <div data-bbox="511 772 1377 861">  <b>Note:</b> If the start date of the <b>Date Range</b> is from a historical period, the SuiteApp uses the current system date as the start date to load results.         </div> <ul style="list-style-type: none"> <li>If date range is not selected, the SuiteApp defaults to a time frame from the current system date to seven days in the future.</li> </ul> |

To download data from the **All Components Requirements** subtab, click **Export** button. This will generate an .xls file containing the relevant subtab data.

# Available To Build

The Available To Build (ATB) SuiteApp helps manufacturers see how many assemblies and kits they can make with their current on-hand inventory. The SuiteApp evaluates based on end assembly and sub-assembly component inventory, helping manufacturers to determine the quantity they can provide to customers. Manufacturers can analyze how many assemblies or kits they can assemble right now with current component inventory, using either the top-level or fully exploded Bill of Materials (BOM).

You can use the SuiteApp functionality from the Sales Order, and Item 360 SuiteApp pages. You'll see ATB content in a portlet on the Manufacturing and Sales pages of the Item 360 Dashboard. You can also calculate the ATB value for top-level or all-level items from the Sales Order page.



**Important:** Following are the key considerations for the Available to Build SuiteApp:

- ATB value is calculated for assembly item and kit items.
- ATB is calculated by considering either top level or all levels of the BOM.
- For assembly items, the component details are obtained from the BOM. You need the Advanced Bill of Materials feature to calculate ATB for assembly items; otherwise, the SuiteApp uses only the assembly's on-hand quantity.
- The SuiteApp does not consider any other supply and demand types to calculate ATB.
- You must have Item 360 SuiteApp installed in your account to use the Available to Build SuiteApp functionality through Item 360 SuiteApp.

## Installing Available to Build SuiteApp

The Available to Build SuiteApp requires a NetSuite OneWorld account.

Only users with Administrator role can install the SuiteApp. For more information about installing a SuiteApp, see the help topics [Installing from the SuiteApp Marketplace](#) and [SuiteApp Marketplace in NetSuite](#).

## Prerequisites for Installing Available to Build

To use the Available to Build SuiteApp, you'll need to enable the features listed below. For more information, see the help topic [Enabling Features](#).

| Subtab            | Feature Name             |
|-------------------|--------------------------|
| Items & Inventory | Inventory                |
| Items & Inventory | Multi-Location Inventory |
| Transactions      | Sales Order              |

## Creating or Customizing Roles and Permissions to Use Available to Build SuiteApp

If you are an administrator, you can create new roles or customize roles for using the Available to Build SuiteApp. For more information, see the help topics [Assigning Roles to an Employee](#) and [Customizing or Creating NetSuite Roles](#).

Refer to the following table for minimum access levels of permissions required for the SuiteApp deployment. These minimum permissions must be updated to the existing role or when creating a new role.

| Subtab       | Permission        | Minimum Access Level | Purpose                                                                                                                                                                                                                                         |
|--------------|-------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Transactions | Sales Order       | Edit                 | Required for Available to Build SuiteApp.                                                                                                                                                                                                       |
| Transactions | Work Order        | View                 | Required for Available to Build SuiteApp.                                                                                                                                                                                                       |
| Transactions | Find Transaction  | View                 | Required for Available to Build SuiteApp.                                                                                                                                                                                                       |
| Lists        | Items             | View                 | Required for Available to Build SuiteApp.                                                                                                                                                                                                       |
| Lists        | Locations         | View                 | Required for Available to Build SuiteApp.                                                                                                                                                                                                       |
| Lists        | Subsidiaries      | View                 | Required for Available to Build SuiteApp.                                                                                                                                                                                                       |
| Lists        | Bill of Materials | View                 | Required for Available to Build SuiteApp.<br><br> <b>Note:</b> This permission is required only if the <b>Advanced Bill of Materials</b> feature is enabled. |
| Lists        | Units             | View                 | Required for Available to Build SuiteApp.<br><br> <b>Note:</b> This permission is required only if the <b>Multiple Units of Measure</b> feature is enabled.  |
| Setup        | Set Up Company    | Full                 | Required for Available to Build SuiteApp.                                                                                                                                                                                                       |

## Supported Browsers for Available to Build

The Available to Build SuiteApp supports the following browsers and operating systems:

| Browser                         | Operating System                      |
|---------------------------------|---------------------------------------|
| Google Chrome                   | Windows 10<br>Mac OS 10.15 or newer   |
| Microsoft Edge (Chromium-based) | Windows 10 or newer                   |
| Mozilla Firefox                 | Windows 10<br>Mac OS 10.15 or newer   |
| Safari                          | Two most recent stable major versions |

## General Preferences for Available to Build

You can manage on how you want to view the Available to Build values from the Custom Preferences subtab. The **General** subtab is found under Setup > Company > General Preferences > Custom Preferences subtab.

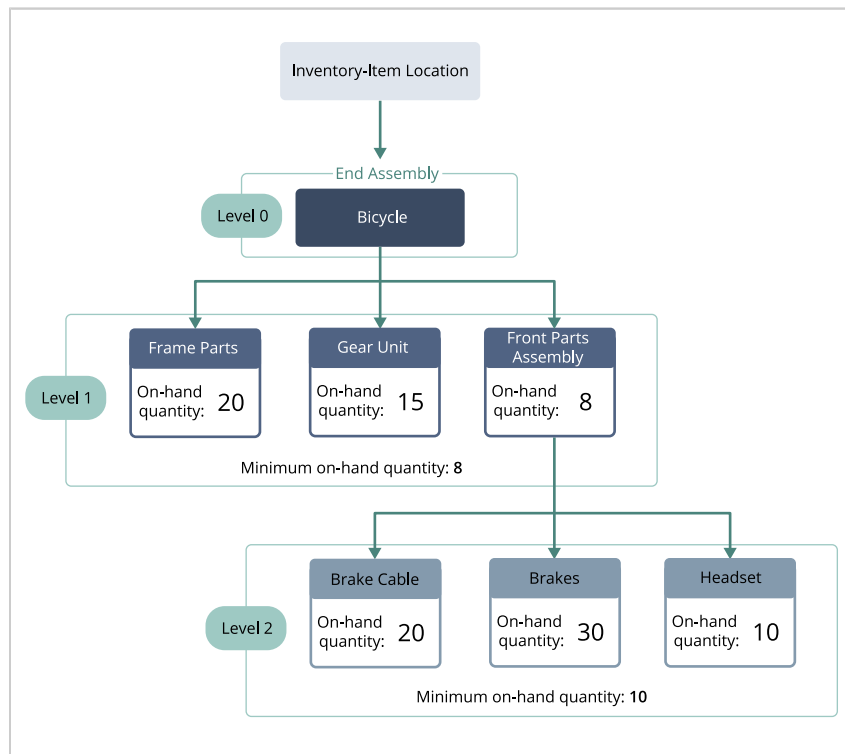
Check the following boxes to choose how you want to see available to build details on the Sales Order Entry page.

- **Show Available to Build Subtab in Sales Order:** Check this box to view Available to Build subtab in the Sales Order Entry page that provides history on the list of items for which ATB is calculated.
- **Allow Available to Build Alerts:** Check this box to see ATB results in a pop-up notification.

During the installation of the SuiteApp, these boxes are checked by default.

## Available to Build Calculation

The following graphic provides a use case on how available to build is calculated:



For each component at each level, the quantity required is 1.

- **ATB value at Top Level = 8**  
Minimum on-hand quantity at level 1 (L1) = 8 (based on Front Parts Assembly)
- **ATB value at All Levels = 15**  
Additional ATB value of front parts assembly = 10 (based on the Headset)  
On-hand of Frames = 20, On-hand of Gear = 15, On-hand of Front Parts Assembly = 8 + ATB of Front Parts = 10 => Total is 18.

Minimum on-hand quantity at L2 = 15 (based on Gear Unit)

For calculating the available to build, note the following:

- All the on-hand quantities are based on the location (LOC) selected for an end assembly.
- The quantity shown in the graphic is on-hand quantity of an item.
- For assembly items, the item hierarchy is fetched from the BOM revision.
- For kit items, the component quantity details are available on the item record.
- The ATB value for an end assembly is displayed according to the base unit specified on the item record. When calculating the ATB value for components in a BOM revision, all conversions for multiple units of measure are considered.
- To calculate ATB for an item, the SuiteApp selects BOM based on the below priority:
  1. The SuiteApp first checks the default BOM location set for the item in its record.
  2. If no default BOM location is set, the SuiteApp uses the master default BOM specified on the item record.
- For a phantom item at any level, the SuiteApp uses the phantom item BOM to check quantities and calculate ATB.
- The SuiteApp does not consider any BOMs with inactive revisions or expired revisions to calculate ATB.

## Calculating Available to Build on Sales Order

The ATB details for assembly items and kits are shown on the Sales Order page. You can calculate ATB for an assembly item or kit at top level or all levels. On the Sales Order page, under **Item** subtab, select the items for which you want to calculate ATB value.

**Note:** The **Calculate Available to Build** option is available only for assembly items and kit items. For other items, the **Calculate Available to Build** option is disabled.

To calculate available to build for an assembly item or kit item:

1. Go to Transactions > Sales > Enter Sales Orders.
2. In the **Custom Form** field, select a sales order form.
3. In the **Customer** field, select a customer.
4. Select the location from one of the following places:
  - In the Classification section, from the **Location** list, select the location by which you want to classify the sales order.
  - In the **Items** subtab, select the location from either the **Inventory Location** list or the line level **Location** list.

**Note:** The SuiteApp uses one of the locations from the following fields:

- The location provided in the **Inventory Location** field under **Items** subtab. If **Inventory Location** is empty, the SuiteApp uses the sales order line level location added in the **Location** field under the **Items** subtab as an input.
- If the **Location** field is missing in the sales order form, then the SuiteApp uses the default sales order location added in the Order Management page.

If the location is missing in the above fields, you'll see an error message prompting you to select a location.

5. In the **Items** subtab, add assembly or kit items to calculate the ATB value.
6. In the **Calculate Available to Build** field, select **Top level** or **All level** to trigger ATB value for that level. This calculates the on-hand inventory for the selected item based on the BOM and provides the available to build details.
  - The following details are displayed as part of ATB results for an assembly item:
    - Item
    - Available to Build Level
    - Location
    - Available to Build
    - On-Hand
    - Available
    - Work Order Quantity
    - Units
  - For kit items, the available to build results include:
    - Item
    - Location
    - Available to Build
7. Add any other required details to the sublist and click **Add**.  
 You'll only see the **Calculate Available to Build** option on the **Items** subtab when you select an assembly or kit item. In the Items > Available to Build subtab, you can view the history of all the items for which ATB is calculated for that single instance.
8. Enter additional information as necessary to complete the sales order entry.
9. Click **Save**.  
 In **View** mode, the **Calculate Available to Build** option is not displayed on the sales order.

## Viewing Available to Build on Item 360 SuiteApp Dashboard

You'll see ATB details for assembly items and kits in the Item 360 SuiteApp dashboard portlets. The **Manufacturing** and **Sales** page of the Item 360 dashboard are enabled with the **Available to Build** portlet.

### Available to Build on Manufacturing Page

On this page, the ATB value is generated for an assembly item. You must select a location on the Manufacturing page to calculate the ATB value.

Go to Reports > Item 360 > Item 360, then click on the **Manufacturing** page. From there, expand the **Available to Build** portlet.

The portlet provides the following details for the selected assembly item at a selected location:

- Location
- On-Hand
- Available

- Available to Build
- Work Order Quantity
- Unit (base unit of the item)

Check the **Top Level** box to see the ATB value at the top level of an assembly item. Clear this box to calculate ATB at **All levels**.

## Available to Build on Sales Page

On this page, the ATB value is generated if the item is a kit item. Go to Reports > Item 360 > Item 360, then click on the **Sales** page. From there, expand the **Available to Build** portlet.

The portlet provides the following details for the selected kit item at a selected location:

- Location
- Available to Build

Check the **Top Level** box to see the ATB value at the top level of a kit item. Clear this box to calculate ATB at **All levels**.

**Note:** If the Available to Build SuiteApp is not installed, you won't see the Available to Build portlet on the Manufacturing and Sales pages of the Item 360 dashboard.

## Using Available to Build Without Advanced Bill of Materials

If the Advanced Bill of Materials feature is disabled, the sublevel items of an assembly item are not considered when calculating ATB. In this scenario, the ATB value for an assembly item is calculated from the on-hand quantity. For more information about how to enable or disable Advanced Bill of Materials feature, see the help topic [Enabling Advanced BOM](#).

**Note:** The following capabilities are affected when Advanced Bill of Materials feature is disabled:

- The Item 360 Manufacturing page does not display **Available to Build** portlet.
- You don't need the Advanced Bill of Materials feature to calculate ATB for kit items, the ATB for kits is calculated based on the item master component list.
- On the Sales Order page, you won't see the **Calculate Available to Build** option for assembly items.

## Available To Build Glossary

### A

#### All Levels



The All levels approach considers the entire BOM hierarchy including sub-assemblies and their respective components. The ATB value is calculated at all levels by considering both immediate components and multilevel components, including sub-assemblies and sub-kit items.

**Assembly Item**

An assembly item is an inventory item made of several components (defined by a Bill of Materials) but is identified as a single item. The available to build is calculated for assembly items.

**Available**

This term represents the available quantity in the inventory search after the commitment of stock in the units of measure (UOM).

**Available to Build**

The ATB value is a metric derived from the Available to Build SuiteApp. The resulted value is a comprehensive calculation of either assembly items or kit item types at top level or all levels.

## K

**Kit Item**

A kit item consists of a group of components, parts, or sub-assemblies. These items are grouped together based on a predefined configuration to form a complete unit.

## O

**On-hand**

This term represents the available quantity of an item as base multiple units of measure (UOM) in the inventory for a selected location.

## T

**Top Level**

The top level refers to the highest level in a BOM hierarchy. It represents the end product that is assembled from various components and sub-assemblies. The ATB value is calculated at the top level based on the on-hand quantity of immediate components.

## U

**Units**

The resulted ATB value is in the format of base unit of an item record.

## W

**Work Order Quantity**

In a work order transaction, the sum of outstanding work order quantities of work orders with the statuses **Released** or **In Progress**.

# Advanced Bill of Materials

A Bill of Materials (BOM) lists the quantities of raw materials, assemblies, sub-components, and parts you need to manufacture a product. A BOM can be used to communicate between manufacturing partners, multiple facilities within the organization, or with a single manufacturing plant.

 **Note:** After your administrator enables the Advanced Bill of Materials feature, the Advanced BOM record replaces the Assembly/Bill of Materials record. When you use Advanced BOM, any existing BOM is automatically converted to a read-only legacy BOM record.

Advanced BOM enables you to do the following:

- Copy existing BOMs and create new BOMs
- Use a single BOM across multiple assemblies  
For example, multiple bicycle frame sizes could all use the same wheel assembly BOM.
- Assign multiple BOMs to a single assembly
- Apply unique BOMs to various stages of the product lifecycle  
For example, an engineering BOM, a production BOM, and a subcontracting BOM.
- Copy existing Manufacturing Routings and create new routings
- Designate a default BOM for an assembly, or select an assembly default BOM for a location  
For example, a bicycle assembly produced in Canada uses BOM CA. When the company starts producing the same bicycle in the U.S., the system uses BOM U.S.
- Define yield at the component level on the BOM revision record  
Component yield enables you to account for material component loss during ordering and planning.
- Create multiple revisions of a BOM with different effective start and end dates

## Enabling Advanced BOM

An administrator must use the following procedure to enable the Advanced Bill of Materials (BOM) feature.

 **Important:** Bill of Materials impacts many areas within NetSuite. Before making Advanced BOM available to your organization, test it in your sandbox or release preview account. Any company customizations and third party integrations could also be affected.

### To enable advanced BOM:

1. Go to Setup > Company > Enable Features.
2. Click the **Inventory & Items** subtab.
3. In the **Inventory** section, check the **Advanced Bill of Materials** box.
4. In the warning box, click **OK**.
5. Click **Save**.

Multiple BOM revisions are created for assemblies that were using BOM revision control. For more information, see [Revision Control BOM Management](#). These revisions appear on the BOM record revisions subtab.



**Important:** If you disable Advanced BOM after using it, NetSuite deletes any BOM and BOM revision records you created when Advanced BOM was enabled. NetSuite saves only legacy BOMs.

## Viewing Legacy BOM Records

After an administrator enables the Advanced BOM feature, NetSuite automatically converts existing BOMs to read-only legacy BOM records. A legacy BOM is a snapshot of an assembly and its components at the time of conversion. The legacy BOM contains a link to the assembly record.

Advanced BOM also links existing routings to corresponding legacy BOMs. NetSuite saves a copy of the link between the routing and the assembly in the event Advanced BOM is disabled. If Advanced BOM is disabled, the routing is linked to the assembly record.

After Advanced BOM is enabled, the **Assembly/Bill of Materials** label changes to **Assembly** on the New Item, Assembly Item, and Saved Search forms. In addition, the **Derived from Member Items Cost Estimate Type** is no longer available for Assembly Items.

### To view a legacy BOM:

1. Go to Lists > Supply Chain > Bill of Materials.
2. Beside the item you want to display, click **View**.
  - To view Bill of Materials – click the link under the **Legacy BOM for Assembly**.
  - To view BOM Revision details – on the **Revisions** subtab, click the revision **Name**.

Legacy BOM records should be used as back up records only. To modify a legacy BOM record, create a copy of the legacy BOM. For more information, see [Copying a BOM Record](#).

## Enabling Legacy BOM Editing

Some manufacturing customers want to use Advanced BOM, but don't want to copy Legacy BOMs. These customers can enable legacy BOM editing.

### To enable legacy BOM editing:

1. Go to Setup > Manufacturing > Manufacturing Preferences.
2. In the Product Definition section, check the **Allow Editing of Legacy BOMs** box,
3. Click **I Agree**.

After you click **Agree**, you cannot disable the Advanced BOM feature. It is important to test this feature in your sandbox account before enabling it in your production account.

4. Click **Save**.

## Create a BOM



Creating Advanced Bills of Materials and BOM Revisions

You can create a BOM using one of the following methods:

- [Creating a New BOM Record](#)
- [Creating a BOM From an Assembly](#)
- [Copying a BOM Record](#)
- [Default BOM Workflow](#)

## Creating a New BOM Record

The following procedure describes how to create a new BOM record.

### To create a new BOM record:

1. Go to Lists > Supply Chain > Bills of Materials > New.
  2. Enter a unique and descriptive BOM **Name**.
  3. In the **Memo** field, enter any information you want to include with this BOM.  
For example, include phase labeling.
  4. To apply component yield to all BOM revisions, check the **Use Component Yield** box.  
For more information, see [Component Yield Preferences](#).
  5. To allow all assemblies to use this BOM, check the **Available for All Assemblies** box.  
Clear the box to limit BOM use to only the assemblies selected in the **Restrict to Assemblies** field.
-  **Note:** Selecting an assembly from the **Restrict to Assemblies** field doesn't create a link between the BOM and the assembly.
6. To permit all locations to use this BOM, check the **Available for All Locations** box.  
Clear this box to limit BOM use to the locations selected in the **Restrict to Locations** field.
  7. Check the **Inactive** box if you don't want this BOM to appear in search lists on records and forms.  
Clear this box if you want this BOM to appear in lists.
  8. If you use NetSuite OneWorld, select the **Subsidiary** this BOM is available for. Subsidiary is inherited by BOM revisions.  
Press and hold CTRL to select multiple subsidiaries.
  9. To make the BOM available for all subsidiaries of the selected parent, check the **Include Children** box.  
If checked, the read-only **Used on Assembly** box indicates that the BOM is associated to an assembly.
  10. Click **Save**.

## Creating a BOM From an Assembly

The following procedure describes how to create a BOM from an assembly.

### To create a BOM from an assembly:

1. To create a new assembly or use an existing assembly, go to Lists > Accounting > Items.
2. Beside the assembly you want to create a BOM for, click **Edit**.
3. Click the **Manufacturing** subtab.
4. In the **Bill of Materials** list, click **New**.
5. Complete the **Bill of Materials** window.
6. Click **Save**.

 **Note:** Assemblies with six or more BOM levels may interfere with the Material Requirements Planning process performance.

## Copying a BOM Record

Copying a BOM enables you to reuse existing (legacy) work to create a new BOM. The following are reasons you might want to copy a BOM:

- Create a new BOM to use the same current and future revisions
- Create alternate BOMs with minor component changes for the same product
- Update a product version with component updates

### To copy a BOM:

1. Go to Lists > Supply Chain > Bill of Materials.
2. Beside the BOM you want to copy, click **View**.
3. Click **Actions** and then click **Make Copy**.
4. Change the copied Bill of Materials **Name**.  
For example, change Engineering V1 BOM to Engineering V2 BOM.
5. Make other changes as necessary.
6. Click **Save**.

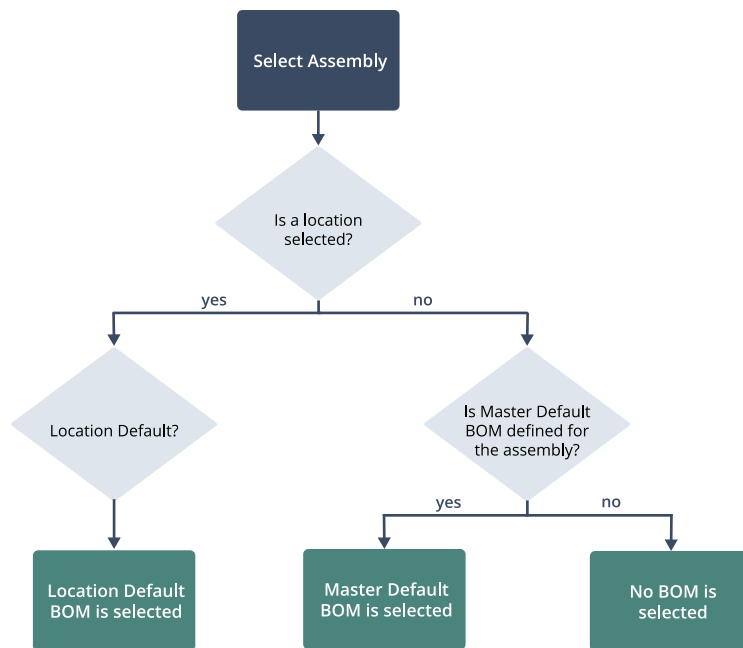
 **Note:** As of NetSuite 2023.2, when you copy an assembly, the corresponding Bill of Materials (BOM) won't be copied.

## Default BOM Workflow

Advanced BOM lets you assign a location-specific default BOM or a master default BOM to an assembly.

For example, you can set one BOM as the master default for the assembly. Then, set a second BOM as the East Coast location default. A BOM cannot be both a master and location default. They must be assigned to separate BOMs within an assembly.

The following diagram displays the workflow an assembly follows to select the appropriate BOM:



For example, a mountain bike work order assembly is fulfilled at Wolfe Manufacturing's Canadian facility. Canadian models of the mountain bike use a different wheel assembly than those manufactured in the United States. To complete the order, the system searches the BOM settings for a location default BOM. If a Canadian default location BOM exists, it is selected.

If a Canadian location default is not defined, the assembly then searches for a master default BOM. If a master BOM is assigned, it is selected. If a default BOM is not assigned, no BOM is selected.

When an assembly has no BOM defined, the **Items** sublist on the Work Order is empty. You must manually add components.

This default BOM workflow logic also applies to the following NetSuite records and transactions:

- Work orders
- Assembly builds and unbills
- Supply plans
- Standard cost roll-ups
- Phantom items
- BOM inquiries

## Creating BOM Revisions

A BOM revision let you update a BOM's details throughout the product lifecycle. A BOM revision also provides an accessible revisions history. Use revisions to compare and track cost savings when many BOM revisions are used in production.

For example, revision Engineering RV2 saves Wolfe Manufacturing \$27,000 dollars a year because it uses a simplified spoke insertion method on its wheel assembly. This results in faster machine run times compared to the previous Engineering BOM.

### To create a BOM revision:

1. Go to Lists > Supply Chain > Bill of Materials.
2. Beside the BOM you want to revise, click **View**.
3. On the **Revisions** subtab, click **Create New Revision**.
4. Enter a revision **Name**.
5. In the **Memo** field, enter any information you want to include with this revision.
6. To set an **Effective Start Date** and **Effective End Date**, click the calendar icon and then select a date from the calendar.

If you don't enter an effective end date. Future BOM revisions may be affected.



**Note:** Only one revision can be active at a time. BOM revision start and end dates cannot overlap. Gaps between revision dates are permitted.

For more information, see [Updating BOM Revision Record Dates](#).

7. Check the **Inactive** box if you don't want this BOM to appear in search lists on records and forms. Clear this box if you want this BOM to appear in lists.

8. The read-only **AM BOM Operations** field:

When you edit or save a Bill of Materials Revision record without making any changes, the AM BOM Operations field Internal and component ID values are updated.

However, when you create, edit, or save a BOM Revision record, the AM BOM Operations field component internal ID value is also updated.

For example, an AM BOM Operations field that displays 508,T,11, represents the following:

- **508** – is the internal id of the AM Operation. For example, the Assemble Table task.
- **T** – the Auto-issue box options, True or False.
- **11** – the internal id of the component. For example, 9MS.

After editing or saving, the AM Operations BOM Field could change to the following:

- **508** – is the internal id of the AM Operation. For example, the Assemble Table task.
- **T** – the Auto-issue box options, True or False.
- **42** – the internal id of the component. For example, 23MN.

9. Click **Save**.
10. On the **Revisions** subtab, click **Edit** beside the revision you created.
11. On the **Components** subtab, select an **Item** to include with this revision.  
All associated revision components appear in this subtab.
12. Accept or edit the **BOM Quantity**.  
You can enter partial quantities up to 5 decimals.
13. Select or enter the **Units**.

Use the unit of measure that is defined on the BOM and consumed in production. The unit improves flexibility, makes BOMs easier to read, and enables you to specify quantity units of measure.

Select an **Item Source**.

The work order item source determines where item quantities are taken from. For example, from stock, work order, or purchase order.

For more information, see the Item **Source** field on the [Phantom Assemblies](#) window.

14. To add more items, click **Add** and then repeat steps 9 through 12.

After you select your components, use the **Move** buttons to change component order.

15. Click **Save**.

## Copying BOM Revisions

Copying a BOM revision is a convenient way to reuse existing work to help you create a new revision. It also reduces input tasks and helps maintain consistency across records.

### To copy a BOM revision:

1. Go to Lists > Supply Chain > Bill of Materials.
2. Beside the BOM containing the revision you want to copy, click **View**.
3. Beside the revision you want to copy, click **Edit**.
4. Click **Actions** and then click **Make a Copy**.
5. Change the **Name** of the new BOM revision.  
For example, change Engineering Rev1 BOM to Engineering Rev2 BOM.
6. Optionally, update the **Memo** text.
7. Accept the default **Bill of Materials**, or select a new one.  
You can copy a revision from one BOM and associate it with another BOM.  
For example, you can move an assembly from an engineering BOM to a production BOM.
8. Accept the default **Start** and **End Dates**, or select new dates.
9. Check the **Inactive** box if you don't want this routing to appear in routing lists on records and forms.  
Clear this box if you do want this routing to show in lists.
10. Accept the default component **Items**, or select a new one.  
To remove a component, select the item and then click **Remove**.
11. Click **Save**.

## Updating BOM Revision Record Dates

Advanced Bill of Materials doesn't permit BOM revision dates on the same BOM to overlap. A current BOM revision must end before a new BOM revision can start. If revision dates overlapped, NetSuite would be unable to identify which revision to select.



**Note:** You can update the current revision **Effective End Date** on a new revision. If you try to save the new BOM revision without an effective end date, NetSuite prompts you to set the date minus one day. After you click OK, the system automatically sets the date and saves your revision.

For example, BOM Revision 1 was created without an effective end date. You create BOM Revision 2 and enter an **Effective Start Date** of February 25, 2020. Attempting a save displays the following prompt. "Effective End Date of the previous revision is not set. Do you want it to be set to current revision Effective Start Date minus one day?" Click **Go Back** in the message box. Set the **Effective End Date** for BOM Revision 1 to February 24, 2020 and then save BOM Revision 2.

### To update a BOM revision date:

1. Go to Lists > Supply Chain > Bill of Materials.
2. Click **View** beside the BOM you want to revise.
3. In the **Revisions** subtab, click **Edit** beside the BOM Revision you want to update.
4. Click the **Calendar** icon beside the **Effective End Date** field and then select a date.
5. Enter a date at least one day before your new BOM Revision **Effective Start Date**.
6. Click **Save**.

## Customizing the BOM Revision Component Sublist

BOM revision component sublist customization lets you define and display component level information specific to your business workflows. You can add custom columns, change column order, update column labels, and hide column fields.

To customize a sublist, go to Customization > List, Records, & Fields > Other Sublist Fields. After you create a custom field, it appears on the BOM Revision Components subtab.

**Note:** The BOM Revision component sublist is not automatically added to the work order.

### To add custom columns to a work order:

1. Go to Customization > Lists, Records, & Fields > Transaction Line Fields > New.
2. Complete the **Transaction Line Field** form:
  - a. Enter a descriptive **Label** name.
  - b. **ID** value is used when scripting to instances of the transaction type.  
Enter a name that begins with custcol and then an underscore. If you don't enter a value, the system generates one. For example, custcoliqity\_vendor\_control\_no.
  - c. Select an **Owner**. Only owners can edit this field.
  - d. Enter a **Description** of this field. For example, capture the vendor's lot/serial number during item receipt.
  - e. Select the **Type** of field you want to create.
  - f. The **Store Value** box is checked by default so that the entered information is stored in the custom field. This enables you to look at data that is stored elsewhere.

Clear this box to not store changes in the custom field. If you don't store the value, changes are discarded. You can make the field read-only.

- g. Optionally, check the Use **Encrypted Format** box to encrypt stored values for this field in the database (values still appear in the UI).

After you save this field, you cannot change this option.

- h. Optionally, check the **Inactive** box. Inactive custom fields don't appear on any forms or in global searches.

3. On the **Applies To** subtab, check the **Work Order** box.
4. On the **Sourcing & Filtering** subtab, in the **Source List**, select **BOM Revision Component**.
5. In the **Source From** list, select the **Bill of Materials Revision** custom field.
6. Click **Save**.

After a BOM revision is selected on a work order, NetSuite automatically retrieves a list of components related to the BOM revision. This ensures that custom field information is included in the work order. For more information, see the help topic [Creating Custom Transaction Line Fields](#).

## Linking Assemblies and BOMs

After you create BOM records and revisions, you can link them to individual assemblies. You can also link BOM records to multiple assemblies. For more information, see [Linking a BOM to an Assembly](#) and [Linking Assemblies to BOM Records](#).

### Linking a BOM to an Assembly

After you create BOM records and revisions, you can link them to individual assemblies. You can also designate a BOM as either a location default or master default for that assembly.

#### To link a BOM to an assembly:

1. Go to Lists > Accounting > Items.
2. Beside the item you want to update, click **Edit**.
3. Click the **Manufacturing** subtab.
4. In the **Current Revision** list, select a BOM to associate with this assembly.  
The displayed **Current Revision** name is the same as the **Bill of Materials** revision name.
5. To designate this as the default BOM for the assembly, check the **Master Default** box.  
Checking this box disables the **Default for Location** option.
6. Alternatively, check the **Default for Location** box to designate a default location. Select a location from the list.  
For example, the West Coast location is the default location BOM. It is also the default assembly location. To see the example illustrated, see [Default BOM Workflow](#).
7. To add more BOMs for this assembly, click **Add** and then repeat steps 4 to 6.  
For example, Wolfe Manufacturing operates from 10 distinct locations. BOM U.S. is designated as the default for 6 locations and BOM CA is the default for the remaining 4 locations.
8. Click **Save**.

 **Note:** NetSuite 2024.2 Manufacturing now displays the Effective Start Date and Effective End Date columns.

## Linking Assemblies to BOM Records

After you create a Bill of Materials (BOM), you can link it to one or more assemblies. All assemblies associated to the BOM appear on the Assemblies subtab.

### To link an assembly to a BOM record:

1. In the BOM record, click the **Assemblies** subtab.
2. In the **Assembly** list, select the assembly you want to link to the BOM.
3. Click **Add**.
4. Click **Save**.

From the BOM record you can assign the same default assembly record validation (master default or location-specific default) to the BOM. A BOM cannot be both master and location default. They must be assigned to separate BOMs within an assembly.

For example, you can set one BOM as the master default for the assembly. Then, set a second BOM as the East Coast location default.

## Creating a Manufacturing Routing

In Advanced BOM, routings are associated with a Bill of Materials. This association enables you to define default routings for multiple BOMs.

For example, The Wolfe Company manufactures mountain bikes its U.S. location. To do this, the U.S. location uses BOM U.S. prompting NetSuite to automatically select the BOM U.S. Routing. Due to increased mountain bike sales, Wolfe starts to manufacture mountain bikes in Canada. To manufacture mountain bikes in Canada, BOM CA is selected and the system automatically uses BOM CA Routing.

 **Important:** You cannot update Bills of Materials for routings that were used before Advanced BOM was enabled.

The Manufacturing Routing page at Lists > Supply Chain > Manufacturing Routing displays the following updates:

- Legacy routings are associated with legacy BOMs.
- Bill of Materials replaces the assembly (item) list.
- The Component per Operation subtab changes to read-only for legacy BOM routings. This subtab displays a line for each component and revision combination.

For example, a bicycle frame component appears in multiple revisions of the Mountain Bike BOM. Each revision in this BOM displays a line each time the frame component appears in Mountain Bike BOM Revisions.

### To create a manufacturing routing:

1. Go to Lists > Supply Chain > Manufacturing Routing > New.

2. If you use NetSuite OneWorld, select a **Subsidiary**.
3. Select a **Bill of Materials** from the list.
4. Select or enter one or more **Locations** where this assembly is performed.
5. Enter a routing template **Name**.  
This name appears in the manufacturing routing field routings list on records and forms. For example, Beta Alternate Supply Routing.
6. Optionally, enter a **Memo** for this routing.  
You can search for the text you enter in this field. For example, enter **Use Beta Alternate when primary widget supply is unavailable**.
7. To use this routing by default for item forms, check the **Default** box.  
This sets the default steps for creating new special work orders and mass created work orders. Clear this box if you don't want this routing to be the default.
8. Check the **Inactive** box if you don't want this routing to appear in routing lists on records and forms.  
Clear this box if you want this routing to appear in lists.
9. If you want NetSuite to calculate lag times for operation tasks, check the **Auto-Calculate Lag** box.  
For more information, see [Operations Overlap](#).
10. Click the **Routing Steps** subtab and then complete the fields, as required.
11. Click **Save**.

## Copying a Manufacturing Routing Record

You can manually copy an existing manufacturing routing record and then reuse it with a new BOM. Reusing existing routings saves time, leverages existing processes, and avoids manual copy errors.

For example, Wolfe Manufacturing builds bicycles in their Denver location using the BOM U.S. Routing. The company then starts manufacturing bicycles at their Ontario location. By copying the BOM U.S. Routing and renaming it to BOM ON Routing, the Ontario location can follow the U.S. manufacturing process.

### To copy a manufacturing routing record:

1. Go to Lists > Supply Chain > Manufacturing Routing.
2. Beside the routing you want to copy, click **View**.
3. From the **Actions** list, click **Make Copy**.
4. Change the manufacturing routing **Name**.  
For example, change BOM U.S. Routing to BOM ON Routing.
5. Make other changes to the routing, as necessary.
6. Click **Save**.

## Running a BOM Inquiry

A Bill of Materials Inquiry enables you to review the build requirements for an assembly item. The BOM displays the assembly components and the number of each component needed for each assembly.

The system checks for an active default or master default BoM for the location and uses that BoM. If a BoM is not default available, the system cannot identify the BoM and so its components won't be displayed.

### To run a Bill of Materials Inquiry:

1. Go to Transactions > Manufacturing > Bill of Materials Inquiry.
2. If you use NetSuite OneWorld, select a **Subsidiary** from the list.
3. Optionally, select an **Assembly** to filter the available bills of materials.  
The assembly supply type needs to be set as Build for particular location. For example, if an assembly is set to purchase, the system doesn't expect to source an assembly through Manufacturing. It will be purchased, so no components are expected to be consumed and cost for the assembly will be determined from Purchase.
4. Optionally, select a **Location** to display data for that location.
5. To limit the details to the top level of member items, check the **Top Level Only** box.  
Sub-assembly information doesn't appear.  
To show details about all levels of member items, clear the **Top Level Only** box.
6. Select a **BOM Display Control** option:
  - **By Date** – the Bill of Materials Revision field is disabled.  
If the date changes, the Bill of Materials revision field displays a new BOM revision.
  - **By Revision** – the Bill of Materials revision field is enabled.
7. Select the **Bill of Materials** to run this inquiry against.  
This selection can automatically populate the Bill of Materials Revision and Effective Date fields.
8. To change the default revision, select a **Bill of Materials Revision**.
9. To change the default date, click the **Effective Date** calendar icon and then select a date from the calendar.

## Costed Bill of Materials Inquiry

The Costed BOM Inquiry report details individual and aggregated assembly costs. It shows how cost components are rolled up to an assembly item based on the BOM. This report includes material and conversion costs (labor and machine costs and their overhead).

 **Note:** Standard costing and assemblies must be enabled to use the Costed BOM Inquiry. You can run this inquiry for a standard cost assembly item only.

### To run a Costed Bill of Materials Inquiry

1. Go to Transactions > Manufacturing > Costed Bill of Materials Inquiry.
2. If you use NetSuite OneWorld, select a **Subsidiary** from the list.
3. Optionally, select an **Assembly** to filter the available Bills of Materials.
4. Select a **Location** to display data for that location.
5. To limit the details to the top level of member items, check the **Top Level Only** box.  
Sub-assembly information doesn't appear.  
To show details about all levels of member items, clear the **Top Level Only** box.

6. Select a **BOM Display Control** option:
  - **By Date** – the Bill of Materials Revision field is disabled.  
If the date changes, the Bill of Materials Revision field displays a new BOM.
  - **By Revision** – enables the bill of materials revision field.
7. Select the **Bill of Materials** you want to run the from the list.  
This selection can automatically populate the **Bill of Materials Revision** and **Effective Date** fields.
8. To change the default revision, select a **Bill of Materials Revision**.
9. To change the default date, click the **Effective Date** calendar icon and then select a date from the calendar.

## Matrix Items

Matrix items consist of multiple combinations of product styles and variations with their respective bill of materials and routings. During production, matrix assemblies can be created using work orders and assembly builds.

If you use the Advanced BOM feature, the Bill of Materials subtab appears on the Manufacturing subtab:

- The Bill of Materials subtab moves from the matrix assistant to the Manufacturing subtab.
- Both parent and child items display a bill of materials sublist.
- BOMs can be copied from parent to child item records.
- The BOM sublist can be updated on child items.
- You can assign a BOM or multiple BOMs to an assembly matrix.
- The Components subtab no longer appears on the Matrix Assistant.

 **Note:** You can add BOMs only from the assembly matrix parent or subitem.

### To assign a BOM to an assembly matrix:

1. Go to Lists > Accounting > Items.
2. Beside the item you want to update, click **View**.
3. In the **Assembly/Bill of Materials** window, click **Edit**.
4. Click the **Manufacturing** subtab.
5. Select a **Bill of Materials** from the list.
6. To assign a BOM as a default for the assembly, check the **Master Default** box.
7. Alternatively, you can check the **Default for Location** box. Select the location from the list.  
For example, if the West Coast location is the default location BOM. It is used as the default for the assembly and the West Coast location. To see an illustration of this example, see [Default BOM Workflow](#).

 **Note:** A BOM cannot be both a master and a location default. Only one default designation is allowed for a BOM.

8. To apply the BOM subtab settings to child matrix items, click **Update Matrix**.
9. In the **Update Matrix Item** window, check the child items to **Include** in this update.

10. To copy the parent BOM to all child BOMs (mass update), check the **Update BOMs of Matrix Sub-items** box.
11. Click **Submit**.

# Bill of Materials Member Control for Assembly Items



**Important:** With the release of NetSuite 2023.1 Bills of Materials, where components are embedded to an Assembly Item, will no longer be supported. Only business critical issues will be fixed. To continue working with this functionality you should transition to the [Advanced Bill of Materials](#) feature, which is free of charge. After you enable this feature, NetSuite will automatically migrate all of your Bill of Materials to the new structure.

When you use the Assembly Items feature, the components needed for assemblies are identified in the Bill of Materials (BOM). Components required for an assembly can change due to engineering changes, vendor supply, availability, or seasonal requirements.

BOM member control helps you ensure that the right components are included in assembly builds at the right time. You can use BOM controls to plan for the utilization and purchase of components that are effective or obsolete within specific time frames.

To use BOM component member control, define effective and obsolete dates for member items on assembly records. NetSuite determines whether a component is valid for an assembly based on these date.

- **Effective Date/Revision** – Defines the first date an item can be used for an assembly. Before the effective date, the item is not included in the BOM.
- **Obsolete Date/Revision** – Defines the last date an item can be used for an assembly. After the obsolete date, the item is not included in the BOM.

After a work order is created, NetSuite determines which components are required based on the transaction date. If you use the Demand Planning feature, NetSuite considers the production start date and determines which components are required on that date.

On assembly item records, choose a BOM control method and set up effective and obsolete dates. NetSuite uses them to determine which member items are needed to create an assembly based on the date the item is produced.

For example, Wolfe Manufacturing assembles bicycles for distribution throughout the year. The mountain bike component item includes the following:

- Disk brake 1: has an effective date of 4/1/2020
- Disk brake 2: has obsolete date of 3/31/2020

Wolfe creates a work order dated 3/31/2020 that includes a Mountain Bike assembly. NetSuite examines the effective and obsolete dates for the components on the assembly record and determines the following:

- The BOM won't include Brake 1 because it is not effective.
- The BOM will include Brake 2 because it is not obsolete.

Wolfe enters a work order dated 4/1/2020 that includes a Mountain Bike assembly, and NetSuite determines the following:

- The BOM will include Brake 1 because it is currently effective.
- The BOM won't include Brake 2 because it is obsolete.



The appropriate assembly BOM items are shown on work orders at the appropriate dates without having to manually change each work order.

For BOM management to track which components are needed at specific times, identify effective and obsolete dates for member items. To do this, select a BOM control method on assembly item records. Select to set dates individually for components or to create revision records to assign to items. When a new work order is created, NetSuite can determine the member items required based on the work order production date. For more information, see [Setting Up BOM Control on Assembly Item Records](#).

You can choose to set an assembly to use revision control. For more information, see [Setting an Assembly to Use Revision Control](#).

If you choose to manage assembly BOMs with revision records, set up revision records. For more information, see [Creating Revision Records for BOM Control](#).

## Setting Up BOM Control on Assembly Item Records



**Important:** With the release of NetSuite 2023.1 Bills of Materials, where components are embedded to an Assembly Item, will no longer be supported. Only business critical issues will be fixed. To continue working with this functionality you should transition to the [Advanced Bill of Materials](#) feature, which is free of charge. After you enable this feature, NetSuite will automatically migrate all of your Bill of Materials to the new structure.

To use Bill of Materials (BOM) component member control, define effective and obsolete dates for assembly record member items. You can enter the effective and obsolete dates individually on assembly records. You can also create revision records that update many assembly records at one time. The following methods enable you to manage which items are used in assemblies based on production dates. Select one of the following methods:

- [Using Effective Date BOM Management](#) – Control the BOM based on dates entered on the assembly record's Components subtab.
- [Revision Control BOM Management](#) – Control the BOM by selecting a version with preset dates that determine components.

## Using Effective Date BOM Management

Use Effective Date BOM management to set the effective and obsolete dates for items on individual assembly records.

### To set up Effective Date BOM Control on assembly records:

1. Go to Lists > Accounting > Items.
2. Click **Edit** next to the existing assembly item record.  
You can also click **New** to enter a new assembly record.
3. On the **Purchasing/Inventory** subtab in the **Effective BOM Control** field, select **Effective Date**.
4. Click the **Components** subtab.
5. Click an existing member item or add a new one.
6. In the **Effective Date** field, enter the date when an item can be included as a member for an assembly.
  - After the Effective Date, the item is included in the BOM.
  - Before the Effective date, the member is not included in the BOM.

 **Note:** A blank effective date indicates the item has always been included.

7. In the **Obsolete Date** field, enter the date when an item can be included as a member for an assembly.
  - After the Obsolete Date, the item is not included in the BOM.
  - Before the Obsolete Date, the member is included in the BOM.

 **Note:** A blank obsolete date indicates that the item will always be included.

8. Click **Add** or **Done**.
9. Repeat steps 3 to 8 to set effective or obsolete dates for additional members of this assembly
10. Click **Save**.

NetSuite can source this item record to determine the correct BOM based on the assembly production date.

## Revision Control BOM Management

To manage the Bill of Materials (BOM) for assembly items, use Revision Control as your Effective BOM Control method. This method simplifies effective and obsolete date management. To use revision control, create revision records that define an effective date or an obsolete date. The revision records you assign to assembly members determine their effective and obsolete dates.

The Revision Control method enables you to set the effective or obsolete date for many items at one time by updating one revision record. When several items use a revision record, date changes can be made on the revision record rather than individually for many member items. Rather than changing the dates on every line item, you change only the effective or obsolete date in the revision record.

For assembly items that use Revision Control, create revision records to define effective and obsolete dates. The assembly effective and obsolete dates are determined by the assigned revision record.

Revision records can be created in two ways:

- As individual records. For more information, see [Creating Revision Records for BOM Control](#).
- From assembly record Members subtab. For more information, see [Setting an Assembly to Use Revision Control](#).

After a revision is assigned to an assembly member, you can enter a new work order and select the assembly. The correct revision defaults to based on the work order production date. The work order item list displays the correct set of member components based on the revision used.

- If you select a new revision, the item list updates to show the correct member items.
- If you change the date, NetSuite updates the revision to the one which is effective for that date.

To use only the assembly default revision, on the work order form, click Customize and then make the field not selectable

- Edited revision record effective or obsolete dates are not retroactive. Previously entered transaction data using that revision remain unchanged.
- BOMs created for individual assembly builds compare the transaction date to the effective and obsolete dates.
- On an assembly unbuild, select a revision to determine the BOM. The default revision for an unbuild is based on the current date.

- BOM costs using Standard Costing are based on the effective date shown on the planned standard cost rollup record.

**Note:** If you use the Matrix Items feature, you cannot set the Effective BOM Control to Revision Control on a matrix parent item. However, you can set the Effective BOM Control to Revision Control on a matrix subitem.

## On Work Orders

NetSuite automatically populates the work order effective revision based on the effective date. If you change the work order, NetSuite changes the components on the top level assembly based on the revision selected.

If you change the work order revision and the Build Subassembly box is checked, top-level components change based on the selected revision. The lower level components are determined based on the effective date.

## Demand Planning

When demand increases for member items from a parent assembly, NetSuite reviews the work order start date to determine demand for those member items. For example, the Mountain Bike component items include the following:

- Brake Item 1: has an effective date of 4/1/2020
- Brake Item 2: has obsolete date of 3/31/2020

Demand for the item requires a work order to be created on 3/20 and one on 4/20. Therefore, the first work order uses the Member Item 2, and the second one will use Member Item 1. This is relevant if the member items are assemblies that need to have work orders created for them.

## Setting an Assembly to Use Revision Control

**Important:** With the release of NetSuite 2023.1 Bills of Materials, where components are embedded to an Assembly Item, will no longer be supported. Only business critical issues will be fixed. To continue working with this functionality you should transition to the [Advanced Bill of Materials](#) feature, which is free of charge. After you enable this feature, NetSuite will automatically migrate all of your Bill of Materials to the new structure.

To use revision records to manage effective and obsolete dates, set the assembly record to use revision control.

### To set an assembly to use revision control:

1. Go to Lists > Accounting > Items.
2. Click **Edit** next to the assembly item record you want to set.
3. On the **Purchasing/Inventory** subtab, in the **Effective BOM Control** list, select **Revision Control**.

**Note:** If you select Revision Control, you must use revision control for this item.

4. The **Default Revision** field displays the default revision for this item.

5. If you are creating a new item record, click **Save** and then click **Edit** to re-open the item.  
If you are editing an item already set to revision control, go to the next step.
6. Click the **Components** subtab.
7. Select an existing member **Item** or add a new one.
8. In the **Effective Revision** field, enter a revision or create a new one. The revision record effective date determines the start time when this item is included as a member for an assembly.  
For each member, the default selection is **Default**. The item will then be included in builds by default.
  - a. To define a non-default date revision record, in the **Effective Revision** field, select **New**
  - b. In the **New Item Revision** popup window, define the following for the assembly item:
    - **Name** (for example, Version 2)
    - **Effective Date** (for example, 4/1/2020)
    - **Memo** – Optionally enter a memo.
    - **Inactive** – Check this box to not display this revision in lists.

Alternatively, enter new revision records at Lists > Accounting > Item Revisions. For more information about revision records, see [Creating Revision Records for BOM Control](#).

For more information about using effective fields, see [Setting Up BOM Control on Assembly Item Records](#).
9. Select an **Obsolete Revision**. The obsolete revision record date determines the end time that an item is to be used for an assembly.  
When you select an obsolete revision, the correct obsolete date displays in that field.
10. Click **Done** or **Add**.
11. Repeat the steps 1 to 4 for each member item you want to assign a revision to.
12. Click **Save**.

The BOM for this assembly is determined by referencing the production date of each work order against the revision record dates for member items.

## Creating Revision Records for BOM Control



**Important:** With the release of NetSuite 2023.1 Bills of Materials, where components are embedded to an Assembly Item, will no longer be supported. Only business critical issues will be fixed. To continue working with this functionality you should transition to the [Advanced Bill of Materials](#) feature, which is free of charge. After you enable this feature, NetSuite will automatically migrate all of your Bill of Materials to the new structure.

When using the Revision Control method for Bill of Materials (BOM) management, create revision records to assign to assembly items. These revision records define effective and obsolete dates and can be assigned to many items.

Update effective and obsolete dates on the revision record one time to change the dates for many items.

### To create a revision record:

1. Go to Lists > Accounting > Item Revisions > New.
2. Select an **Item** to be associated with this revision.

Required fields display a red asterisk (\*).

3. Enter a revision **Name**. For example, Version One.
4. Enter the revision **Effective Date**.
5. Enter a **Memo** about this revision. You can later search for this version by memo text, if needed.
6. Check the **Inactive** box to not display this revision in lists.
7. Click **Save**.

## Obsolete Dates

Obsolete dates on a revision record cannot be edited. You can set the effective date and then save the revision. NetSuite then determines the obsolete date to avoid gaps or overlaps in dates covered by revisions.

The obsolete date field does display a value if you try to insert a revision record between two others.

For example, you have a default revision and one with an obsolete date of 1/1/2017. The default revision has an obsolete date of 12/31/2017. Next, you create a revision with an effective date of 1/1/2017. In this case, the obsolete date is set to 12/31/2017. After you save, NetSuite changes the obsolete date of the Default revision to 12/31/2016. If you create another revision with an effective date of 1/1/2018, it has no obsolete date. This is because it is the one with the latest effective date. When you save it, it still has no obsolete date and the 2017 revision has its obsolete date set to 12/31/2017.



**Note:** You can use the Import Assistant to add or update item revision records based on CSV file data. For more information, see the help topic [Item Revision Import](#).

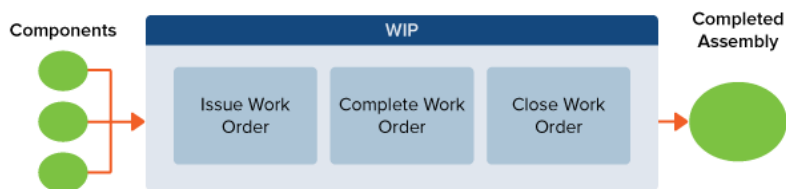
# Manufacturing Work In Process (WIP)

Manufacturers often use work orders to assemble the materials needed to produce an item. Manufacturing Work In Process lets manufacturers track work orders through the production process, from gathering materials, through shop floor assembly, to stocking finished goods.

Assembly items built using Manufacturing Work In Process (WIP) are divided into the following processes:

1. **Issue Work Order:** Move raw materials to an assembly area (work center)
2. **Complete Work Order:** Assemble raw materials and stock finished goods
3. **Close Work Order:** Reconcile variances

This separation lets you enter work order transactions that define the completion of each step, from material consumption to assembly and completion.



To know where materials are in the manufacturing process, use WIP to track your materials and record the number of:

- Component materials in stock
- Component materials consumed against a work order
- Finished assemblies in stock
- Assemblies you can build

Use WIP to track assembly component to identify the materials not used in the process.

Tracking goods and materials enables you to control your inventory and raw materials. Controlling stock can help you avoid the following:

- Tying up funds in stock or on the production floor
- Unanticipated raw material stock-outs

NetSuite supports production processes with a long lead time and can track item assemblies as a work in process.

To learn when materials are issued or removed from inventory, see [Entering Work Order Issues](#).

To use WIP, enable the feature and set up item records, see [Enabling the WIP Feature](#).

**Note:** On transactions using WIP, you must identify a WIP location for line items. The WIP line location must match on all work order issue, work order completion, and work order close transactions.

## Enabling the WIP Feature

Before you can work with Manufacturing Work in Process (WIP), you must also enable Assemblies and Work Orders. For more information, see [Enabling Assembly Items](#) and [Assembly Work Orders](#).

### To enable the feature:

1. Go to Setup > Company > Enable Features (Administrator).
2. Click the **Items & Inventory** subtab.
3. Check the **Manufacturing Work In Process** box.
4. Click **Save**.

After the feature is enabled, you can set up items for WIP processing. For more information, see [Setting Up Items as WIP Assemblies](#).

## Setting Default WIP Accounts

NetSuite WIP enables you to set default preferences for new item records. To set preferences, , and then

### To set default WIP accounts:

1. Go to Setup > Accounting > Accounting Preferences.
2. Click the **Items & Transactions** subtab.
3. Set the following accounts:
  - **Default WIP Cost Variance Account** – The selected account appears by default in the **WIP Cost Variance Account** field on item records. This is an expense account for actual or average cost assemblies when the reconciliation amount cannot be returned to the asset account. This happens when the amount has been shipped. This account is required if WIP is checked for any location.
  - **Default Scrap Account** – The selected account appears by default in the **Scrap Account** field on item records. This is an expense account for scrapping that occurs during the work order completion. This account is required if WIP is checked for any location.
  - **Default WIP Account** – The selected account appears by default in the **WIP Account** field on item records. This is an asset account used when a work order component issue is entered. This account is required if WIP is checked for any location.
4. Click **Save**.

## Setting Up Items as WIP Assemblies

After your administrator enables the Manufacturing Work In Process (WIP) feature, you can set up assembly items to use WIP.

### To set up an assembly item to use WIP:

1. Go to Lists > Accounting > Items.
2. Click **Edit** next to the existing assembly item record.  
Alternatively, click **New** to enter a new assembly record.  
Only items that use standard or average costing can be used on a WIP work order.
3. On the **Basic** subtab, make a selection for the following accounts:
  - a. **WIP Cost Variance Account** – The expense account for actual cost or average cost assemblies. This is used when the reconciliation amount cannot be returned to the asset

account because the amount has been shipped. This account is required when WIP is checked for any location.

- b. **Scrap Account** – The expense account for scrap that occurs during work order completion. This account is required if WIP is checked for any location.
- c. **WIP Account** – The asset account used when a work order component issue is entered. This account is required if WIP is checked for any location.

 **Note:** If you change the selected WIP account, the new WIP account affects only future transactions. Existing transactions continue to show the WIP account when the transaction was created.

- 4. To create a new assembly, complete assembly item form fields. For more information, see the help topic [Creating Item Records](#).
- 5. Click **Save**.

When the assembly is added to a work order, you can use WIP to process the assembly.

Work orders that are only partially completed don't account for unused WIP components. They are moved into the cost of the finish goods account, even if they haven't been used.

## Using WIP on Work Orders

The following procedure provides instructions for using WIP on work orders.

### To use WIP on a work order:

- 1. Go to Transactions > Manufacturing > Enter Work Orders.
- 2. Complete the fields, as required.
- 3. To use WIP **rather than a standard assembly build** on this work order, check the **WIP** box.

 **Note:** You cannot clear the WIP box after a posting assembly transaction has been attributed to this work order.

- 4. Select a **Location**.  
This field is required for WIP work orders.
- 5. Complete the necessary form fields. For more information, see [Entering an Individual Work Order](#).
- 6. Click **Save**.

After you designate a work order as WIP, complete the build process using the following:

- **Work Order Issue** – Issue components to track material consumption or log service against a work order.  
For more information, see [Entering Work Order Issues](#).
- **Work Order Completion** – Identify the quantity completed and stocked.  
For more information, see [Entering Work Order Completions](#).
- **Work Order Close** – Generate reconciliation entries to post variances that may occur during the manufacturing process.  
For more information, see [Entering Work Order Closes](#).



The following table shows an example of the general ledger impact for WIP transactions:

|                    |                 |          |              |
|--------------------|-----------------|----------|--------------|
| <b>Consumption</b> |                 |          |              |
|                    | Component Asset | Location | CR- (credit) |
|                    | Component Asset | WIP      | DR+ (debit)  |
| <b>Completion</b>  |                 |          |              |
|                    | Assembly Asset  | Location | DR+          |
|                    | Assembly Asset  | WIP      | CR-          |
| <b>Close</b>       |                 |          |              |
|                    | Assembly Asset  | WIP      | DR+          |
|                    | Component Asset | WIP      | CR+          |
|                    | Variances       | WIP      | DR+          |

## Entering Work Order Issues

When you use the Manufacturing Work In Process (WIP) feature, you must issue components to start work order processing. When you issue components, you are logging only the material consumption, not any work on the assembly process.

The work order issue indicates the material is consumed against the work. Issue assembly member components to track material consumption or log service against a work order.

After components are issued, the material value is recorded in the assigned WIP account. You can issue components for many work orders at one time.

If you use the Multiple Units of Measure feature, the issued component quantity is recorded in base units.

### To bulk enter work order issues:

1. Go to Transactions > Manufacturing > Issue Components.
2. If you use NetSuite OneWorld, select a **Subsidiary** from the list.
3. Optionally select the following to filter the list of work orders:
  - **Item**
  - **Customer**
4. Select the **Posting Period**.
5. Select the **Date** for this issue transaction.
6. Select a **Location** from the list.  
Location is displayed in the location field in the header of all work order issues.
7. Select a **Filter By** option:
  - **Some Items Committed** – Displays the committed quantity
  - **All Items Fully Committed** – Displays the committed quantity
  - **Ignore Commitment** – Displays the remaining quantity
8. Check the **Issue** box next to all work orders you want to create issues for.

9. Click **Submit**.

After the page is submitted, the work order issues are generated for all marked orders.

After you issue components for an order, you can enter a work order completion against it to log the finished assemblies. For more information, see [Entering Work Order Completions](#).

## Issuing Components for an Individual Work Order

You can issue components for a single work order.

### To issue components for an individual work order:

1. Go to Transactions > Manufacturing > Enter Work Orders > List.
2. Click **View** next to the work order you want to issue components for.
3. Click the **Issue Components** button.
4. In the **Reference #** field, you can enter a reference number to track this transaction.
5. In the **Assembly** field, select the assembly item you want to complete. A completion can be entered only for assembly items on record.  
To create new assembly item records, click Assembly at Lists > Accounting > Items > New .
6. Select the **Revision** to use for this completion. For more information, see [Revision Control BOM Management](#).  
The **Manufacturing Routing** field displays the associated routing.  
The **Created From** field identifies the work order that this issue is created from. The work order is linked to the issue record.
7. Select an issue **Date**.
8. Select the transaction **Posting Period**.
9. Optionally, enter a **Memo** for this transaction.  
You can search for text entered here to find the transaction later.
10. Select a **Department** or **Class**, if you use them.
11. Select a **Location**.
12. Select a **Starting Operation** and **Ending Operation** to define a range.  
The default component quantity shows for components to be issued within the operation range.
13. For each component, verify or enter the quantity being issued. The remaining committed component quantity shows by default.
14. Click **Save**.

## Entering Work Order Completions

To use the Manufacturing WIP, enter a completion transaction to log the number of assemblies you completed and stocked for a work order. This transaction indicates the number of finished goods produced.

Entering a completion doesn't record the consumption of materials. It only records the work done on the assembly process and journals the value of the assembly out of the assigned WIP account.

You can enter a work order completion with backflush to issue components and complete the assembly at one time.

**Note:** A Work order completion with Backflush is created for backflush orders. As of NetSuite 2024.1 work order completions and work order issues are created separately. Both transaction records are updated in the Manufacturing Mobile production record.

### To enter a work order completion:

1. Go to Transactions > Manufacturing > Enter Completions.
2. Select an assembly **Item** to filter the list and show only work orders for the item.
3. Select the **Customer** or vendor providing this item.  
Alternatively, select an employee to associate with this transaction.
4. Select the **Posting Period** to post this transaction to.
5. Accept today's **Date** as the transaction date, or select another date.
6. Select a **Location** to use to build orders, issue components, or enter completions.  
This location appears in the location field in the header of all work order completions created.
7. To enter completions with backflush, check the **Backflush** box.  
Backflush records component consumption at the same time. Component consumption is based on the proportion designated on the original work order and the build quantity.  
Clear this box to enter the completion without backflush and not record consumption.
8. Enter a **Sales Order Number** or scan a transaction bar code.
9. Check the box in the **Complete** column for each order to be completed.
10. Verify or enter the quantity completed for each order.  
The field shows the quantity remaining on the order by default.
11. Click **Submit**.  
After a completion is entered against a work order, that quantity of assemblies is recorded as being built and stocked in inventory.

**Note:** The work order completion **Projected Value** field displays the cost of the assembly. It doesn't display individual components (unless you use backflush). A Work Order Close for the work order shows the difference between the cost of the assembly and the components.

## Entering Work Order Closes

Use Manufacturing WIP to enter a work order close to finalize the accounting for that order. The work order close reconciles the accounting by reviewing all issues and completions associated with that work order.

### To bulk enter work order closes:

1. Go to Transactions > Manufacturing > Close Work Orders.
2. Select the **Posting Period**.
3. Select the transaction **Date**.
4. Select a **Location**.
5. Optionally, select the following to filter the list of work orders:

- **Item**
- **Customer**

6. To include work orders with In Process or Built status, check the **Include In Process Work Orders** box.
7. This list includes work orders in Released and Planned status.

To show only work orders in Built and In Progress status:

- a. Go to Transactions > Manufacturing > Close Work Orders
  - b. In the **Close Work Orders** page, click **Customize**.
  - c. In the **Customize Sublist** page, click the **Additional Filters** subtab.
  - d. Beside the **Status** field, check the **Include** checkbox.
  - e. Click **Save**.
8. In the **Under-Produced Variance Tolerance (%)** field, enter a percentage to close only orders that produced less than planned.

This helps determine whether some orders should not be closed because enough finished products haven't been produced.

**Order Variance Calculation:** quantity ordered - quantity built/quantity ordered = variance percentage

A percentage in this field filters the list to show only orders that have a variance lower than the percentage entered.

For example, if order quantity is 10 and built quantity is 5, then the variance percentage is  $(10-5)/10 = 50\%$ .

If you enter a tolerance of 60%, this order shows in the list for closing.

If you enter a tolerance of 10%, this order doesn't show in the list.

9. In the **Production Variance Tolerance (%)** field, enter a percentage to filter out orders that have a specific value variance.

The order variance calculation is: absolute value (remaining WIP value / WIP of assembly) = variance percentage

A percentage in this field filters the list to show only orders that have a variance lower than the percentage entered.

For example, the WIP account value is \$10, and the cost of building the assembly is \$20. Therefore, the variance percentage is  $(10/20) = 50\%$ .

If you enter 10% in this field, this order doesn't show in the list for closing.

If you enter 60% in this field, this order shows in the list.

10. You can filter the list order by selecting a range of dates.

If you use demand planning, optionally filter the list by selecting a production start date and end date.

1. To filter by an order date range, enter a beginning date in the **Order Date From** field. Enter an end date in the **Order Date To** field.
  2. To filter by a production start date range, enter a beginning date in the **Production Start Date From** field. Enter an end date in the **Production Start Date To** field.
  3. To filter by a production end date range, enter a beginning date in the **Production End Date From** field. Enter an end date in the **Production End Date To** field.
11. Next to all orders you want to close, check the **Close** box.
  12. Click **Submit**.



**Important:** Cost discrepancies may occur because of currency decimal precision and rounding off values. To lessen the probability of substantial variances, reduce the decimal precision used for the cost of components to 2.

## Entering a Close for an Individual Work Order

You can also enter a close for a single work order.

### To enter an individual work order completion:

1. Go to Transactions > Manufacturing > Enter Work Orders > List.
2. Beside the work order you want to close, click **View**.
3. Click the **Close** button.
4. Complete additional fields on the Work order Close page as necessary.
5. Click **Save**.

## Associating Components with Operations

Use the Manufacturing Work In Process (WIP) feature to assign and issue components at the operation level for items you assemble.

Associating components with specific operations can benefit operations that take a long time to complete. Some components may not be required when the work order starts. Therefore, it isn't ideal to issue all assembly components during the first operation. If you associate components with specific operations for a routing, components are issued on the day the corresponding operation begins. For example, a work order is set up as follows:

- The routing includes 4 operations
- Each operation takes 1 week
- Operation 4 requires a special component

|                           |               |
|---------------------------|---------------|
| Work Order - Start Date:  | July 1, 2020  |
| Work Order - End Date:    | July 28, 2020 |
| Operation 1 - Start Date: | July 1, 2020  |
| Operation 2 - Start Date: | July 7, 2020  |
| Operation 3 - Start Date: | July 14, 2020 |
| Operation 4 - Start Date: | July 21, 2020 |

- If the special component is linked to Operation 1, the component issue date is July 1, 2020.
- If the special component is linked to Operation 4, the component issue date is July 21, 2020.

### To associate a component with an operation:

1. Go to Lists > Accounting > Items.
2. Click **Edit** next to the WIP assembly item you want to work with.
3. Click the **Manufacturing** subtab.
4. Click the **Components Per Operation** subtab.

The list displays all components and their corresponding quantities, as defined on the component list of the item record.

 **Note:** To display the Components Per Operation subtab, create a new or edit an existing Manufacturing Routing.

5. Click a **Component Line**.
6. In the **Operations** field, select an operation to associate the component with.
7. Click **Save**.

After you save your work, the operation associations you set up are applied only to new work orders you create.

Associations can also be defined for existing routings on assembly item records. Based on the specified routing, NetSuite updates the associations to components on work orders.

- One operation can be associated with multiple components.
- Each component line can be associated with only one operation.
- One component can be associated with multiple operations if the component is defined on multiple lines.
- Not all components are required to be associated with an operation. If no operation is defined for a component, that component is issued at the start date for the routing.

 **Warning:** Changing or deleting an operation number breaks all associations to the original operation number. For example, any component associated with the previous operation number won't be associated with the new operation number.

Changing or deleting a component also breaks the association to an operation.

## Operation Associations and Sub-Assemblies

If you check the Mark Sub-Assemblies Phantom box, the association between the sub-assembly and the operation is broken. The sub-assembly is replaced with its components on the list of available components. The same operation association is applied to the sub-assembly components. You can create new associations for these sub-assembly components. However, you cannot edit the list of components.

## Entering a Completion for an Individual Work Order

You can enter a completion for a single work order.

### To enter individual work order completions:

1. Go to Transactions > Manufacturing > Enter Work Orders > List.
2. Click **View** next to the work order you want to complete.
3. Do one of the following:
  - To create a work order completion without backflush, click the **Enter Completion** button.

The completion records the number of assemblies completed against a work order. It doesn't record component consumption.

- To create a work order completion with backflush, click the **Enter Completion with Backflush** button.

This records completion and component consumption.



**Important:** If you use Manufacturing Routing and Work Centers, when you enter a completion with backflush, you must complete additional steps. See [Entering a Completion With Backflush](#).

4. On the **Work Order Completion** page, enter a **Reference #** (number).
5. To enter the completions with backflush, check the **Backflush** box.  
Backflush enables component consumption to be recorded at the same time. Component consumption is based on the proportion designated on the original work order and the build quantity.
6. Select the issue transaction **Date**.
7. Select the **Posting Period**.
8. Optionally, enter a **Memo**.
9. After you enter a quantity to build, click **Inventory Detail** to specify a bin, serial or lot number for the items being processed.  
If you use Multiple Units of Measure, the Units field on the Inventory Detail record defaults to base units and cannot be changed.
10. Select a **Department** or **Class** if you track them.
11. Select a **Location**.
12. If you are entering a completion with backflush, verify or enter the **Completed Quantity** of each component consumed.
13. Click **Save**.

## Entering a Completion With Backflush

If you use Manufacturing Routing and Work Centers, when you enter a completion with backflush, complete the following additional steps.

### To enter a completion with backflush:

1. Go to Transactions > Manufacturing > Enter Work Orders > List.
2. Access the work order that requires additional steps.
3. In the **Operation Completion** section, select the completed operation tasks.
  - To mark a single operation task complete, select the task in both the **Starting Operation** and **Ending Operation** lists.
  - To mark a range of operation tasks complete, do the following:
    1. To identify the first operation task complete, enter the **Starting Operation**.
    2. To identify the last operation task complete, enter the **Ending Operation**.

Operation tasks logged as completed include the starting operation, the ending operation, and all operation tasks in between the starting and ending operation tasks.
4. Enter the **Completed Quantity**.  
After you enter the quantity, the **Components** subtab and **Operations** subtab are automatically populated based on requirements.

For more information, see [Routing Completion Labor and Machine Time Entry](#).

5. Optionally, modify the default quantity issued for components.

When entering a routing completion with backflush, if the starting operation is the first operation, the quantity issued defaults based on the completed quantity. For example, there are 5 components in each unit. The completed quantity is 60 units, therefore, the form displays 300 of the component item issued.



**Note:** If the Allow Overage on Work Order Transactions preference is enabled, then you can complete/issue a quantity larger than planned.

6. If the ending operation is the last operation, you can enter a scrap quantity.

The quantity of assemblies scrapped, posts a value to the scrap account indicated on the item record.

7. Verify the **Quantity to Build**:

When the ending operation is the last operation, the **Quantity to Build** field is populated with the completed quantity. The quantity is marked completed and moved to inventory.

You cannot modify the quantity in this field. It updates dynamically based on the starting and ending operation.

8. If this is a lot numbered or serial numbered assembly, click **Inventory Detail** to enter lot or serial numbers.

If you use Multiple Units of Measure, the Units field on the Inventory Detail record defaults to base units and cannot be changed.

9. Click **Save**.

## WIP and Inventory Costing

Only assemblies using standard cost and average cost are compatible with the Manufacturing Work In Process (WIP) feature. Assembly component members can use LIFO/FIFO, standard, average, or actual costing. Assemblies using LIFO/FIFO/specific costing aren't compatible with this feature.

Standard cost assemblies have costs calculated based on the WIP account indicated for the order, not by date.

## Standard Cost Assembly Revaluations

The standard cost of raw materials can change when a work order is still in process causing a change in the WIP valuation.

For example:

- A bicycle wheel is made of the following component parts: rims, tires, and spokes.
  - Rims cost \$5.
  - Tires cost \$6.
  - Spokes cost \$7.
  - The total bicycle wheel cost is \$18.
- When a work order is still in process, the standard cost of components changes to the following:
  - Rims cost \$4.



- Tires cost \$5.
  - Spokes cost \$6.
  - The total bicycle wheel cost is \$15.
- The change in the bicycle wheel cost from \$18 to \$15 requires the following WIP cost adjustment:
  - (-1) for each component issued
  - (-3) for each assembly taken out of the WIP location

When the standard cost for an inventory or assembly item in a location changes, note the following. NetSuite creates a separate standard cost revaluation transaction for each partially built work order. NetSuite creates standard cost revaluation transactions for each work order that doesn't have a status of Released or Built.

## Posting GL Lines for a WIP Assembly Completion with Backflush

When you enter an assembly completion with backflush, you log the assembly completion and component consumption at the same time. This transaction posts to the general ledger as follows:

### Cost of the Assembly

- For an average cost item, an estimate of the last purchase price is used for the assembly posting.
- For a standard cost item, the standard cost is posted.

### Cost of a Component

- For an actual cost item or average cost item, the average or actual cost from the warehouse location is posted.
- For a standard cost item, the standard cost for component is posted.



**Note:** If you use WIP, an average cost assembly item generates a variance based on the assembly cost and the component cost. If an adjustment needs to be made on the average cost based on the variance, you should use an inventory adjustment worksheet.

# Manufacturing Routing

Manufacturing Routing and Work Center lets you schedule and record manufacturing operational activities against a work order that requires multiple employees, or work centers. For example, you may have a set of operations for the following: a preparation team, an assembly run team, and a quality assurance team.

You can use the Manufacturing Routing and Work Center feature to record quantity assembly completions and team resource costs. You can also use it to process overhead costs against individual work order operations.

The following roles represent members of your organization who can benefit from using Manufacturing Routing.

## Operational Planner or Production Manager

Your Operational Planner or Production Manager can benefit from this improved planning efficiency:

- Set up a routing record that defines multiple steps for building a complex assembly.
- Assign default scheduling parameters against each step.
- Use backward scheduling to establish a supply plan based on manufacturing scheduling requirements.

## Production Manager

Your Production Manager can refine the shop floor tracking activities using the following:

- Facilitate scheduling by assigning work center groups to operation steps.
- Record progress of activities such as completion and component issue against multiple tasks or one task at a time
- Record actual machine and labor times against anticipated times.

## Cost Accountant

Your Cost Accountant can identify opportunity areas:

- Assign labor and machine overheads against completion activities.
- Develop a costing template for standard rates used in multiple routings.
- Track cost variances between actual and standard at a per service item and cost category level.



**Important:** Be aware of the following:

- Manufacturing Routing can be used only with assembly items using standard costing or average costing.
- Manufacturing Routing can be used only with work orders that are marked as Work In Process (WIP).
- Manufacturing Routing creates variances based on per-service item and cost categories.

To use the Manufacturing Routing and Work Center feature, a user with sufficient permission must do the following:

- enable the feature
- define cost categories
- define charge items
- create cost templates
- define resources
- create routings
- set up standard costing, if required

The following tasks must be completed:

- Complete the [Setting Up Manufacturing Routing](#) procedures.

This lets you use routings on WIP designated work orders.

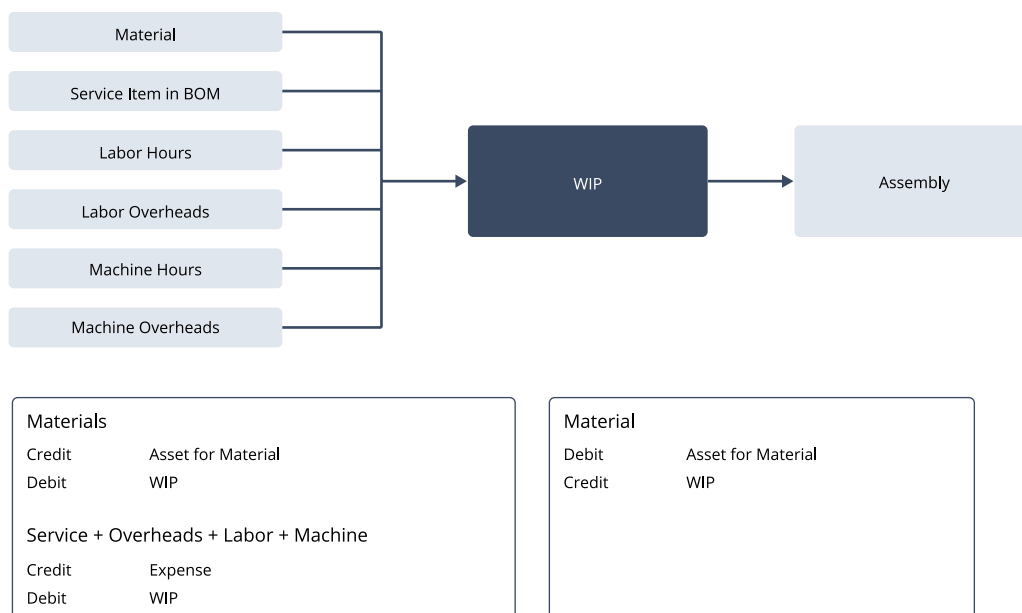
- The steps required to complete the assembly are detailed in operation task records. For more information, see [Manufacturing Routing and Work Orders](#).

These task records designate what needs to be done and when, how much has been done, and how much remains to be done. They define how much time you expect to spend on the task and the rates to be charged for it. Task records designate work centers to assign tasks to certain labor resources.

- During the assembly process, NetSuite logs time against tasks to show progress towards completion. By entering data on the completion form you determine the following:
  - **Starting and Ending Operation** – operation tasks that are done
  - **Quantity Completed** – time logged against each operation

Completion records show the time machines and labor used. When time is entered against an operation task, the scheduling for all tasks related to the work order automatically update. This update accurately portrays progress against each operation. For more information, see [Manufacturing Routing Completions and Time Entry](#).

Values for assets and expenses associated with a routing work order post to the designated Work In Process (WIP) account during the assembly process.



Values are added to the WIP account based on time logged against operation tasks or quantity produced in a run. After the assembly process completes, the values are removed from the WIP account and added to the Asset for Assembly account. For more information, see [Manufacturing Routing Costing](#).

## Setting Up Manufacturing Routing

### To set up manufacturing routing:

1. To enable the feature, see [Enabling the Manufacturing Routing Feature](#).
2. To set preferences, see [Setting Routing Preferences](#).
3. To define cost categories, see [Defining Cost Categories for Manufacturing Routing and Work Center](#).
4. To define manufacturing charge items, see [Defining a Manufacturing Charge Item](#).
5. To group manufacturing charge items into a cost template, see [Creating Manufacturing Cost Templates](#).
6. To define resources, see [Creating Manufacturing Work Centers or Groups](#) and [Work Center Calendars](#).
7. To create routings, see [Creating a Manufacturing Routing](#).
8. To set up standard costing, (for accounts using the Standard Costing feature), see [Standard Costing for Manufacturing Routing](#).

## Enabling the Manufacturing Routing Feature

Before you can use the Manufacturing Routing and Work Center feature, an administrator must enable it.

### To enable manufacturing routing:

1. Go to Setup > Company > Setup Tasks > Enable Features.
2. Click the **Items & Inventory** subtab.
3. Check the **Manufacturing Routing and Work Center** box.
4. Click **Save**.

The following features must be enabled to use the Manufacturing Routing and Work Center feature:

- Manufacturing Work In Process
- Multi-Location Inventory
- Work Orders
- Assembly Items
- Project Management

## Setting Routing Preferences

With Manufacturing Routing and Work Center, you can use the **Show Planned Capacity on Work Orders** preference to help manage work order planning. This preference lets production managers and planners monitor planned completion times for work order builds against work center capacity.

Setting this preference enables NetSuite to automatically create planned time entries when processing manufacturing work orders. When you use this preference, work orders display a new Planned Time

subtab. NetSuite generates planned time entries showing the amount of time being allocated to each work center per day. The Planned Time subtab displays the duration of each operation and the associated work center. When you examine this planned time data, you can determine if the resource capacity is enough to meet needs, providing efficient manufacturing planning.

**Note:** You cannot edit the generated planned times.

When you use this preference, NetSuite automatically recalculates planned time after each completion. For example, your process requires Operation 10, which produces a completed quantity of 30. Next, you report the completion of Operation 10 with a quantity of 15 and NetSuite recalculates the planned time for the remaining quantity.

### To set routing preferences:

1. Go to Setup > Manufacturing > Manufacturing Preferences.
2. To display the Planned Time subtab on work orders, in the Production Exception section, check the **Show Planned Capacity on Work Orders** box.

This preference permits NetSuite to automatically create planned time entries.

3. In the **Default Scheduling Method** field, choose either **Forward** or **Backward** scheduling. Your selection appears by default in the **Scheduling Method** field on new work orders you enter. For more information, see [Production Scheduling Methods Overview](#).
4. In the **Create Work Orders in Supply Planning** field, select one of the following to define the default for supply planning work orders:

- **Do Not Generate**
- **Generate in Planned Firm Status**
- **Generate in Planned Open Status**
- **Generate in Released Status**

Your selection defines the default status of new work orders generated by a planning engine run.

**Note:** If you use the Manufacturing Routing and Demand Planning features, note the following. A selection to generate orders provides the option to define production scheduling methods on work orders. For more information, see [Supply Planning and Routing](#).

5. Click **Save**.

**Note:** You can activate the Planned Time subtab on work orders that were entered prior to using the **Show Planned Capacity on Work Orders** preference. To do so, open the work order in edit mode, and then save. The Planned Time subtab and planned time entries appear on that work order.

## Defining Cost Categories for Manufacturing Routing and Work Center

You can create a cost category to use the Manufacturing Routing and Work Center feature. Cost categories help define expenditures associated with a work order. For example, you have a warehouse and employ workers to assemble widgets that you sell. You need to track costs associated with employee labor, warehouse machines, and overhead associated with each work order.

You can create one of the following eight cost categories to help track costs:

- **Direct Cost** – These costs are calculated when you record time for these items.
  - **Labor Setup** – Cost of time for labor to set up a run  
For example, this is the cost of paying an employee to set up the machine that will be used to paint the widget.
  - **Labor Run** – Cost of time for labor to run an assembly  
For example, this is the cost of paying an employee to use the machine that paints the widget during the assembly run.
  - **Machine Setup** – Cost of time spent to set up a machine to be used in a run  
For example, each time you set up the machine that paints the widget, wear and tear costs you .02 cents.
  - **Machine Run** – Cost of time spent to run a machine during assembly  
For example, each time you use the machine that paints the widget for an assembly run, wear and tear costs you .04 cents.
- **Overheads**
  - **Labor Setup Overhead** – Cost of overhead associated with labor to set up a run  
For example, this is the safety training expense incurred for an employee who sets up the machine used to paint the widget.
  - **Labor Run Overhead** – Cost of overhead associated with labor to run an assembly  
For example, this is the safety training expense incurred for an employee completes an assembly run.
  - **Machine Setup Overhead** – Cost of overhead associated with setting up a machine used in a run  
For example, this is the water expense incurred by running a machine during the machine set up.
  - **Machine Run Overhead** – cost of overhead associated with running a machine during assembly  
For example, this is the water expense incurred by running a machine during an assembly run.

### To set up a cost category for Manufacturing Routing and Work Center:

1. To add choices to this list, go to Setup > Accounting > Setup Tasks > Accounting Lists..
2. Click **New**.
3. Click **Cost Category**.
4. Enter a name for the cost category. For example, US Labor Run Standard.
5. Select a cost type for this category. Choose from the following:
  - **Labor Run**
  - **Labor Run Overhead**
  - **Labor Setup**
  - **Labor Setup Overhead**
  - **Machine Run**
  - **Machine Run Overhead**
  - **Machine Setup**
  - **Machine Setup Overhead**
6. Check the **Inactive** box if you do not want this category to show in lists.

For information about creating a new cost category, see the help topic [Creating Cost Categories](#).

## Defining a Manufacturing Charge Item

With Manufacturing Routing and Work Center, when a specific routing operation is recorded, you can use items to define charges for the activity. For example, you can define the hourly cost of activities being performed, and the expense account the charges are logged against. To do so, you must set up the item record to define the item as a manufacturing charge item.

### To define an item as a manufacturing charge item:

1. Go to Lists > Accounting > Items > New.
2. Click a link to create a charge item. You can track routing charges and expenses using the following item types:
  - Other Charge for Purchase
  - Service (for Purchase or for Resale)
3. Enter an **Item Name**.  
For example, Machine Run Time Cost.
4. If you use NetSuite OneWorld, select a **Subsidiary**.  
A manufacturing charge item can be associated with only one subsidiary.
5. On the item record, check the **Manufacturing Charge Item** box.  
This box cannot be cleared if the item is included in a cost template.  
You cannot check the **Include Children** box on the item record when the **Manufacturing Charge Item** box is checked.
6. Select a **Cost Category**.  
The cost category cannot be changed if the manufacturing charge item is included in a cost template.  
Only manufacturing charge items can use the labor and machine cost categories.
7. Enter an hourly rate up to 7 decimal places in the **Purchase Price** field.  
Charges are based on hourly rate (amount per hour).
8. Click the **Accounting** subtab.
9. Select the expense account these hourly charges should be logged against.
10. Complete any additional necessary fields.
11. Click **Save**.

A manufacturing charge item cannot be associated with a unit type even if you use the Multiple Units of Measure feature. For information, see the help topic [Creating Item Records](#).

## Creating Manufacturing Cost Templates

A manufacturing cost template is a list of rates that can be associated with completing a specific operation. The template defines the activities that occur and related costs to be recorded each time this step is completed.

For example, an employee works 10 hours on an assembly activity. A manufacturing operator needs to record the hours worked as a completion for this step. The cost template defines costs associated with the step completed: the rate for each activity, and what accounts these amounts should post to.

A manufacturing cost template streamlines tracking assembly process costs. It provides which rates and accounts are commonly used for each step in an assembly process.



**Important:** Including too many cost types on a manufacturing cost template may degrade NetSuite performance. Performance is compounded when multiple cost items aren't related to production.

### To create a manufacturing cost template:

1. Go to Lists > Supply Chain > Manufacturing Cost Template > New.
2. Enter a name for the template.
3. Optionally enter a memo. You can search for text you enter here to find this template later.
4. Check the **Inactive** box if you do not want this template to show in lists on forms and records. Clear this box if you do want this template to show in lists.
5. Select a cost category.  
For information about creating a cost category, see [Defining Cost Categories for Manufacturing Routing and Work Center](#).
6. Select an item.  
Only items that have been marked as **Manufacturing Charge Items** show on this list.  
For more information, see [Defining a Manufacturing Charge Item](#).
7. Enter rates up to 7 decimal places for this line item.
  - If this is a Setup category, enter a fixed rate. This is a one-time charge for a setup activity.
  - If this is a Run category, enter a run rate. This is an amount charged for each run completed.



**Note:** Template creation performance is negatively affected when the number of lines on the cost templates isn't kept to a minimum.

8. Click **Add**.
9. Repeat steps 5 through 8 for each category. Enter one category for each activity associated with this operational step.  
You can add only one of each of the following cost category types: Labor Run, Machine Setup, or Machine Run. However, you can add multiple categories for Overhead cost category types.
10. When all necessary categories have been added, click **Save**.

## Cost Template Examples

A manufacturing cost template shows rates for many possible activities that are associated with an assembly step, such as the examples below:

| Activity Type                                   | Example                       | Rate Type | Rate Amount   | Cost Category          |
|-------------------------------------------------|-------------------------------|-----------|---------------|------------------------|
| <b>Manufacturing Labor Setup Service</b>        | Warm up molding machine       | Fixed     | \$10 per run  | Labor Setup            |
| <b>Manufacturing Labor Setup Overhead</b>       | Facility rental               | Fixed     | \$16 per run  | Labor Setup Overhead 1 |
| <b>Manufacturing Labor Run Service</b>          | Costs to complete one run     | Run       | \$14 per hour | Labor Run              |
| <b>Manufacturing Labor Run Overhead Service</b> | Electric utility cost per run | Run       | \$13 per hour | Labor Run Overhead 1   |



## Creating Manufacturing Work Centers or Groups

A work center is a group of people that perform a specific step in the manufacturing process. After you define a work center group, that work center can be assigned to cover specific steps in the manufacturing process. For example, your assembly process might require the following groups: manufacturing, quality assurance, and packing machine.

You create work centers by creating a static group, and then identifying it as a manufacturing work center.

### To create a manufacturing work center:

1. Go to **Go to Lists > Relationships > Groups > New**.
2. Select **Static** as the kind of group.
3. Select **Employee** as the kind of members.
4. Click **Continue**.
5. On **Create Static Employee Group** page, enter a name for this group.  
For example, enter **Packing Machine Group**. This name shows in work center lists on records and forms.
6. Select the owner of this group.



**Note:** You are selected by default. Only the owner of a group can add or remove members or delete the group.

7. Check the **Manufacturing Work Center** box. This enables this group to be used as a work center with routing records.
8. Click the **Manufacturing Work Center Settings** subtab.
9. Select one or more locations to associate with this work center.  
If you use NetSuite OneWorld, you can select multiple locations within a subsidiary.
10. Enter the number of machine resources for this work center.  
If this machine is used in multiple work centers, enter a decimal amount. For example, if this machine is used by another work center for half the day, enter 0.5.
11. Enter number of labor resources for this work center.  
If labor is used by multiple work centers, enter a decimal amount. For example, if a welder spends only two hours in this work center in a day, enter 0.25.
12. Complete any additional fields necessary for this group record.
13. Click **Save**.

NetSuite uses the associated work calendar to schedule the operation tasks associated with each work center.

For more information about creating a static group record, read the help topic [Creating a Static Group](#).

## Work Center Calendars

For each work center, you can create a need-specific work calendar that represents the times the center is available to process work orders. Specified work center calendars enable you to manufacture products efficiently within time and resource constraints. For example, your work centers operate as follows:

- Work Center 1: Operates five days a week for eight hours
- Work Center 2: Operates four days a week for ten hours

Your production manager can set up a work calendar for each work center that specifies the hours available for each. Based on this work center calendar information, you have real-time visibility into total scheduling capacity. You also have intelligent manufacturing routing that calculates the processing time.

After you create the work calendar, specify that calendar on the work center record. Then, NetSuite can use the designated work calendar to determine the time when resources should be scheduled for associated manufacturing operation tasks.

Based on the requirement date, the supply planning engine determines the start date of a planned order. The supply planning engine uses backward scheduling, based on the work calendar assigned. The supply planning engine also considers work center calendars for forward scheduling when creating work orders.

To use work center calendars, complete the following steps:

- [Setting Up a Work Center Calendar](#)
- [Assigning a Work Calendar to a Work Center](#)

## Setting Up a Work Center Calendar

Use the following procedure to set up a work center calendar.

### To set up a work center calendar:

1. Go to Lists > Employees > Work Calendars > New.
2. Enter a name for the calendar.
3. On the **Working Days** subtab, define the working hours and days.
4. On the **Non-working days** subtab, define exceptions to the working days rules.
5. Click **Save**.

For additional details, see the help topic [Setting Up a Work Calendar](#).

## Assigning a Work Calendar to a Work Center

Use the following procedure to assign a work calendar to a work center.

### To assign a work calendar to a work center:

1. Go to Lists > Relationships > Groups > New.
2. Click the **Static** button.
3. In the members list, select **Employee**.
4. Click **Continue**.
5. Check the **Manufacturing Work Center** box.  
This enables the **Work Calendar** field on the **Manufacturing Work Center Settings** subtab.
6. Click the **Manufacturing Work Center Settings** subtab.
7. In the **Work Calendar** field, select the appropriate work calendar from the list.
8. Click **Save**.

Enter additional information, see the help topic [Creating a Static Group](#).

## Creating a Manufacturing Routing

A manufacturing routing is a template that contains a list of steps required to build an assembly item. Each step is in the order necessary to complete the operational sequence for completing the assembly.

After you create a routing record, that routing can be selected on a work order to direct the completion of the assembly. The routing determines the work center, cost template, labor resources, and machine resources that will be used during the assembly.

Routings are unique for each assembly item. However, routings can be shared across multiple locations.

### To enter a manufacturing routing:

1. Go to Lists > Supply Chain > Manufacturing Routing > New.
2. Select the **Item** you are creating an assembly sequence for.
3. Enter one or more locations where this assembly will be performed.
4. Enter a routing template **Name**.  
This name appears in the list of routings in the **Manufacturing Routing** field on records and forms. For example, enter **Beta Alternate Supply Routing**.
5. Optionally, enter a **Memo**.  
You can search for the text you enter in this field. For example, enter **Use Beta Alternate** when primary widget supply is unavailable.
6. Check the **Default** box to use this routing by default for this item on forms.  
When you check this box, these routing steps will be used when new special work orders and mass created work orders are created.  
Clear this box if you do not want this routing to be used by default.
7. Check the **Inactive** box if you do not want this routing to show in routing lists on records and forms.  
Clear this box if you do want this routing to show in lists.
8. Check the **Auto-calculate Lag** box if you want NetSuite to calculate lag times for operation tasks.  
For more information, see [Operations Overlap](#).
9. Click the **Routing Steps** subtab.
10. Enter the **Operation Sequence** number for the step you are entering.
11. For example, if you are entering the first step to be performed to build this assembly, enter 1.  
Sequence number determines dependencies between different operations. For example, operation 1 comes before operation 3 in the assembly process.
12. Enter the **Operation Name**.  
For example enter Assembly Setup.
13. Select a **Manufacturing Work Center**. This is the labor team that will complete this step.  
After you select a work center, the labor resources and machine resources are automatically entered from the work center record.
14. Select a **Manufacturing Cost Template** for this operation.
15. Enter the operation **Setup Time** in minutes.  
This is the amount of time required (fixed time per step) to prepare for this step in the sequence. For example, this could represent the time in minutes required to warm up a molding machine to bring the mold to the proper temperature.

There is one setup time per order.

16. Enter the operation **Run Rate** in minutes.

This is the amount of time required to complete a run and produce one unit.

- NetSuite uses the **Setup Time**, **Run Rate** and the default calendar to schedule the completion of each step for work orders using this routing.
- If you use the Demand Planning feature, NetSuite uses backward scheduling to determine the appropriate start date. There is one run time per base unit.
- The **Setup Time** + **Run Rate** = total manufacturing task time.

17. Select the **Connection Type**.

Define operations overlapping by selecting **Finish-To-Start** or **Start-To-Finish** connection type.

For more information, see [Operations Overlap](#).

18. If you did not check the **Auto-calculate Lag** box, set **Lag Type**, **Lag Amount**, and **Lag Unit**, as necessary.

19. Click **Save**.



**Tip:** You can create a new routing by clicking the **New Manufacturing Routing** button on the **Manufacturing** subtab of an assembly item record.

## Standard Costing for Manufacturing Routing

If you use the Standard Costing and Manufacturing Routing and Work Centers features, NetSuite performs calculates the assembly cost. NetSuite incorporates the labor and machine costs, based on the default routing.

Standard Costing with routings requires the following:

1. **Cost Version**

Verify that you have created a cost version.

For more information, see Defining Cost Versions [Defining Cost Versions](#).

2. **Planned Standard Cost Rollup**

Run a cost rollup to calculate assembly cost. When you run a cost rollup, NetSuite checks for a default routing to calculate costs for the assembly. If you have no defined default routings, NetSuite uses the first routing created as the default routing to calculate the assembly cost.

Planned Standard cost is a consolidation of cost based on the component and cost category.

For subassemblies, each of the cost categories are rolled up to the next level in the Bill of Materials (BOM) hierarchy. The difference between the cost categories of this level and lower levels in the roll up depend on the items associated with the rollup.

Note the following when you review the cost of an assembly item based on the cost rollup. The lower level routing cost of building the subassembly is denoted with the subassembly item as a component on the planned standard cost record. The routing cost of building this level final assembly is denoted as follows. The Service and Other charge items appear as a component on the planned standard cost record.

For more information, see the help topic [Standard Cost Rollup](#).

3. **Inventory Revaluation**

When you use Standard Costing features, run update production cost to establish a standard cost in production.

For more information, see the help topic [Revalue Standard Cost Inventory](#).

## Manufacturing Routing Cost Calculation

NetSuite calculates the cost of each step in a routing as follows:

- Part 1: Definition of Time / Quantity

Total Setup Time = (number of resources x setup time)

Total Run Time = (number resources x run time)

- Part 2: Rate

Based on the manufacturing charge item, the quantity is the total hours required.

The total unit cost is derived at a component level per cost category and per operation sequence.

- Number of resources (from the work center) x Setup Time (from the routing record) x Manufacturing Charge Item Unit Cost (from the item record)
- Number of resources (from the work center) x Run Rate (from the routing record) x Manufacturing Charge Item Unit Cost (from the item record)

For more information about costing, see [Manufacturing Routing Costing](#).

## Manufacturing Routing and Work Orders

If you use manufacturing routing and work centers, you can use routings on work orders to manage your assembly process.

The routing you use on a work order is a template that describes the required steps to build an assembly item. The routing determines the work center, cost template, labor resources, and machine resources to use during assembly.

For more information, see [Setting Up Manufacturing Routing](#).

### To use routing on a work order:

1. Create a new work order.

For more information, see [Entering an Individual Work Order](#).

2. In the Classification section, select a **Location**.
3. If you use NetSuite OneWorld, select a subsidiary.
4. Check the **WIP** (Work In Process) box.

When you designate a routing as WIP, NetSuite uses WIP accounting to issue materials in the designated WIP account.

For more information, see [Manufacturing Work In Process \(WIP\)](#).

5. Select a **Manufacturing Routing** option.

NetSuite displays the default assembly routing based on the location. However, you can select a different routing.

For more information, see [Creating a Manufacturing Routing](#).

6. Click **Save**.

The following subtabs are displayed:

- The **Items** subtab shows components that are issued.
- The **Operations** subtab shows all operation tasks required for a particular assembly run.  
Operation tasks are created based on the routing. These tasks define the list of steps that must be completed to finish the assembly process. Tasks can be viewed and edited from the work order Operations subtab.  
For more information, see the following help topics:
  - [Manufacturing Operation Tasks](#)
  - [Manufacturing Operation Task](#)

## Work Order Quantity Changes

If you change the assembly item quantity on the work order and then re-save it, the Operations subtab information updates to reflect the changes. For example, entering a higher quantity on the work order results in more required time to complete the run.

## Routing and Time Zones

The associated work calendar applies for all routing work orders created.

- If you **do not use** NetSuite OneWorld, the schedule time zone is based on the time zone selected for the company.  
For more information, see the help topic [Configuring Company Information](#).
- If you **do use** NetSuite OneWorld, the schedule time zone is based on the time zone of the subsidiary selected on the work order.

## Planned Time Subtab

When you use Manufacturing Routing and Work Center and set the **Show Planned Capacity on Work Orders** preference, note the following. Work orders show a Planned Time subtab. This subtab details work allocated to each work center. In addition, NetSuite automatically generates planned time entries. For more information, see [Setting Routing Preferences](#).

## Manufacturing Operation Tasks

After you save a WIP work order that has a designated routing, manufacturing operation tasks are created based on the routing.

Each of these tasks is a step that must be done for the assembly process to finish. After work for the task has been done, manufacturing operators can log progress against each task on a work order completion form.

### To view a list of operation tasks:

1. Go to Transactions > Manufacturing > Enter Work Orders > List.
2. Click **View** next to the work order.
3. Click the **Operations** subtab.

The following links are displayed in the **Work Order Operations** table:

- **Edit** – opens the editable Manufacturing Operation Task popup window.
- **Operation Sequence** – opens the viewable Manufacturing Operation Task popup window.
- **Operation Name** – opens the viewable Manufacturing Operation Task popup window.
- **Manufacturing Work Center** – opens the viewable Static Employee Group popup window.
- **Predecessor** – opens the viewable Manufacturing Operation Task popup window.



**Note:** You can also modify tasks from the work order. Go to Transactions > Manufacturing > Enter Work Orders > List. Click Edit next to the work order to be edited. Then click the Operations subtab to view a list of operation tasks.

## Add or Delete Tasks

From the work order Operations subtab, you can do the following:

- You can add a new task with any sequence number.
- You can edit existing operations with the "Not Started" status to change their sequence numbers to any value.
- You can delete any operation task with the "Not Started" status.

To delete a task, click **Edit**. Then, select **Delete** under **Actions** and click OK in the confirmation message to remove the task.

After task records are created, you can open and modify them individually. For more information, see [Manufacturing Operation Task](#).

## Operations Overlap

When you use the Manufacturing Routing feature, manufacturing planners can use the operations overlap function to schedule overlapping manufacturing operations. Overlapping of manufacturing operations can reduce work order lead times and allow more efficient utilization of manufacturing resources. When you set up operations to overlap, work order operations can be processed in a staggered method through the production cycle.

To define operations overlap, select Connection Type and Lag Amount as needed. The following options are available:

- **Start-To-Start** - the subsequent operations start at the same time (with minimal lag calculated automatically).
- **Start-To-Start** - the subsequent operation starts after a specified delay (lag) from the start of the first operation. For more information about defining lag on a routing, see [Define Lag on a Routing](#).
- **Finish-To-Start** - the subsequent operation starts only after the previous one finishes.
- **Finish-To-Start** - the subsequent operation starts after the first operation finishes, with an additional specified delay (lag). For more information about defining lag on a routing, see [Define Lag on a Routing](#).

To use operations overlap, you must enable these features:

- Assembly Items
- Work Orders
- Manufacturing Work In Process
- Manufacturing Routing and Work Center

You can view and edit the connection type and lag amount in the **Predecessors** subtab in the Manufacturing Operation Task popup window. For more information, see the [Manufacturing Operation Tasks](#) help topic.

You can also define the connection type and lag amount in the **Routing Steps** subtab in the Manufacturing Routing popup window. For more information, see the [Creating a Manufacturing Routing](#) help topic.

## Define Lag on a Routing

The Lag Amount defines what portion of an operation needs to be completed before the next operation can start. It defines lag between beginning of an operation (run time) and beginning of consequent operation (setup time).

For each sequence of the routing, you can define a lag type using the following types:

- Time (in minutes)
- Quantity (in assembly units)
- Time Percentage
- Quantity Percentage

## Autocalculate Lag

If you choose to autocalculate the routing, NetSuite can automatically calculate the optimal lag amount. The Optimal (Minimal) Lag Amount is the shortest possible lead time for each work order. Optimizing lag means that operations on a work order are scheduled strategically. This minimizes the lead time of the order and maximize the utilization of work centers within the work order.

To autocalculate lag amount, check the Auto-Calculate Lag box on the routing record. When you check this box, NetSuite disables the following fields in the grid:

- Lag Type
- Lag Amount
- Lag Unit of Measure



**Important:** If you want to be able to edit the fields noted above, do not check the Auto-Calculate Lag box. If you choose to define these values, they default on work orders.

If lag settings are changed on the operation record, NetSuite reschedules the operation when the task is saved. The task reflects the new settings and reschedules all affected subsequent tasks.

When a work order is scheduled, NetSuite calculates the Optimal (Minimal) Lag Amount and Maximal Lag Amount for each operation. This is true if the operation has a preceding operation and defined lag type. Then, NetSuite does not allow the lag amount to be defined outside of this range.

The autocalculation setting can be changed only when the status of a work order is Planned or Released.

- When Auto-calculate Lag is enabled, lag amounts on operations cannot be edited.
- When Auto-calculate Lag is enabled, the only lag type available is Quantity.

After creation, click the Operations subtab on work orders to view the Start Date/Time and End Date/Time of each operation.

On the operation task record under the **Predecessor** subtab, the Lag Type and Lag Amount can be edited. After save, NetSuite validates that each lag amount entered falls between the Optimal (Minimal)



Lag Amount and Maximal Lag Amount. If the lag amount entered is outside this range, an error displays the valid range.

## Operations Overlap and Supply Plans

Because a supply plan generates supply plan lines for assemblies or work orders, calculations do account for lag related settings on the assembly routing. NetSuite verifies that the defined lag amounts fall in the valid range between the Optimal (Minimal) Lag Amount and Maximal Lag Amount. If not, the supply plan is automatically adjusted.

- If the lag amount defined on the routing is less than the Optimal (Minimal) Lag Amount, it is adjusted to the optimal lag amount.
- If the lag amount defined on the routing is greater than the Maximal Lag Amount, it is adjusted to equal the maximal lag amount.

## Manufacturing Operation Task

For any work order with a status of **Pending Build**, you can make changes to operation tasks on the order. For example, due to specifications particular to one order, you may need to change information defaulted on tasks from the routing template.

After work is logged against the order and the status is In Process, you can no longer edit the operation tasks.

On individual task records, you can view the following:

- **Manufacturing charge items** – Charge items are derived from the routing template, but can be modified as necessary for individual orders.
- **Estimated time required for completion**  
When you enter a setup time or run time on the task record, these times are planned estimates and are used for scheduling. It is only after completion time is entered against a task is the actual time updated in the Actual Hours field.

Task dependencies are assigned based on the numeric order of the operation sequence and aren't editable.

### To edit operation tasks:

1. Go to Transactions > Manufacturing > Manufacturing Operations Tasks.
2. Click **Edit** next to the operation you want to modify.
3. You can enter a new **Operation Name**.

 **Note:** When you modify task settings, labor and machine scheduling reflects calculations based on the new entries after you save the task changes.

The following fields **Primary Information** aren't editable:

- The **Work Order** identifier related to this operation task.
  - The **Insert Before** field identifies the next step in the sequence.
  - The operation sequence **Status**. For example, Not Started, Started, Completed.
4. You can enter any additional **Comments** that you want to accompany this operation task. For example, "use Shimano spokes when Bontrager spokes aren't available."

The following **Operation Overview** fields aren't editable:

- NetSuite calculates **Estimated Hours** as the expected time to complete a task based on total setup time + quantity x run rate.
  - **Actual Hours** displays the time in hours it took to complete this task.
  - The **Remaining Hours** of work needed to complete this task are displayed.
  - **Input Quantity** represents the number of items entered or created during this task.
  - **Completed Quantity** is the total number of items created during this task.
5. You can change the operation **Setup Time** (in minutes) for the working day as defined in the work calendar.
  6. You can change the operation **Run Rate** (in minutes). This is the amount of time required to complete a run and produce one unit.
- The following **Operation Schedule** fields aren't editable:
- The **Start Date** represents the date you expect to begin assembly production. This field defaults to today's date.
  - **End Date** is the date you expect to complete assembly production. This field defaults to today's date plus lead time.
  - Check the **Auto-Calculate Lag** box if you want NetSuite to calculate lag times for operation tasks. For more information, see the [Autocalculate Lag](#) help topic.
7. Unless it is the first operation, you can change operations overlapping in the **Predecessors** subtab. For more information, see the [Operations Overlap](#) help topic.
  8. Click **Save**.

## Work Order Operation Start and End Dates

**Operation Start Date** is set to the start time for the working day as defined in the working calendar.

To set an **Operation End Date**, the remaining operation time is added to the operation start of the next working day after the completed time.

When the remaining time exceeds the available working time, any outstanding time is added to the beginning of the next working calendar working day.

## Completions on Non-Working Days

**Operation Start Date** is set to the start time for the working day as defined in the working calendar.

To set an **Operation End Date**, the remaining operation time is added to the start of the next working day as defined in the working calendar.

You can also modify tasks from the work order. Go to Transactions > Manufacturing > Enter Work Orders > List. Click **Edit** next to the work order to be edited. Click the **Operations** subtab to view a list of operation tasks.

## Work Centers and Manufacturing Operations Tasks

You can view the Manufacturing Operations Tasks list to process work orders based on work centers. This lets you see which centers have completed tasks that are predecessors for other tasks to be worked on.

You can view the operation tasks list and filter the list to show only tasks associated with Work Center 2. Customize the view to show the Predecessor and Predecessor Completed Quantity. These fields let you see which tasks Work Center 2 is due to work on next. (Create a custom view for the task list to display specific columns and information by clicking **Edit View**.)

For example, the task list shows that two work orders have tasks for which the predecessors are completed. You know that Work Center 2 needs to work on those tasks. When Work Center 2 finishes their requirement for the tasks, click **View** next to those tasks in the task list to enter completions. The completions entered then update associated work orders with new predecessor data. The work center task lists display the updated task statuses.

After completions are entered for tasks, scheduling is updated for labor and machines to reflect calculations based on the new entries.

For example, you complete work earlier than anticipated. Therefore, subsequent tasks move up and start and end dates reschedule to be earlier. However, if a completion reflects that work is falling behind, subsequent tasks move out and start and end dates reschedule to be later.

### To view the operation tasks list:

1. Go to Transactions > Manufacturing > Manufacturing Operations Tasks.
2. Optionally filter the list by selecting a work center.

## Supply Planning and Routing

If you use the Demand Planning and Manufacturing Routing and Work Center features, routings on work orders can affect your supply planning. This is because supply planning uses backwards scheduling to meet manufacturing due dates.

**Note:** Procurement lead times do not affect these time requirement calculations.

The supply planning method used depends on whether a default routing is identified.

### Without a Default Routing

When you generate a supply plan on the Generate Supply Plan page, NetSuite calculates the order date (or release date). NetSuite bases the calculation on the due date using the following formula for assembly items that are required to be built:

$$\text{Order Date} = \text{Due Date} - (\text{Quantity} \times \text{Work Order Lead Time})$$

- Quantity = Quantity of items required
- Work Order Lead Time = Number of days required to build, per unit
- Due Date = Date when the additional supply is required



**Note:** Without a default routing, NetSuite makes calculations without reference to any calendar or resource requirements.

### With a Default Routing

When you use Manufacturing Routing and Work Center, the supply planning engine considers the default routing and associated work calendar for scheduling purposes.

When an assembly build is required and a default routing is defined, NetSuite calculates the cumulative lead time across all operation sequences. NetSuite uses the following formula:

Total time =

[Sum of Setup Time for all operation tasks + (Sum of Run Rate x Quantity)] x Total Hours per day

**Note:**

- Setup Time = Total cumulative setup time across all operation sequences in the default routing
- Run Rate = Total cumulative run time across all operation sequences in the default routing
- Total Hours per day = Total number of hours available on the associated work calendar

NetSuite calculates the order by backward scheduling from the due date. NetSuite does consider the associated work calendar for days available, including holidays.



If changes are made to the associated work calendar or to the routing record, you should regenerate the supply plan. These changes can include:

- modifying the work calendar (such as to increase/decrease the working days or add/remove holidays)
- editing the routing to increase/decrease a setup time or run rate

After regeneration, NetSuite modifies the order dates to reflect the new requirements and still meet the due date deadline.

## Routings and the Generate Work Order in Supply Plan Preference

NetSuite supply plan processing for an assembly with a routing defined depends on your setting for the Generate Work Order in Supply Plan preference.

| Generate Work Order in Supply Plan Setting | Default Scheduling Method | Result                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Do Not Generate                            | —                         | The supply plan uses backward scheduling to determine order date. In this case, NetSuite creates only the supply plan line, not the Work Order.                                                                                                                                                                                                                                                                           |
| Not Do Not Generate                        | Forward                   | The supply plan uses backward scheduling to determine the order date. Within the supply plan run, NetSuite automatically creates a work order, and sets the production start date the same as the order date. When the work order is created, NetSuite uses forward scheduling to calculate the work order production end date and sets the production start date.<br><br>In this case, the supply plan creates an order. |
| Not Do Not Generate                        | Backward                  | NetSuite creates a work order using the receipt date from the demand plan. The production end date is set at one day prior to the receipt date. This allows time to complete production and be available on the required date. The work order uses backward scheduling to calculate the work order production start date.                                                                                                 |

| Generate Work Order in Supply Plan Setting | Default Scheduling Method | Result                                          |
|--------------------------------------------|---------------------------|-------------------------------------------------|
|                                            |                           | In this case, the supply plan creates an order. |

For information about setting the Generate Work Order in Supply Plan preference, see [Setting Routing Preferences](#).

## Production Scheduling Methods Overview

When you use the Routing and Demand Planning features to generate supply work orders, you choose the method NetSuite uses to calculate production requirements. These calculations assess the time, materials, and resources required to complete an order and set a start or end date for the production run.

### Forward Scheduling

Forward scheduling lets you set a production start date. Then, NetSuite calculates the time, materials, and resources required to complete all necessary operations to finish the task. The production end date is determined based on these calculations.

When using the Forward Scheduling method on work orders, the Production Start Date field is required and defaults to the current date. The Production End Date field is dimmed because it will be calculated.

### Backward Scheduling

When you use Backward Scheduling, you set the production end date, which is the date you need to have the completed items. Based on data from the associated routing and related work center calendar, NetSuite calculates the time, materials, and resources required to complete all operations. NetSuite determines the production start date based on these calculations.

When you use Backward Scheduling on work orders, the Production End Date field is required. The Production Start Date field is dimmed because it is automatically calculated.

**Note:** This calculated start date may be a date in the past, prior to the current date.

When Backward Scheduling is set as the default scheduling method, different factors determine the production end date on generated work orders. For more information, see [Backward Scheduling](#).

These scheduling methods can be used when generating individual work orders and by generating work orders using supply planning.

When a work order is saved, or generated, the supply planning engine calculates requirements and then generates necessary work orders. On the work order Operations subtab, click an operation name to view or edit details about the operation.

To choose a production scheduling method, you must first set these preferences:

## Generate Work Orders in Supply Planning

To set production scheduling methods on work orders, you must first set the preference to Generate Work Orders in Supply Planning. For information about this preference, see [Automatically Generating](#)

**Planned Work Orders.** After you set this preference, you select a scheduling method on orders and set a default scheduling method. When you set this preference, the supply plan schedules work orders based on the default scheduling method.

## Setting a Default Scheduling Method

A default production scheduling method appears by default in the Scheduling Method field on work orders created manually and by automated supply planning.

### To set default scheduling method:

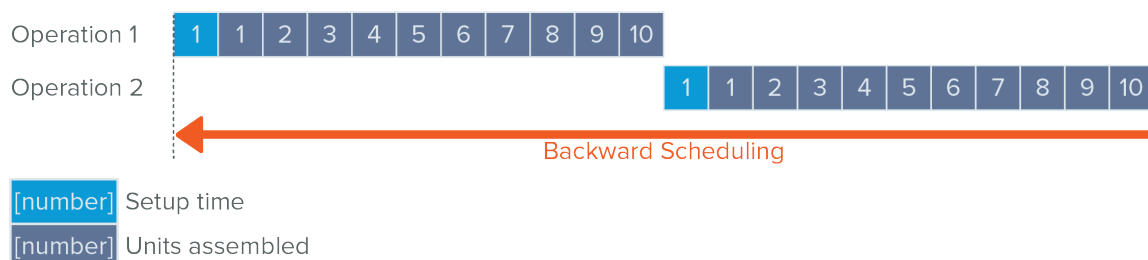
1. Go to Setup > Manufacturing > Manufacturing Preferences.
2. In the **Production Planning and Scheduling Allocation** section, **Default Scheduling Method** field, choose either **Forward** or **Backward**.  
This field defaults to the Forward scheduling.
3. Click **Save**.

This method you select automatically completes work orders. However, you can change the method on individual orders with a status of Planned or Released.

## Backward Scheduling

When you use the backward scheduling method for production planning, you set the production end date. This is the date you need to have the completed items. Based on data from the associated routing and related work center calendar, NetSuite calculates the time, materials, and resources required to complete all operations. NetSuite determines the production start date and time based on these calculations.

For example, you need to schedule an order of widgets that requires two operations to complete a production run. Each operation requires 1 day of setup time and 10 days of production time. Operation 1 must be complete before Operation 2 can start. These production requirements appear in the following illustration:



When you enter the items completed date, NetSuite calculates the day work must begin to complete production on time. When a work order is created by a supply plan, NetSuite schedules work so that the last operation is completed before the due date. The due date time is 00:01 AM. Therefore, the last operation on the work order is scheduled for completion by 11:59 PM on the day before the due date.

The time displayed for each operation is based on work hours set for each work center. For more information, see [Creating Manufacturing Work Centers or Groups](#).

Planners who use the Backward Scheduling method can reduce waste of manufacturing resources because NetSuite calculates start dates automatically. Work order production is scheduled to start as late as possible, giving planners flexibility to cancel or change an order.

 **Note:** The NetSuite planning engine calculations may set a production start date in the past, depending on the end date entered.

Production work isn't automatically reallocated based on resource capacity. After work orders are created, you may need to assess resource assignments.

When Backward Scheduling is used as the default scheduling method, the production end date on generated work orders is determined using the following factors:

- **Sales Order** - The box in the **Create WO** column on the **Items** subtab is checked. This is true for an item that uses the WIP multi-step production process. A work order is generated for your assembly item. The production end date on the work order is set to one day before the expected ship date on the work order. If no expected ship date is entered, the production end date is set to the same day as the sales order date.
- **Work Order** - The box in the **Create WO** column on the **Items** subtab is checked. A work order is generated for a sub-assembly. The sub-assembly's production end date is set to the same date as the parent item's production start date.
- **Supply Plan** - A work order is generated during a supply plan. The production end date for a sub-assembly is set to the same date as the parent item's production start date.



**Important:** The **Generate Work Orders in Supply Planning** preference must be set to any option except **Do not Generate**.

- **Mass Creating Work Orders - Reorder point** is the Replenishment Method. NetSuite uses Forward Scheduling for the work orders regardless of the default scheduling method set in the account preferences.

## Manufacturing Routing Completions and Time Entry

For work orders that use manufacturing routings, enter time and completions against the operation tasks to track assembly process and monitor scheduling and costs. The following help topics provide details about completions and time entry:

- [Routing Work Order Completions](#)
- [Routing Completion Labor and Machine Time Entry](#)
- [Time and Status Updates on Tasks](#)

### Routing Work Order Completions

After an assembly task completes, record a completion against the operation. The completion logs time and activities against the operation. When you update work order records helps you track costs and expenditures up to the current point in time. You can enter a completion in the following ways:

- [Enter a Completion from a Work Order](#)
- [Enter a Completion on a Task Record](#)
- Bulk enter completions (See [Entering Work Order Completions](#)).

After you enter an operation task completion, on the record, click Actions > GL Impact to see completion effects on the general ledger. The GL Impact page shows the overheads and labor expenses recorded against the WIP account.

When you mark the final task work for an assembly completed, saving the completion records the items as put into inventory. When you view the GL Impact page, the value is removed from the WIP account and added to the inventory account.

You can set a preference for completion validation. For more information, see [Completion Validation Preference](#).

## Enter a Completion from a Work Order

When you use the completion buttons on a routing work order, you can choose from the following options:

- **Enter Completion** (completion only)  
Records one of the following:
  - completion of a single operation or a range of operations
  - completion of entire assembly
- **Enter Completion with Backflush** (completion + issue components)  
Records one of the following:
  - completion of a single operation or a range of operations **AND** issue components
  - completion of entire assembly **AND** issue components

 **Note:** When a completion and issue occurs and the status isn't closed or built, the operation status automatically changes to in-progress.

For detailed steps on entering a completion from a work order, see [Entering a Completion for an Individual Work Order](#).

## Enter a Completion on a Task Record

You can open the complete list of task records to enter a completion for an operation task.

### To complete an operation task record:

1. Go to Transactions > Manufacturing > Manufacturing Operations Tasks.
2. Click **View** next to the completed task.
3. On the operation task record, enter the completed quantity.  
After you enter the quantity, NetSuite automatically enters data on the **Components** subtab and **Operations** subtab based on the necessary requirements.  
For more information, see [Routing Completion Labor and Machine Time Entry](#).
4. Click **Save**.

## Completion Validation Preference

You can set a preference to validate that routing operation sequences are always followed in accordance with the work order. NetSuite uses this validation to ensure that the correct quantity is completed for each operational step before permitting the work order to continue processing. This preference restricts the completed quantity amount you can enter on a work order completion for a particular operation. Such verification helps prevent problems due to out-of-sequence processing.



Choose one of the following settings for the **Check Completed Quantity in Prior Operations During Operation Completion** preference:

- **No Verification** – Choose this setting if you do not want to receive verification warnings.
- **Require Confirmation before Saving** – Choose this setting to receive verification warnings. NetSuite permits the completed quantity to be greater than the predecessor completed quantity after the warning is acknowledged.
- **Do Not Allow Saving** – Choose this setting to require that the completed quantity isn't greater than the predecessor completed quantity.

 **Note:** This field defaults to **No Verification**. If you do not change this default setting, unverified completions can be entered. For more information about setting this preference, see the help topic [Order Management Accounting Preferences](#).

For example, completing Operation B requires a set amount of items that are generated during Operation A. You can verify that the requirements of the predecessor have been met before saving the completion for Operation B.

- Creating 5 units during Operation A is a predecessor for creating 5 units during Operation B.
- You choose the preference setting **Do Not Allow Saving**. The total completed quantity of Operation B cannot be greater than the total completed quantity of Operation A.
- Operation B requires 5 units that are created during Operation A. NetSuite verifies that 5 units are completed during Operation A before Operation B can begin.

For orders completed across multiple days or shifts, partial quantities can be logged over time.

| Operation   | Qty Day 1 | Qty Day 2 | Qty Day 3 | Qty Day 4 |
|-------------|-----------|-----------|-----------|-----------|
| Operation A | 50        | 30        | 15        | 5         |
| Operation B | 40        | 40        | 5         | 15        |
| Operation C | 35        | 45        | 0         | 20        |

## Routing Completion Labor and Machine Time Entry

Enter a routing work order completion enables you to record labor and machine time completed against an operation task. For more information, see [Routing Work Order Completions](#).

Entering data on the completion form enables you to define the following information:

- **Starting and Ending Operation** - The completed operation tasks.
- **Quantity Completed** - The amount of time to be logged against each operation.

The Operations subtab shows which operations are being completed and the amount of labor and machine time to record against each operation.

### Identify Completed Operations

1. Enter the **Starting Operation**.  
The first operation task you want to identify as being completed.
2. Enter the **Ending Operation**.  
The last operation task you want to identify as being completed.

Operation tasks logged as completed include the indicated starting operation, the ending operation, and all operation tasks in between the starting and ending tasks.

3. Enter the **Completed Quantity**.

NetSuite uses this quantity to calculate the amount of labor and machine time for completed sequences .

 **Note:** Enter the completed quantity in the Operation Completion section of the form. The Completed Quantity field on the Operations subtab is read-only.

For example, enter a completed quantity of 1, with the starting operation of 10, and ending operation of 30.

Operations 10, 20, and 30, the completed quantity is multiplied against the labor and machine time requirements set on the operation task record. The labor and machine time fields on the Operations subtab are populated.

## Labor Time and Machine Time for Completed Operations

After the operations marked as complete are identified, you can enter details about labor and machine time for each operation.

The Operations subtab can be used as follows for each operation completed:

- Completed Quantity displays the amount entered in the Completed Quantity field.
- To record Setup Time:
  - If you haven't recorded an operation setup time against an operation, the setup columns display the full operation setup time. Time is based on the setup time defined on the operation task record.
  - Check the **Record Setup Time** box.
 

For each operation, you have the following options:

    - Modify the default setup time quantity.
    - Clear the Record Setup Time box.
- NetSuite automatically enters machine and labor run times for each operation based on the quantity completed using the following formula:

Default run time = Qty completed x Run rate on the operation record

After the system enters labor and machine times for all completed operation tasks, NetSuite calculates the progress and costs of the assembly.

- For more information, see [Time and Status Updates on Tasks](#).
- For more information, see [Manufacturing Routing Costing](#).

## Time and Status Updates on Tasks

Time entered against an operation task ensures scheduling for all related tasks are updated to accurately portray progress against each operation.

Often machine and labor resources working concurrently are fully used against a certain operation task. In such cases, both resource types are weighted equally to determine the true time recorded on the operation task record.

Sometimes, one resource may have a higher value than the other resource for a certain task. For example, on a task, the machine run time is less than labor run time. In such a case, the critical path is the labor time because it is the greater of the two. NetSuite updates the production schedule based on the larger requirement of the two. This applies to setup time and run time.

In another example, the recorded machine time is larger than the labor time because the machine can run without constantly being overseen by labor. In this case, the machine time is the critical path, and NetSuite uses it to update scheduling. NetSuite uses the labor time for costing purposes.

For an operation, if the labor run time is larger than the machine run time, NetSuite uses the labor run time as the true hours.



**Note:** The default values that show for machine run and labor run times can be modified to accommodate individual run times.

After the appropriate time values are determined, the amount appears in the **Actual Hours** field on the operation task record. Based on the hours recorded, NetSuite adjusts the schedule of subsequent tasks to provide a realistic view for completions.

## Operation Task Status Updates

NetSuite updates the status of an operation task based on data entered for the work order or tasks. Possible status options include the following:

- **Not Started**

- No time is recorded against the task.
- No quantity completed is recorded on the task.

- **In Progress**

- Some time is recorded against the task.
- Some quantity completed is recorded on the task **and** the completed quantity is less than the input quantity required.

- **Completed**

The completed quantity is equal to or greater than the input quantity required **or** the work order is closed.

## Mark a Routing Work Order as Built or Closed

For some orders, you may want to show the items as being assembled without finishing all the individual steps for each operation task. In such a case you can do one of the following:

- **Mark an Order Built**

When you mark an order as built, the required items are marked built and added to inventory. Note that associated variances **aren't** created.

To mark a work order built, go to Transactions > Manufacturing > Mark Work Orders Built.

For more information, see [Marking Work Orders Built](#).

- **Mark an Order Closed**

When you mark an order as closed, the required items are marked built and added to inventory. Note that associated variances **are** created.

To mark a work order closed go to Transactions > Manufacturing > Mark Work Orders Built..

For more information, see [Marking Work Orders Closed](#).

After you mark an order as built or closed, the operational task record associated with that order displays a task status displays as Completed.

## Manufacturing Routing Costing

Values for assets and expenses associated with a routing work order post to the designated Work In Process (WIP) account during the assembly process.

Values are added to the WIP account based on time logged against operation tasks or quantity produced in a run. After the assembly process completes, the values are removed from the WIP account and added to the Asset for Assembly account.

## Time Updates and Costing

When time is logged against an operation task for an assembly, NetSuite uses this time to calculate costs associated with the assembly. For example, when completion time is logged against an operation task, NetSuite generates the following accounting entries:

|    | Account                  | Formula                  |
|----|--------------------------|--------------------------|
| CR | Labor                    | Rate x Resources x Hours |
| DR | WIP account for assembly | Rate x Resources x Hours |

- **Hours** - time logged on a completion record
- **Resources** - sourced from the completion record
- **Rate** - sourced from the operation task record

The result of these calculations posts to the WIP account for the assembly. Costs can be one of two types: setup costs or run costs.

## Setup Costs

Costs need to be logged for expenses related to setting up for an assembly run. These costs are defined on the operation task record and are based on time logged against the task.

 **Note:** Setup costs are only time dependent, not based on quantity produced.

For example, the operation task **Staging** defines costs on the Cost Detail subtab. One cost category defined is Labor Setup. Using this category on a task defines the following:

- The Labor Setup cost category record indicates the item **Other Charge for Purchase - Labor Setup**.
- The task record defines the fixed rate for using the item **Other Charge for Purchase - Labor Setup** at \$30.
- The item record for **Other Charge for Purchase - Labor Setup** indicates the **Assembly Staging Expense** expense account.

Therefore, when you log time against the **Staging** task, the appropriate amount posts to the **Assembly Staging Expense** account. This amount is calculated as Expense amount = Setup time logged x Labor setup fixed rate.

If 2 hours of time are logged against the **Staging** task, \$60 is logged to the **Assembly Staging Expense** account.

| Labor Setup | Account                  |                          |
|-------------|--------------------------|--------------------------|
| CR          | Labor Setup              | Rate x Resources x Hours |
| DR          | WIP account for assembly | Rate x Resources x Hours |

Similar calculations are made for other categories that may be defined on a task record. For example, overheads (Overhead expense amount = Setup time logged x overhead rate).

| Labor Setup Overhead | Account                      |                          |
|----------------------|------------------------------|--------------------------|
| CR                   | Labor Overhead Setup Account | Rate x Resources x Hours |
| DR                   | WIP account for assembly     | Rate x Resources x Hours |

## Run Costs

Costs need to be logged for expenses related to processing an assembly run. These costs are defined on the operation task record and are based on quantity completed during the assembly run.

For example, the operation task **Staging** defines costs on the Cost Detail subtab. One cost category defined is Labor Run. Using this category on a task defines the following:

- The Labor Setup cost category record indicates the item **Other Charge for Purchase - Labor Run**.
- The task record defines the run rate for using the item **Other Charge for Purchase - Labor Run** at \$65.
- The item record for **Other Charge for Purchase - Labor Setup** indicates the **Assembly Staging Expense** account.

Therefore, when you log time against the **Staging** task, the appropriate amount posts to the **Assembly Staging Expense** account. This amount is calculated as Expense amount = Setup time logged x Labor setup fixed rate.

If 10 units are completed for the **Staging** task, \$650 is logged to the **Assembly Staging Expense** expense account.

| Labor Run | Account                  |                          |
|-----------|--------------------------|--------------------------|
| CR        | Labor Run                | Rate x Resources x Hours |
| DR        | WIP account for assembly | Rate x Resources x Hours |

Similar calculations are made for other categories that may be defined on a task record. For example, overheads (Overhead expense amount = Run quantity logged x overhead rate).

| Labor Run Overhead | Account                    |                          |
|--------------------|----------------------------|--------------------------|
| CR                 | Labor Overhead Run Account | Rate x Resources x Hours |

| Labor Run Overhead | Account                                                |
|--------------------|--------------------------------------------------------|
| DR                 | WIP account for assembly      Rate x Resources x Hours |

## Costing Lot Size

If you use Standard Costing and Manufacturing Routing and Work Center, the Costing Lot Size field appears on the assembly item record's Locations subtab. The default value for this field is 1, and the minimum value for this field is 0.01.

If you also use Multiple Units of measure, the value is in base units of measure.

During the cost rollup process, NetSuite calculates the routing cost of assemblies as follows:

| Setup Cost                                                                                                                          | Run Cost                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| # of Resources x Setup Time x Manufacturing Charge Item Unit Cost / Standard Cost Lot Size (assembly item - item location map)      | # of Resources (work center) x Run Rate (routing record) x Manufacturing Charge Item Unit Cost (item record) |
| Resources are work centers. The Setup Time is from the routing record. Manufacturing Charge Item Unit Cost is from the item record. |                                                                                                              |

## Work Instructions and Traveler

Work Instructions and Traveler SuiteApp helps you to generate traveler reports of required work orders. Using these reports, the shop floor personnel can perform day to day assembly operations. Each report includes details like required components, work order info, and step-by-step work instructions to complete each operation. This report travels from one work center to another for every operation in the work order until the work order is completed.

Traveler reports are printed only for work order with **Released & In Process** status either using the delivered predefined templates or any customized templates defined in the account.



**Note:** This SuiteApp enables you to generate traveler reports which is also provided by the Advanced Manufacturing SuiteApp. The accounts with Advanced Manufacturing SuiteApp by default will continue to use the existing work instructions and traveler feature.

However, when you install the Work Instructions and Traveler SuiteApp, you can override the default setup and copy existing instructions from the Advanced Manufacturing SuiteApp using CSV Import/Export. For more information, read the help topic [CSV Imports](#).

Watch the following help video for an overview of the Work Instructions and Traveler SuiteApp:

 [Work Instructions and Traveler SuiteApp Overview](#)

## Installing Work Instructions and Traveler SuiteApp

The Work Instructions and Traveler SuiteApp requires a NetSuite OneWorld account. Only users with Administrator role can install the SuiteApp. For more information about installing a SuiteApp, see the help topics [Installing from the SuiteApp Marketplace](#) and [SuiteApp Marketplace in NetSuite](#).

**Note:** After you install the SuiteApp, it creates a preferences record with a default template. You can also set up your traveler preferences for the work order instructions and traveler reports. Setting up traveler preferences helps you to generate the report in the right template format and directory location within the file cabinet. For more information, read [Updating Traveler Preferences](#).

If you want to reinstall Work Instructions and Traveler SuiteApp after uninstalling it, you must perform the following activities:

- You can take a backup of **Work Instructions & Traveler** folder from the File Cabinet page. For more information about how to download a file or folder from the File Cabinet, see the help topic [Downloading Files from the File Cabinet](#).
- Delete the **Work Instructions & Traveler** folder from the File Cabinet page, then reinstall the SuiteApp. For more about how to delete a file or folder from the File Cabinet, see the help topic [Deleting Folders from the File Cabinet](#).

## Prerequisites for Installing Work Instructions and Traveler SuiteApp

To use the Work Instructions and Traveler SuiteApp, enable the features in the following table. For more information, see the help topic [Enabling Features](#).

| Subtab            | Feature Name                          |
|-------------------|---------------------------------------|
| Items & Inventory | Work Orders                           |
|                   | Advanced Bill of Materials            |
|                   | Manufacturing Work In Progress        |
|                   | Manufacturing Routing and Work Center |
| Analytics         | SuiteAnalytics WorkBook               |
| SuiteCloud        | Advanced PDF/HTML Templates           |
|                   | Custom Records                        |
|                   | Client SuiteScript                    |
|                   | Server SuiteScript                    |

## Supported Browsers for Work Instructions and Traveler SuiteApp

The Work Instructions and Traveler SuiteApp supports the following browsers and operating systems:

| Browser       | Operating System |
|---------------|------------------|
| Google Chrome | Windows10        |

| Browser                         | Operating System                   |
|---------------------------------|------------------------------------|
|                                 | Mac OS 10.15 or newer              |
| Microsoft Edge (Chromium-based) | Windows10                          |
| Mozilla Firefox                 | Windows10<br>Mac OS 10.15 or newer |
| Safari                          | Mac OS 10.15 or newer              |

## Setting Up Work Instructions and Traveler SuiteApp

This section requires you to perform some setups after installing the Work Instructions and Traveler SuiteApp.

### Verifying the Custom Form of Work Instructions Subtab

The **WOIT Manufacturing Routing Form** custom entry form and the **WOIT\_CS\_WorkInstructions\_main.js** script must be connected to ensure that the **Work Instructions** subtab functions properly.

To verify the custom form of Work Instructions subtab:

- Check if the **WOIT Manufacturing Routing Form** is preferred. For more information, see [Enabling Work Instructions Subtab in Custom Entry Forms](#).
- Check if the **Sequence No** column in the **Work Instructions** subtab gets autopopulated. If not, see [Attaching a Script in WOIT Manufacturing Routing Form](#).

#### Enabling Work Instructions Subtab in Custom Entry Forms

**Work Instructions** subtab is a custom record in the custom routing form (**WOIT Manufacturing Routing Form**). This form is selected by default as preferred during the SuiteApp Installation.



**Note:** You must enable the **WOIT Manufacturing Routing Form** as a preferred form to ensure that the **Work Instructions** subtab works well.

#### To enable work instructions subtab:

1. Go to Customization > Forms > Entry Forms.
2. On the Custom Entry Forms page, search for **WOIT Manufacturing Routing Form**.
3. Check the **Preferred** box.

#### Attaching a Script in WOIT Manufacturing Routing Form

When you edit a manufacturing routing record, the **Sequence No** column in the **Work Instructions** subtab must be autopopulated, as you add a new line. If not, follow the below procedure to re-attach the script to WOIT Manufacturing Routing Form custom entry form.

#### To attach a script in WOIT Manufacturing Routing Form:

1. Go to Customization > Forms > Entry Forms.
2. Click **Edit** on the **WOIT Manufacturing Routing Form** custom entry form.
3. In the **Custom Code** subtab, select the empty value in the **Script File** list.



4. Click **Save**.
5. Edit the **WOIT Manufacturing Routing Form** custom entry form again (follow steps **1** and **2**), and select **WOIT\_CS\_WorkInstructions\_main.js** in **Script File** list.
6. Click **Save**.

You can now go to any manufacturing routing record and verify if the sequence number is autopopulated in the **Work Instructions** subtab.

## Creating or Customizing Roles and Permissions to Use Work Instructions and Traveler SuiteApp

If you're an administrator, you can create new roles or customize existing roles to use the Work Instructions and Traveler SuiteApp. For more information, see the help topics [Customizing or Creating NetSuite Roles](#) and [Assigning Roles to an Employee](#).

Refer to the following table for minimum access levels for different permissions required to use the Work Instructions and Traveler SuiteApp:

| Subtab         | Permission                          | Minimum Access Level | Purpose                                                                                                                                             |
|----------------|-------------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Transactions   | Work Orders                         | View                 | Required for: <ul style="list-style-type: none"> <li>■ <b>Print Traveler</b> button in Work Order page</li> <li>■ Generate Traveler page</li> </ul> |
|                | Find Transaction                    | View                 | Required for Generate Traveler page                                                                                                                 |
| Lists          | Documents and Files                 | View                 | Required for: <ul style="list-style-type: none"> <li>■ <b>Print Traveler</b> button in Work Order page</li> <li>■ Traveler Preferences</li> </ul>   |
|                | Items                               | View                 | Required for Generate Traveler page                                                                                                                 |
|                | Locations                           | View                 | Required for Generate Traveler page                                                                                                                 |
|                | Manufacturing Routing               | View                 | Required for <b>Work Instructions</b> subtab                                                                                                        |
|                | Perform Search                      | View                 | Required for <b>Work Order Saved Search</b> field in Generate Traveler page. If you don't have this permission, that field will be disabled.        |
|                | Custom Record Entries               | Edit                 | Required for <b>Work Instructions</b> subtab                                                                                                        |
| Setup          | SuiteScript                         | View                 | Required for all pages                                                                                                                              |
| Custom Records | Traveler Preferences                | View                 | Required for all pages                                                                                                                              |
|                | Traveler Template                   | View                 | Required for: <ul style="list-style-type: none"> <li>■ Generate Traveler page</li> <li>■ Traveler Template</li> </ul>                               |
|                | Additional Work Instructions Subtab | Full                 | Required for <b>Additional Work Instructions</b> subtab in work order record                                                                        |

You can create new roles or customize existing roles using the Administrator role to use the SuiteApp. For more information about minimum permissions required, see the help topic [Customizing and Creating Roles](#).

## Updating Traveler Preferences

The traveler preferences enable you to print traveler reports in a predefined template to the desired directory.

**Note:** You can only create one preferences record. It is automatically created by the SuiteApp during the installation. However, you can modify the preferences as per business needs.

### To edit a preferences record:

1. Go to Setup > Manufacturing > Traveler Preferences.
2. Click **Edit** on the preferences record.  
The Traveler Preferences page opens.
3. From the **Template** list, select the desired traveler template for the work order instructions and traveler report.

**Note:** When you install the SuiteApp, a default template appears in the field. You can also make your own templates. For more information, read [Creating Custom Template for Traveler Report](#).

4. In the **Report Directory** field, select the directory path for the selected custom template.  
For the default template, the report directory populates with a default path.
5. (Optional) Check the **Use Advanced Manufacturing** box to use the Advanced Manufacturing SuiteApp. Clearing the box will enable the Work Instructions and Traveler SuiteApp.

This action will enable:

- The **Work Instructions** subtab in the manufacturing routing record.
- The **Additional Work Instructions** subtab in the work order record.
- The **Print Traveler** button in the work order record.

**Note:** Follow the below points for **Use Advanced Manufacturing** box.

- **Use Advanced Manufacturing** box is checked by default for Advanced Manufacturing customers. So, the **Work Instructions** subtab is hidden in the routing record. However, if you clear the box, you'll see the subtab with any work instructions you've added to the routing record.
- When you enable the **UseAdvanced Manufacturing** box and save the preferences record, all the work instructions associated with the manufacturing routing will be deleted.

## Creating Custom Template for Traveler Report

You can create a custom template for your traveler report.

### To create a custom template for traveler report:

1. Go to Documents > Files > File Cabinet.

2. Create a folder to store the custom template. For more information, read the help topic [Creating File Cabinet Folders](#).
3. Go to the **SuiteApps** folder on the left and select the **com.netsuite.woinstructionsandtraveler** folder.
4. Then go to the **Templates** folder and click **Download** to download the default template present in each of the XML files.
5. Make the necessary customizations in the XML files and upload in the folder you created (as per step 2). For more information, read the help topic [Uploading Files to the File Cabinet](#).

 **Note:** Do not rename the XML files.

6. Go to Setup > Manufacturing > Traveler Template to create your custom template.
7. Click on the **New Traveler Template** button.  
The Traveler Template page opens.
8. In the **Name** field, enter the name of the template.
9. In the **Folder Path** field, enter the path of the file cabinet folder that contains the XML files for the custom template.

 **Note:** The path must be separated by (/).

10. In the **Transaction Dataset ID** field, enter the routing dataset ID to pull data into the custom template. The field is populated with a default dataset; however, you can use your own dataset.

 **Note:** You can create your own dataset or use an existing dataset (you cannot modify the existing dataset). For more information, read the help topic [Defining a Dataset](#).  
You must share the newly created dataset. The dataset type should be **Transaction**. For more information, read the help topic [Accessing and Sharing Workbooks and Datasets](#).

11. (Optional) Check the **Inactive** box if you want to disable a template.
12. Click **Save**.

## Providing Work Instructions Subtab in Manufacturing Routing Record

As part of the Work Instructions and Traveler SuiteApp, the **Work Instructions** subtab is enabled in the manufacturing record. You can add work instructions for every operation, and include any related images or documents. You can print work instructions and associated images when you generate the traveler report. These reports help shop floor teams to perform the operation of the work order well.

### To define work instructions:

1. Go to Lists > Supply Chain > Manufacturing Routing.  
The Manufacturing Routing page opens.
2. Click **Edit** on a manufacturing routing record.

 **Note:** The subtab only shows up when you're editing the routing record.

3. Check the manufacturing routing record to ensure all the required information in the **Routing Steps** subtab is available. For more information, read [Creating a Manufacturing Routing](#).
4. In the **Work Instructions** subtab:

1. From the **Operation Name** list, select the manufacturing routing operation.

This selection automatically populates the **Operation Sequence**.

 **Note:** The information in the fields is retrieved from the **Routing Steps** subtab

2. In the **Process Notes** field, provide instructions for every operation step to manufacture an assembly item based on manufacturing routing.
3. In the **Work Instructions** column, enter the instructions as per the sequence number to complete the operation.
4. In the **URL** column, provide a URL link for the operator's reference. You can enter a URL that contains audio, video, or image.
5. Check the box in the **Print Image** column to print the image (if URL provided in step 4 contains image) in the traveler report.
6. (Optional) Check the box in the **Inactive** column if you don't want to include a sequence.
7. Click **Add**.
8. Repeat steps 3 to 7 to add multiple steps.
9. Click **Save**.

 **Note:** There are certain limitations for the images to be printed in the traveler report.

- You can upload an image URL in two ways:
  - Upload the image to the file cabinet and add that file URL in the URL column.
  - Copy the image address from any website (right click on image's copy image address).
    - You must copy the proper address. When you click on the URL link, it should directly open the image.
    - If a URL link has multiple images, the images will not display.
- Image type must be .png or .jpg format.
- The SuiteApp supports 24-bit color depth images in the PDF report while rendering the images from the file cabinet. NetSuite does not support images with other color depth.
- When you upload images to the file cabinet, check the **Available Without Login** box to display the images in the report.

## Providing Additional Work Instructions in Work Order

In certain business cases, you can provide specific work instructions related to a work order. You can add these instructions in the **Additional Work Instructions** subtab of the work order record. However, to enable this subtab, check the **WIP** box under the Primary Information section.

### To define additional work instructions:

1. In the **Additional Work Instructions** subtab:
  1. In the **Work Instructions** column, enter the instructions as per the sequence number to perform the steps to complete an operation.

2. In the **URL** column, provide a URL link for the operator's reference. You can enter a URL that contains audio, video, or image.
3. Check the box in the **Print Image** column to print the image (if URL provided in step 2 contains image) on the traveler report.
4. (Optional) Check the box in the **Inactive** column if you don't want to use a particular sequence.
5. Click **Add**.
6. Repeat steps 1 to 4 to add multiple steps.
7. Click **Save**.



**Note:** There are certain limitations for the images to be printed in the traveler report.

- You can upload an image URL in two ways:
  - Upload the image to the file cabinet and add that file URL in the URL column.
  - Copy the image address from any website (right click on image's copy image address).
    - You must copy the proper address. When you click on the URL link, it should directly open the image.
    - If a URL link has multiple images, the images will not display.
- Image type must be .png or .jpg format.
- The SuiteApp supports 24-bit color depth images in the PDF report while rendering the images from the file cabinet. NetSuite does not support images with other color depth.
- While uploading images in the file cabinet, check the **Available Without Login** box to display the images in the report.



**Note:** Following are the points applicable to the **Additional Work Instructions** subtab:

- To access the **Additional Work Instructions** subtab, check the permissions provided for the **Additional Work Instructions Subtab** custom record. For more information, read [Creating or Customizing Roles and Permissions to Use Work Instructions and Traveler SuiteApp](#).
- The extra work instructions and their images will be included when you print the Work Instructions and Traveler report.

## Import Work Instructions

You can import the work instructions by mapping the fields in the manufacturing routing record of the Work Instructions and Traveler SuiteApp with the mappings in the CSV file. For more information about importing CSV files, read the help topic [Importing CSV Files with the Import Assistant](#).

While importing the CSV file, you must select the following options on the Import Assistant- Scan & Upload CSV File page:

- In the **Import Type** dropdown list, select **Custom Records**.
- In the **Record Type** dropdown list, select **WOIT Work Instructions Subtab**.

**Note:** You must check the **Run Server SuiteScript and Trigger Workflows** box on the Import Options page of the Import Assistant. For more information, read the help topic [Set Advanced CSV Import Options](#).

Refer the following table to map the fields.

| CSV Column Names                                                                                        | Work Instructions and Traveler SuiteApp Fields |
|---------------------------------------------------------------------------------------------------------|------------------------------------------------|
| BOM (Name of the Bill of Materials)                                                                     | Bill of Materials (BOM)                        |
| Inactive                                                                                                | Inactive                                       |
| Routing (Name of the routing record)                                                                    | Manufacturing Routing                          |
| Operation Seq (Operation sequence column value under <b>Routing Steps</b> subtab on the routing record) | Operation ID (Required)                        |
| Print Image                                                                                             | Print Image                                    |
| Process Notes                                                                                           | Process Notes                                  |
| Sequence (Sequence number of work instructions)                                                         | Sequence Number (Required)                     |
| <p><b>Note:</b> The sequence numbering for list of work instructions must be a positive number.</p>     |                                                |
| Work Instructions                                                                                       | Work Instructions (Required)                   |
| URL                                                                                                     | URL                                            |

## Generating Traveler Reports

The Generate Traveler page generates the work instructions and traveler report. Use the fields under the **Filter by** section to narrow down which work orders you want in your report.

### To generate a traveler report

1. From the **Location** list, select a location for which you want to filter the work orders.
2. (Optional) To filter down the work orders, provide values for the required fields:
  - From the **Assembly Item** list, select the assembly item for which you want to filter the work orders.

**Note:** The SuiteApp does not work with item names that contain:

- Double quotation marks- You can use two single quotation marks instead of double quotation marks.
- Accented characters- You can use usual English alphabets instead of accented characters.

- From the **Status** list, select the status for which you want to filter the work orders.

- In the **Work Order** field, enter the work order number.

**Note:** Entering a work order number in this field will disable other fields except **Select All Orders**.

- In the **Start Date** field, enter the start date.
- In the **End Date** field, enter the end date.
- From the **Template** list, select a template for the report. By default, this uses the template from Traveler Preferences page, but you can choose a different one if you want.
- In the **Work Order Saved Search** list, select a saved search to filter your work orders.

**Note:** Follow the below points for **Work Order Saved Search** list.

- Selecting a saved search in this field will disable other fields, except **Select All Orders**.
- The SuiteApp provides a work order saved search **Traveler Work Order Search** by default.
- If the saved search criteria type is not of work order, then the saved search results will be blank.

- Check the **Include Printed Orders** box to include all work orders including the work orders that have been printed.

3. Click **Search**.

The **Filter by** section collapses automatically, and you can now view the filtered work orders and their associated details in the **Work Orders** sublist based on the selection in the above fields.

- 4. (Optional) Check the **Select All Orders** box to include all work orders in the **Work Orders** sublist.
- 5. Click **Generate & Print** to generate and print the report for the selected work orders.

After the report generation process is completed, it'll open in a new browser tab with the selected work orders from the **Work Orders** sublist.

## Viewing Report of Each Work Order

You can view the traveler report for each work order individually using the below steps.

### To view report of each work order

1. Go to Documents > Files > File Cabinet.
2. Go to the folder that matches the location listed in the **Report Directory** field of your traveler preferences.

## Printing Traveler Report in Work Order Transaction

You can also generate a work instructions and traveler report from a work order.

### To generate a report from a work order

1. Go to Transactions > Manufacturing > Enter Work Orders > List.
2. On the Work Order page, click **View** on a work order.
3. Click **Print Traveler**.

After the report generation process is complete, the report opens in a new browser tab. For more information, read [Viewing Report of Each Work Order](#).



**Note:** The **WOIT MR Traveler Report** script generates the PDFs for work orders in release status and not for the work orders that are printed already. This MR script is unscheduled by default, and you can schedule it.



# Manufacturing Scheduler

The Manufacturing Scheduler SuiteApp offers a graphical interface, enabling you to efficiently oversee and manage operational tasks assigned to a work center. This SuiteApp facilitates production planners and operations managers with real-time visibility and control over work center activities. On the Manufacturing Scheduler page, each work order displays the operations defined in the manufacturing routing, which can be directly associated with a designated work center. Consequently, all manufacturing operations for a given work center are grouped together in its section, making it easier to monitor and manage.

The SuiteApp enables quick identification and resolution of any operational issues, ensuring efficient task management. The SuiteApp makes the whole process easier, so you can address any issues promptly.

The SuiteApp enables you to:

- Identify work centers that are overloaded or underloaded.
- Drag and drop to reassign or reschedule operations from an overloaded work center to a different day or work center.
- Update the operation details of a specific work order on the Manufacturing Operation Task Details page.

## Installing Manufacturing Scheduler

The Manufacturing Scheduler SuiteApp requires a NetSuite OneWorld account.

Only users with the Administrator role can install the SuiteApp. For more information about installing a SuiteApp, see the help topics [Installing from the SuiteApp Marketplace](#) and [SuiteApp Marketplace in NetSuite](#).

## Prerequisites for Installing Manufacturing Scheduler

To use the Manufacturing Scheduler SuiteApp, enable the features in the following table. For more information, see the help topic [Enabling Features](#).

| Subtab            | Feature Name                          |
|-------------------|---------------------------------------|
| Items & Inventory | Manufacturing Work In Process         |
|                   | Manufacturing Routing and Work Center |



**Note:** You cannot install the SuiteApp if any of the following roles are inactive.

- CEO
- CFO
- Sales Vice President
- Accountant
- Accountant (Reviewer)
- Bookkeeper
- Warehouse Manager

## Creating or Customizing Roles and Permissions to Use Manufacturing Scheduler

If you are an administrator, you can create new roles or customize roles for using the Manufacturing Scheduler SuiteApp. For more information, see the help topics [Assigning Roles to an Employee](#) and [Customizing or Creating NetSuite Roles](#).

Refer to the following table for minimum access levels of permissions required for SuiteApp deployments. These minimum permissions must be updated for the existing role or when creating a new role.

| Subtab                  | Permission          | Level |
|-------------------------|---------------------|-------|
| <b>Find Transaction</b> | Work Order          | View  |
| <b>Transactions</b>     | Work Order          | Edit  |
| <b>Lists</b>            | CRM Group           | View  |
| <b>Lists</b>            | Work Calendar       | View  |
| <b>Lists</b>            | Locations           | View  |
| <b>Lists</b>            | Documents and Files | View  |
| <b>Lists</b>            | Subsidiaries        | View  |
| <b>Custom Record</b>    | MTS Task View       | Full  |
| <b>Custom Record</b>    | MTS User Display    | Full  |
| <b>Setup</b>            | Set Up Company      | View  |

By default, the following standard roles are granted full permissions to the two custom records that are listed in the table:

- CEO
- CFO
- Sales Vice President
- Accountant

- Accountant (Reviewer)
- Bookkeeper
- Warehouse Manager

In addition to the two custom records, ensure you assign the other permissions to standard or custom roles. To edit or customize a role, go to Setup > Users/Roles > Manage Roles. On the Manage Roles page, click the Role name to open the record and verify that the permissions are set to the required level. For more information about editing or customizing roles, see the help topic [Customizing or Creating NetSuite Roles](#).

Based on your role, you can access the SuiteApp by following the appropriate path:

- Warehouse Manager  
Inventory > Manufacturing > Manufacturing Scheduler
- Administrator  
Transaction > Manufacturing > Manufacturing Scheduler
- Executive or Accounting  
Financial > Manufacturing > Manufacturing Scheduler

## Supported Browsers for Manufacturing Scheduler

The Manufacturing Scheduler SuiteApp supports the following browsers and operating systems:

| Browser                         | Operating System                      |
|---------------------------------|---------------------------------------|
| Google Chrome                   | Windows 10                            |
|                                 | Mac OS 10.15 or newer                 |
| Microsoft Edge (Chromium-based) | Windows 10 or newer                   |
| Mozilla Firefox                 | Windows 10                            |
|                                 | Mac OS 10.15 or newer                 |
| Safari                          | Two most recent stable major versions |

## Manufacturing Scheduler Dashboard

The Manufacturing Scheduler page shows the work order data associated with a work center. You can watch the current status of the manufacturing operations that are mapped to a work order. Each operation status is shown with color coding. The chart includes a data grid with task bars showing the planned work schedule planned for each operation.

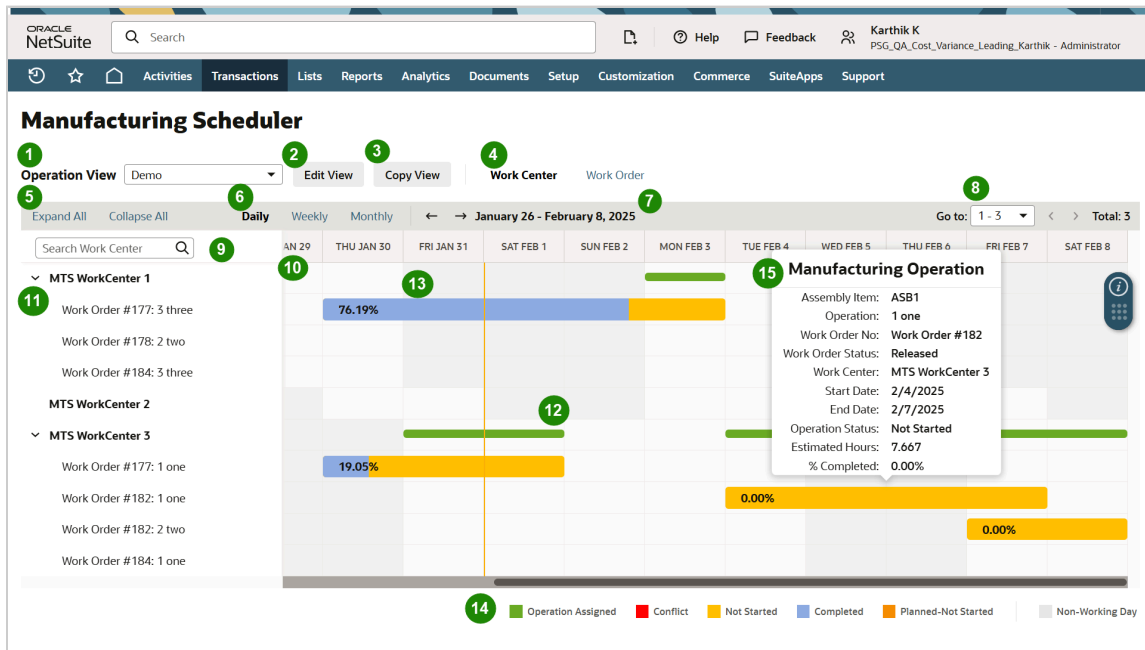
On the Manufacturing Scheduler page, you can:

- Create and manage operation views and load the relevant data on the grid.
- View work order or work center data based on the time period. The time period is categorized to Daily, Weekly, and Monthly.
- Search for a specific work center to see its manufacturing operations.
- View the list of work centers or work orders through available work views.
- Determine the schedule, duration and status of a work order from its task bar.
- Use the legend at the bottom right to understand the color-coded status of summary and task bars.

View the following topics to understand the usage of the SuiteApp:

- [Operation View](#)
- [Time Periods](#)
- [Manage Work Center and Work Order Operations](#)
- [Searching for Work Centers](#)
- [Understanding Legend Colors](#)

The following graphic provides an understanding to the sections and tools of the SuiteApp:



| No. | Control Name                                                  | Description                                                                                                                                           |
|-----|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Operation View Filter                                         | Enables selection of the default or custom view                                                                                                       |
| 2   | Add View                                                      | Add a new custom view<br><br>This option appears when you are on default view. After you created a custom view, you will see <b>Edit View</b> option. |
| 3   | Copy View                                                     | Copy an existing custom view attributes that can be used to create a new custom view                                                                  |
| 4   | Work View options                                             | Switch between the work view: <b>Work Center</b> or <b>Work Order</b>                                                                                 |
| 5   | Expand All/Collapse All                                       | Shows all or hides all operations within a work center or work order                                                                                  |
| 6   | Time Period                                                   | Change the time period to see the relevant data on the chart: <b>Daily</b> , <b>Weekly</b> , and <b>Monthly</b>                                       |
| 7   | Date Range                                                    | Shows the start and end date of a time period                                                                                                         |
|     | Date Range icons (Forward arrow and backward pointing arrows) | Moves the start date from the previous or next date                                                                                                   |
| 8   | Pagination                                                    | Select the preferred number of work orders or work centers to show on the left pane of the Manufacturing Scheduler page                               |

| No. | Control Name                                  | Description                                                                                     |
|-----|-----------------------------------------------|-------------------------------------------------------------------------------------------------|
|     | Page Range icons (Forward and backward icons) | Shows the previous or next page                                                                 |
|     | Page Count                                    | Shows the total number of pages                                                                 |
| 9   | Search Work Center                            | Enables searching of a work center                                                              |
| 10  | Chart Header                                  | Shows the specific day, week, or month within the current date range                            |
| 11  | Manufacturing Operation                       | Shows the list of operations within a work center or work order                                 |
| 12  | Summary Bar                                   | Shows all operations on the chart assigned to a work center or work order                       |
| 13  | Task Bar                                      | Shows a specific operation on the chart                                                         |
| 14  | Legend                                        | Provides a short description of each color on the summary and task bar                          |
| 15  | Operation Details                             | Shows the operation details such as assembly item, start and end dates, and operation sequence. |

## Operation View

The default operation view displays all work orders with the status as **In Process** or **Released**. It also shows the work centers linked to the subsidiary of the currently logged-in user. Use the default view or create a custom view to handle specific set of operations. When creating a custom view, set your preferences for any of the following filters:

- Production Date From and To
- Subsidiary
- Location
- Work Center
- Work Order Status

**Note:** The SuiteApp can display a maximum of 2000 operations per page. Creating a custom view helps you limit the number of operations shown on the chart.

For example, if there are 20 work centers with a total of 2500 work orders, the Work Center view lists the 20 work centers in the left pane. But only the first 2000 work orders are shown under their corresponding work centers. If the 2000 work orders reach to work center 18, the work centers 19 and 20 are listed but no work orders are shown under them.

- On the Operation View Filter bar, the **Add View** button appears only on the **Default** view.
- The **Edit View** and **Copy View** buttons appear only on a custom view.

## Adding Views

You can create multiple custom views to handle manufacturing operations on the chart.

### To add a view:

1. On the Manufacturing Scheduler page, click **Add View**. The **Add View** button appears only when you select **Default** as **Operation View** type.
2. In the **Add View** popup window, fill the following fields:

| Field                | Description                                                                                     |
|----------------------|-------------------------------------------------------------------------------------------------|
| Operation View       | Add a name for the custom view.                                                                 |
| Production Date From | Select the production start date.                                                               |
| Production Date To   | Select the production end date.                                                                 |
| Subsidiary           | Select the subsidiary to view related work order and work center details.                       |
| Location             | Select a location.                                                                              |
| Work Center          | Select the work centers.                                                                        |
| Work Order Status    | Select the status of work order. You can select multiple statuses to view relevant work orders. |

3. Click **Save**. You can see the custom work order on the **Operation View** filter.
4. To make changes to a custom view, select the name of the view and then click **Edit View**.
5. In the **Edit View** popup window, you can change or update the required preferences.
6. To create another view type with similar preferences or slight variations, use the **Copy View** option.

For custom views with a specified date range for the production start date, the start and end of the range appear with vertical bars. All production start dates or end dates that fall within the range appear inside the bars.

## Time Periods

On the Manufacturing Scheduler page, you can view manufacturing operations for different time periods, such as Daily, Weekly, and Monthly. Click any of the period options to change the current chart view. Changing the time period adjusts the entries on the date header above the chart.

- **Daily:** Shows the day, month, and date on the date header. A total of 14 days appears at a time.  
On the daily chart, task bars cover the entire day, regardless of the number of work hours.
- **Weekly:** Shows the month and date of the first day of the week on the date header. A total of 12 weeks appear at a time.
- **Monthly:** Shows the month and year on the date header. A total of 6 months appear at a time.

Based on your selected time period, the date range automatically adjusts. For daily and weekly, it shows the start and end dates. For monthly, it shows the starting and ending month with the year.

Use the forward and backward arrow icons (beside the date range) to move the start and end date or month before or after the current selection, respectively. This automatically adjusts the date header and chart view.

## Manage Work Center and Work Order Operations

On the Manufacturing Scheduler page, you can view the list of work centers or work orders through work views. Based on the selected work view, the left pane shows the operations mapped under that view. By

default, the left pane displays the Work Center view where operation tasks appear under their assigned work centers. You can switch to the Work Order view to see all operations that belong to each work order.

In Work Order view, you can also see each work order's assembly item and the assigned work center for every operation. On the chart, the summary bar across each work order represents all operations it includes.



- You cannot reassign and reschedule operations by dragging and dropping of the task bars. You must switch back to Work Center view to update the operations directly on the chart.
- The SuiteApp can display a maximum of 20 work centers or work orders per page.

## Searching for Work Centers

In the Work Center view, use the search field to search for a work center that does not appear on the current page.

### To search for a work center:

1. On the Manufacturing Scheduler page > Work Center view, in the search field, enter minimum of three characters of a work center name.
2. Click the search icon to start your search. You'll only see the work center found and its assigned work orders on the chart.
3. To return to the initial list of work centers, remove search criteria from the search field, and click the search icon.

Use the **Expand All** or **Collapse All** links to show or hide the work orders of all work centers at the same time. Work centers expand or collapse on all pages, and your selection stays the same as you switch pages.

When work centers or work orders are collapsed, the chart displays only summary bars. Use the summary bars to compare work centers and spot overloaded or underloaded resources so you can make adjustments.

## Understanding Legend Colors

The color of the task bar indicates the current status of an operation as seen in its operation task record. The bar updates automatically to reflect any status changes based on the work order and task statuses. The legend bar provides the color code indication for the different type of statuses. The following table describes which color represent which operation status.

| Color                             | Definition                                                                                                                         |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Green</b> (Operation Assigned) | Summary bar color that indicates an assigned operation.                                                                            |
| <b>Red</b> (Conflict)             | Summary bar color that indicates one or more operations with overlapping schedules.                                                |
| <b>Yellow</b> (Not Started)       | Task bar color that indicates a operation has not started, but the work order status can be <b>In Process</b> or <b>Released</b> . |
| <b>Blue</b> (Completed)           | Task bar color that indicates a operation's percentage of completion.                                                              |

| Color                               | Definition                                                                                                   |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Orange</b> (Planned-Not Started) | Task bar color that indicates a operation has not started, but the work order status can be <b>Planned</b> . |
| <b>Gray</b> (Non-Working Day)       | Column color that indicates a non-working day.                                                               |

## Operation Conflicts

In the Work Center view, the summary bar displays all the operations assigned to a work center. The summary bar can display an operation conflict in red, if two or more overlapping operations exceed the total working hours per day. If you see a red summary bar, you can review the work orders, reschedule them, or move them to another work center to balance the workload.

On the chart, you can change an operation's schedule or reassign it to a different work center to resolve conflicts.

## Manage Operation Conflicts

By looking at the summary bar of a work center, you can check for overlapping operations, indicating that a work center is over-utilized. In this case, you can schedule a work order to a later date, or assign it to another resource. You can perform any of the following changes to an operation:

- [Reassigning Operations](#)
- [Forward Scheduling Operation](#)
- [Backward Scheduling Operation](#)
- [Updating Manufacturing Operation Records](#)

**Note:** You can update operations directly on the chart only when you are in the Work Center view. Use the operation view filters to focus on the specific work centers where you want to make changes.

## Reassigning Operations

To reassign an operation, drag the task bar upward or downward within the same column, toward the new work center. The reassigned operation retains all its details, except for the new work center number.

**Note:** You cannot reassign an operation to an inactive work center, or one that belongs to a different subsidiary.

## Forward Scheduling Operation

To perform forward scheduling on the chart, change the start date of the first operation in the work order. This applies to work orders set to the **Forward** scheduling method. For more information about scheduling methods, see [Production Scheduling Methods Overview](#).

To initiate forward scheduling, switch your chart to the Work Center view. Identify the first operation within a work order by locating the sequence number next to the work order number. For rescheduling, drag the task bar horizontally to the desired date, either left or right from the original start date, and release it when the new start date appears.



**Note:** Select the appropriate time period option to locate a new date on a different week or month. Use the previous and next date range arrows to adjust the dates in the view.

After you move the date of the first operation, NetSuite automatically adjusts the schedules of the succeeding operations. Switch to the Work Order view to review the new schedule of operations within the work order. NetSuite applies the changes made on the chart to the work order and operation task records.

## Backward Scheduling Operation

Backward scheduling on the chart applies to work orders set to the **Backward** scheduling method. To perform backward scheduling, move the end date of the last operation in the work order. Changes to the schedule appear on the corresponding work order and operation task records. For more information about scheduling methods, see [Production Scheduling Methods Overview](#).

To start backward scheduling, set your chart to the Work Center view by clicking its link located above the chart header. On the left pane, look for the last operation of the work order to be rescheduled. On the chart, drag and drop the task bar of the last operation to the new date.

- Drag to the right of the current date to move the schedule to a later date.
- Drag to the left of the current date to move the schedule to an earlier date.

When the last operation task bar has a new date, NetSuite automatically adjusts the preceding operations to accommodate the new schedule. You can review the new operation schedules by switching to the Work Order view.

## Updating Manufacturing Operation Records

You can update a manufacturing operation record, but after you change an operation to **In Progress** status, you cannot edit its details. To view the manufacturing record, double-click its corresponding task bar on the chart.

You can edit the following details on the manufacturing record:

- Setup Time
- Run Time
- Work Center
- Machine Resources
- Labor Resources

After you change a record, any adjustments made to the time or date, automatically appear on any dependent fields. For more information about editing the operation details, see [Manufacturing Operation Task](#).

# Outsourced Manufacturing

Outsourced Manufacturing helps customers manage their subcontracted manufacturing processes by purchasing outsourced assembly production from vendors. This feature enables NetSuite to automatically consume production components and transfer them to a specified location after production. Automating Outsourced Manufacturing synchronizes NetSuite procurement and production transactions.

Outsourced Manufacturing automates the following features:

- Creating subcontracting orders for assemblies
- Managing ordering components for subcontracted production
- Simulating the manufacturing sub-contractor production
- Transferring half finished assemblies to other sub-contractors
- Drop-shipping the assembly to the customer, or shipping it to the brand-owner's warehouses

## Setting up Outsourced Manufacturing

Before the Outsourced Manufacturing feature can be enabled, an administrator must enable prerequisite features, if they are not already enabled. After the prerequisite features are enabled, an administrator can then enable Outsource Manufacturing.

### To enable Outsourced Manufacturing:

1. Go to Setup > Company > Enable Features (Administrator)
2. Ensure that the following features are enabled:
  - **Purchase Orders - Transactions** subtab, Basic Features section
  - **Advanced Receiving - Transactions** subtab, Shipping & Receiving section
  - **Multi-Location Inventory - Items & Inventory** subtab, Inventory section
  - **Assembly Items - Items & Inventory** subtab, Inventory section
  - **Advanced Bill of Materials - Items & Inventory** subtab, Inventory section
  - **Work Orders - Items & Inventory** subtab, Inventory section
3. Click **Save**.
4. Go back to Setup > Company > Enable Features (Administrator)
5. Click the **Items & Inventory** subtab.
6. In the **Inventory** section, check the **Outsourced Manufacturing** box.
7. Click **Save**.

Now you can define an outsourcing location for the vendor who will outsource manufacturing for you.

## Configuring Outsourced Manufacturing

After Outsourced Manufacturing is enabled, you must define an outsourcing location for the vendor who will outsource manufacturing for you.

### To define an outsourced manufacturing location:

1. Go to Lists > Relationships > Vendors.



**Important:** You must have locations defined before you can configure your outsourced manufacturing location. For more information, see the help topic [Creating Locations](#).

2. On the Vendors page, click **Edit** next to the vendor who will outsource manufacturing for you.
3. Click the **Outsourced Manufacturing** subtab.
4. Beside the **Manufacturing Locations** field, click the arrow icon.
5. In the **Choose Manufacturing Location** list, select the location or locations, for the outsourcing vendor.



**Note:** Each location can be used by one vendor, but a vendor can use multiple locations.

6. Click **Done**.
7. Click **Save**.

## Configuring Outsourced Assembly Bins for Locations

### To setup outsourced assembly for locations:

1. Go to Setup > Accounting > Accounting Preferences.
2. Click the Items/Transactions subtab.
3. Check the **Use Preferred Bin on Item Receipts** box.
4. Click **Save**.
5. Create a new **Locations** for each bin.  
To learn more, see [Configuring Outsourced Manufacturing](#).
6. Create an **Outsourced Vendor**.  
Add your new locations to this vendor.
7. Create an **Outsourcing Charge Item**.  
To learn more, see [Creating an Outsourced Manufacturing Item](#).
8. Create an **Inventory Item**.
  - a. On the **Purchasing/Inventory** subtab, check the **Use Bins** box.
  - b. Click **Save**.
 To learn more, see the help topic [Creating Item Records](#).
9. Create an **Assembly Item**.
  - a. On the **Purchasing/Inventory** subtab, check the **Use Bins** box.
  - b. In the **Bin Numbers** section, select a **Location**.
  - c. Add the new **Bin Number**.
  - d. Click **Save**.
 To learn more, see [Assembly Items](#).
10. Create a **Bill of Materials (BOM)**.  
To learn more, see [Create a BOM](#).
  - a. Assign the BOM to the assembly item.

To learn more, see [Linking a BOM to an Assembly](#).

11. To ensure a bin is available on the Item Receipt Component line, adjust inventory for the new inventory item.

To learn more, see the help topic [Entering an Inventory Adjustment](#).

## Creating an Outsourced Manufacturing Item

Outsourced manufacturing items are the goods and services you sell to customers that have been produced in cooperation with your outsourcing vendor. You need to define the charge item used in the purchasing transaction.

### To create an outsourcing charge item:

1. Go to Lists > Accounting > Items > New.
2. On the New Item page, click **Other Charge for Purchase** or **Service For Purchase**.
3. Enter a unique **Item Name/Number**.  
For example, Bike Production charge.
4. Complete the other fields in the page.
5. On the **Purchasing** subtab, enter a **Purchase Price**.
6. In the **Cost Category** field, select **Outsourcing Charge**.  
NetSuite recognizes this as an outsourcing charge item.



**Note:** When you select **Outsourcing Charge**, the **Can Be Fulfilled/Received** box on the **Preferences** subtab is checked and cannot be cleared.

7. In the Vendors section, select a **Vendor**.
8. To designate this vendor as your default vendor, check the **Preferred** box.
9. On the **Accounting** subtab, select a **Tax Schedule**.
10. Click **Save**.

After you create the outsourcing charge item, you must perform the following tasks:

- Create a **Bill of Materials (BOM)**. For example, Outsourced BOM. For more information, see [Create a BOM](#).
- Create a **BOM Revision**. For more information, see [Creating BOM Revisions](#).
  1. On the Bill of Materials Revision page, in the **Item** list, select the components you keep in inventory and will provide to the vendor. For example, bike seats and bicycle wheels.
  2. Select the **Outsourcing Charge** item.
  3. Enter the **BOM Quantity** per assembly. For example, 1 seat, 2 wheels, and 1 charge item.
  4. Click **Save**.
- Create an **Assembly Item**. For more information, see the help topic [Creating Item Records](#).
  1. On the **Manufacturing** subtab, select the **Bill of Materials**.
  2. Optionally, click the **Master Default** box to designate this as your default BOM for this assembly.
  3. Click **Save**.

# Outsourcing Production

NetSuite enables customers to outsource their manufacturing efforts to vendors. Customers can outsource from either a purchase order or from a work order.

## To outsource production from a purchase order:

1. Go to Transactions > Purchases > Enter Purchase Order.
  2. On the Purchase Order page, in the **Custom Form** field, select the **Standard Outsourced Purchase Order**.
  3. Select the outsourcing **Vendor**.  
The **Subsidiary** field is populated.
  4. Complete the remaining fields.  
For more information, see the help topic [Entering a Purchase Order](#).
  5. On the **Items** subtab, click **Items**.
  6. From the **Assembly** list, select the outsource item.
  7. Select the outsourcing **Location**.
  8. In the **Item** field, select the item charge. For example, Bike Production Charge.
  9. Enter the Item **Quantity**.  
NetSuite automatically completes the **Rate**, **Amount**, and **Production Dates** fields, but you can modify the values.
  10. Select the **Bill of Materials**.
  11. Select the **Bill of Materials Revision**.
  12. Click **Add**.  
Continue to add lines based on what you are building in-house, and what you are outsourcing.
  13. To automatically create work order, check the **Create WO** box.
  14. Click **Save**.  
After you save the purchase order, a link to the new work order appears on the **Items** subtab.
  15. Click the **Work Order** number link to display the work orders details.  
The **Outsourcing** subtab displays the following outsourcing information:
    - A checked outsourced box denotes that this order is outsourced
    - The vendor name
    - Outsourcing charge type
    - A link to the original purchase order
  16. You can also click the **Create WO** link to let NetSuite automatically create a work order.
  17. On Purchase Order page, click **Receive**.  
On the Item Receipt page, you can receive all the items, or some of the items.
  18. On the **Items** subtab, in the **To Location** field, select where you want to receive the final product.
-  **Note:** NetSuite automatically creates an Inventory Transfer and not a Transfer Order.
19. Click **Save**.  
NetSuite creates assembly build transactions for the received number of items.

Click an **Assembly Build** link to review assembly build details.

## The Assembly Build Helper

NetSuite enables you to use the Outsourced Manufacturing feature with the Assembly Build Helper to override existing production data.

### To change data using the Assembly Build Helper:

1. On the Purchase Order page, on the **Items** subtab, click the **Items** subtab.
2. Click the link in the **Assembly Build Helper** column.  
The Assembly Build Helper page displays the assembly items with editable fields. You can change the field values, as needed. For example, the vendor built 5 bicycle seats in production, therefore, twenty-seats were consumed. The **Quantity** field shows that twenty seats were consumed in the assembly build. You can change that number to twenty-five.
3. In the **Quantity** field, accept the default number, or enter a new number.  
If you use serial lot numbers for consumed or produced final production, you can edit these field values on the Assembly Build Helper page.  
When you override a purchase order rate for a line, NetSuite automatically uses the new rate in the final assembly charge.
4. Click **Save**.

## Outsourcing Production from a Work Order

Customers who rely on supply planning, or are in a manufacturing environment usually create outsourced items from work orders.

### To outsource production from a work order:

1. Go to Transactions > Manufacturing > Enter Work Orders.
2. Complete the required fields on the Work Order page, and other fields, as needed. For more information, see [Entering an Individual Work Order](#).
3. On the **Items** subtab, select the item to be outsourced.
4. Select an outsourcing location.
5. Click the **Outsourcing** subtab.  
Depending on the outsourcing BOM, NetSuite checks the **Outsourced** box and completes the following selections:
  - Outsourced Vendor
  - Outsourcing Charge
  - Linked Purchase Order
6. Click **Save**.

## Outsourced Manufacturing Consolidator

The Outsourced Manufacturing Consolidator aggregates outsourced work orders into purchase orders.

 **Note:** Before the Outsourced Manufacturing feature can be enabled, an administrator must enable prerequisite features. To learn more, see [Setting up Outsourced Manufacturing](#).

### To manage the outsourced manufacturing consolidator:

1. Go to Setup > Manufacturing > Outsourced Manufacturing Management.  
This page lets you to learn work order specific information, such as:
  - Last Successful Run
  - Next Scheduled Run
  - Pending Work Orders
2. To manually start the consolidation process for pending work orders, click **Run Consolidator Now**.
  - The Consolidator is scheduled to run every day at 2:00 a.m. in your organization's timezone. Currently, there is no option to set a custom schedule or to turn off the consolidator.
  - Outsourced work orders that are not linked to a purchase order in Firm status will be consolidated into purchase orders by Vendor and Subsidiary.
  - Purchase Order Lines for work orders will be ordered by the work order creation time.
  - Due to performance expectations, a single purchase order is limited to 30 work orders.
  - Purchase orders are created on the Standard Outsourced Purchase Order form. Currently, there is no option to use a different form. For more information, see [Outsourcing Production](#).

# SuiteAnalytics Manufacturing Workbook

SuiteAnalytics Workbook offers many workbook and dataset templates, each with predefined source data, criteria, pivot tables, and charts.

This chapter contains the information for the SuiteAnalytics Manufacturing workbook in NetSuite. For more information about SuiteAnalytics Workbook, see the help topic [Workbook and Dataset Templates](#).

- [Manufacturing Dataset](#)
- [Manufacturing Workbook](#)
- [Manufacturing Analytical Record Types](#)

The SuiteAnalytics Workbook user interface enables customers with little record schema and query language knowledge to create complex workbooks using drag-and-drop editing. New customization options provide ways to display your data using rich formatting. For more information, see the help topic [SuiteAnalytics Workbook Overview](#).

## Manufacturing Dataset

This Manufacturing dataset combines fields from the Manufacturing Transaction record type and one custom formula. This dataset enables you to analyze the status of bills received during the last month. It forms the source data for the [Manufacturing Workbook](#).

## Dataset Configuration

The Manufacturing dataset combines fields from one record type, one custom formula, and multiple criteria filters. To edit the dataset, see the help topic [Defining a Dataset](#).

| Root Record Type          | Joined Record Type | Custom Formula Field                                                                                                                                                                                                                                       | Data Grid                                                                                                                                                                                                                                                                                                                                              | Criteria Filters                                                                                                                                                                                                                                                                      |
|---------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Manufacturing Transaction | (none)             | <p>The following custom formula is included in the dataset:</p> <ul style="list-style-type: none"> <li>■ <b>cost/unit</b> - CASE WHEN {quantity} &gt; 0 THEN TO_NUMBER({transaction^transaction.transactionlines.foreignAmount})/{quantity} END</li> </ul> | <p>The following fields are included in the dataset.</p> <p>Manufacturing Transaction:</p> <ul style="list-style-type: none"> <li>■ Assembly</li> <li>■ Date</li> <li>■ End Date</li> <li>■ Location</li> <li>■ Memo</li> <li>■ Posting Period</li> <li>■ Quantity</li> <li>■ Start Date</li> <li>■ Transaction</li> <li>■ Transaction Type</li> </ul> | <p>The following criteria is used to filter the dataset:</p> <ul style="list-style-type: none"> <li>■ <b>Transaction Type</b> any of Assembly Build, Work Order Close</li> <li>■ <b>Transaction Line: Main Line</b> is true</li> <li>■ <b>Date</b> on or after 2 Years ago</li> </ul> |

## Manufacturing Workbook

NetSuite provides a Manufacturing Transaction data source, which introduces the analytics transactions concept for manufacturing-specific transactions. The following transaction record types are available in the Manufacturing Transaction data source:



- Transaction/Manufacturing Transaction – Transaction Header information
- Transaction Line/Manufacturing Component – Standalone lines related to transaction

For more information, see [Manufacturing Transaction](#) and [Creating a Manufacturing Workbook](#).

## Manufacturing Transaction

Manufacturing Transaction is a modified transaction record that improves manufacturing data searches for the following manufacturing transactions:

- Work Order
- Assembly Build
- Assembly Unbuild
- Issue Components
- Work Order Completion
- Work Order Completion with Backflush
- Work Order Close

These searches are filtered so that specified transaction types return results faster than generic transaction searches. The data set is optimized for manufacturing by making the assembly field available in the manufacturing transaction.

When you join a manufacturing component transaction line, the results return only components as displayed on the form with positive quantities. Joining generic transaction lines can cause the main line to behave as a component. Providing results with negative quantities (consumed during the production).

If you need a field from the larger generic transaction data set, you can join it to a Manufacturing Transaction/Component. This join provides an expanded list of available fields.

## Creating a Manufacturing Workbook

The following procedure provides details for creating a manufacturing workbook.

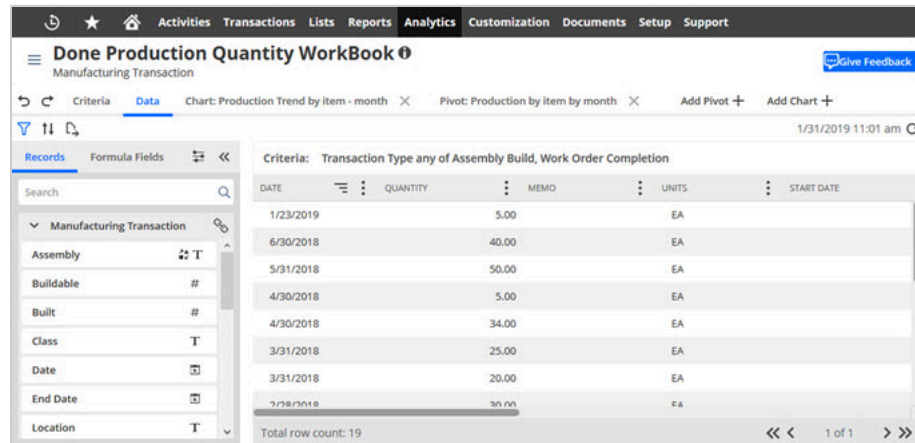
### To create a manufacturing workbook:

1. Enable SuiteAnalytics Workbook.  
For more information, see the help topic [Enabling SuiteAnalytics Workbook in Your NetSuite Account](#).
2. Select a workbook root record:
  - a. Click the **Analytics** tab.
  - b. Click **New Workbook**.
  - c. In the **Search Record Types** field, enter **Manufacturing**.



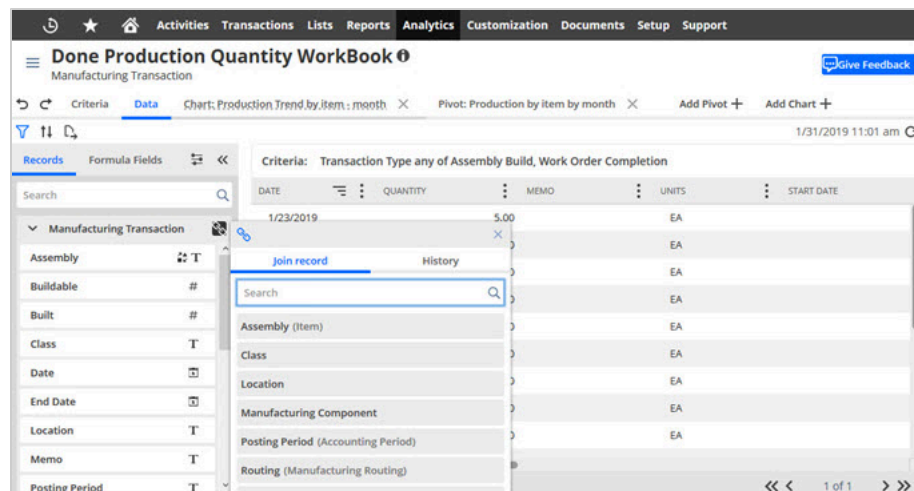
- d. Click the **Manufacturing Transaction [Beta]** record.

The Data tab opens with preselected fields on the Data Grid based on the manufacturing record.



You can use only those fields that you add to the Data Grid to generate a pivot table or chart. By default, the Data Grid displays preselected fields, based on the root record selected for the workbook.

3. To select source data, use one of the following options to add fields from the Manufacturing Transaction record root to the data grid:
  - From the Manufacturing Transaction record, drag a field to the Data Grid
  - Double-click a field
  - In the **Search** field, enter the field name and then drag or double-click the field
    - a. To add fields to the data grid from the Manufacturing Transaction related record, click the **Join Record** icon (🔗).



- b. Select a **Join record**.

The **Fields** list updates to include the related record fields. Double-click or drag fields to the data grid.

4. To create formula fields, click the **Formula Fields** tab.
  - a. Enter the **Formula Field Name**.

For example, **Buildable - Enhanced**.

- b. Select an **Output Type**.

For example, **String**.

- c. In the **Formula** field, enter the field IDs and SQL formula functions to use in the formula expression.

Alternatively, in the **Functions and Fields** sections, double-click the field IDs or formula functions to add them to the expression.

- d. Click **Validate**.
- e. If there are no errors, click **Apply** to add the formula field to the workbook.
- f. To filter the data source, click the **Criteria** tab.
- g. Drag a record or formula field.

- h. In the **Filter** window, select the filter conditions to apply to the field.

- i. Repeat these steps for each filter you want to define.

By default, filters are added using an AND operator. To change the relationship between filters and filter groups, click the AND link and select OR from the popup window.

- j. Click **OK**.

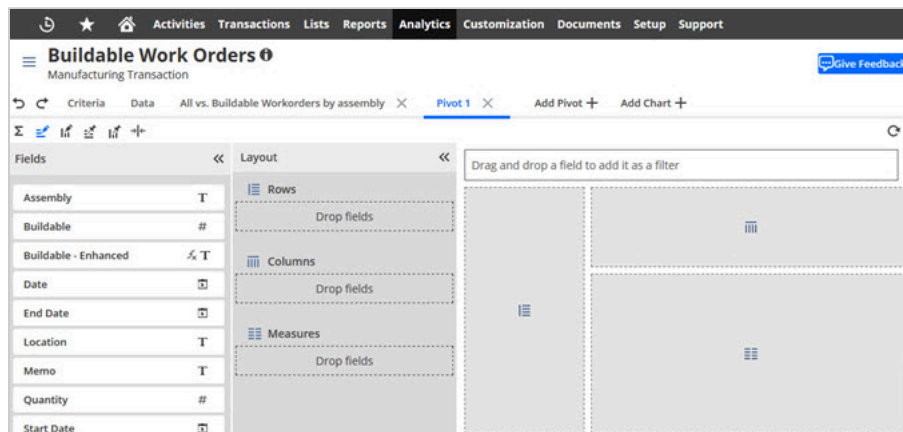
- k. When you finish defining filters, click the **Data** tab.

The Data Grid displays only values that match your selected filter conditions, and a Criteria Summary appears above the grid.

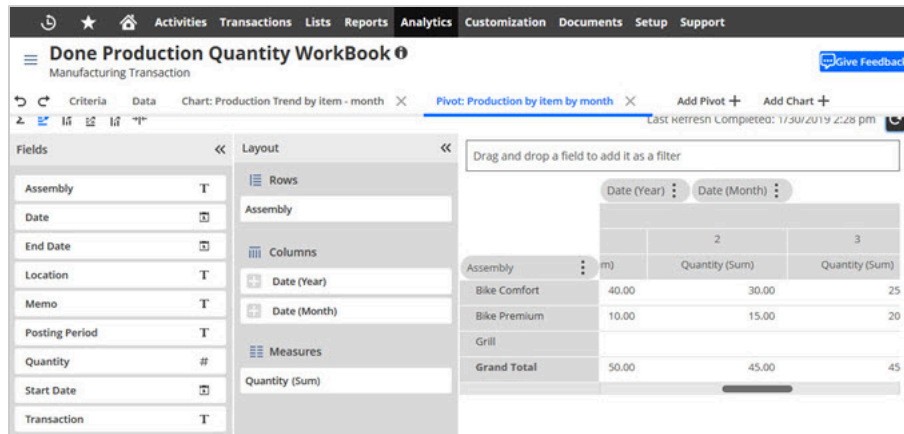
5. To create a pivot table, click **Add Pivot**.

Drag fields into the **Rows**, **Columns**, or **Values** section in the **Layout** panel.

For example, the **Assembly** row, **Date (Year)** and **Date (Month)** columns, and **Quantity (Sum)** measure.



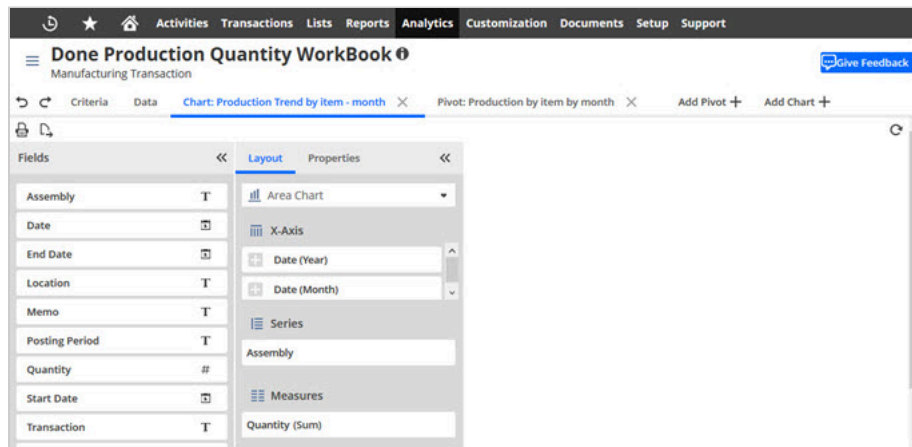
- a. Select the summary type and format options for any date or numerical fields.
  1. Beside the next to the field you want to format, click the **Field Menu** icon (⋮).
  2. Select **Summary Types**.
- b. Add totals and grand totals to the pivot table:
  1. In the menu bar, click the totaling icon (Σ).
  2. In the **Totaling** window, for each field, select where you want the totals or grand totals to appear.
  3. Check the **Set Individually** box to select where the totals for each field will appear on the pivot table.
  4. Click **OK**.
- c. Click **Refresh** (↻) to generate the pivot table.



6. To create a chart, click **Add Chart**.

a. Drag fields into the **X-Axis**, **Series**, or **Measure** section in the **Layout** panel.

For example, create an **Area Chart**, with **Date (Year)** and **Date (Month)** on the **X-Axis**, for the **Assembly Series**, with **Quantity (Sum)** Measures.



b. Select the summary type and format options for any date or numerical fields.

1. Beside the next to the field you want to format, click the **Field Menu** icon (⋮).
2. Select **Summary Types**.

c. In the **Layout** panel select a chart type.

d. Click **Refresh** (↻) to generate the chart.

e. In the Workbook Menu (⋮), click **Save As**.

f. In the **Save Workbook As** window, enter a name and description for the workbook.

g. Click **Apply**.

## Manufacturing Analytical Record Types

The Manufacturing workbook offers two analytical record types:

- Manufacturing Transaction Analytical Record Type
- Manufacturing Component Analytical Record Type

## Manufacturing Transaction Analytical Record Type

The manufacturing transaction analytical record type combines fields from the transaction and transaction line record types. This record type includes data for transactions that meet the following criteria:

- 1 Transaction type is one of WorkOrd, Build, Unbuild, WOCompl, WOIssue, WOClose
- 2 AND
- 3 Transaction line ID is 0

This record type is the root record type on the [Manufacturing Dataset](#). To access the record type or the dataset, you must have the Assembly Items feature enabled. Some fields in the record type are also feature dependent, or display values based on custom formula definitions.

For more information about the fields included in the manufacturing transaction analytical record type, see the following table:

| Field                        | Source Record Type | Source Field                            | Custom formula or feature dependency                                                                          |
|------------------------------|--------------------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Actual Production Start Date | Transaction        | Date                                    | N/A                                                                                                           |
| Actual Production End Date   | Transaction        | Date                                    | N/A                                                                                                           |
| Assembly                     | Transaction Line   | Item                                    | N/A                                                                                                           |
| Bill of Materials            | Transaction        | Bill of Materials                       | Displayed only with the Advanced BOM feature enabled                                                          |
| Bill of Materials Revision   | Transaction        | Bill of Materials Revision              | Displayed only with the Advanced BOM feature enabled                                                          |
| Buildable                    | Transaction Line   | Buildable                               | Displayed only with the Work Orders feature enabled                                                           |
| Built                        | Transaction Line   | Quantity Shipped/Received/<br>Picked Up | CASE WHEN TransactionLine.QUANTITY IS<br>NULL<br><br>THEN NULL ELSE<br><br>TransactionLine.QUANTITY_SHIP_RECV |
| Class                        | Transaction Line   | Class                                   | Displayed only with the Classes feature enabled                                                               |
| Date                         | Transaction        | Date                                    | N/A                                                                                                           |
| Department                   | Transaction Line   | Department                              | Displayed only with the Departments feature enabled                                                           |
| End Date                     | Transaction        | End Date                                | Displayed only with the Manufacturing Routing and Work Center, Demand Planning, or MRP features enabled.      |
| Location                     | Transaction Line   | Location                                | Displayed only with the Locations feature enabled                                                             |
| Memo                         | Transaction        | Memo                                    | N/A                                                                                                           |
| Posting Period               | Transaction        | Posting Period                          | Displayed only with the Accounting Periods feature enabled                                                    |

| Field            | Source Record Type | Source Field          | Custom formula or feature dependency                                                                                                                                              |
|------------------|--------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  |                    |                       |  <b>Note:</b> This field displays no values for work orders.                                     |
| Quantity         | Transaction Line   | Quantity              | N/A                                                                                                                                                                               |
| Reference        | Transaction        | Document Number       | N/A                                                                                                                                                                               |
| Routing          | Transaction        | Manufacturing Routing | Displayed only with the Manufacturing Routing and Work Center feature enabled                                                                                                     |
| Start Date       | Transaction        | Start Date            | Displayed only with the Manufacturing Routing and Work Center OR Demand Planning or MRP features enabled.                                                                         |
| Subsidiary       | Transaction Line   | Subsidiary            | Displayed only with NetSuite OneWorld where subsidiaries are defined                                                                                                              |
| Transaction      | Transaction        | Internal ID           |  <b>Note:</b> This field displays the Transaction Display Name instead of the internal field ID. |
| Transaction Type | Transaction        | Type                  | N/A                                                                                                                                                                               |
| Units            | Transaction Line   | Units                 | Displayed only with the Multiple Units of Measure feature enabled                                                                                                                 |
| WIP              | Transaction        | WIP                   | Displayed only with the Manufacturing Work In Process feature enabled                                                                                                             |

## Manufacturing Component Analytical Record Type

The manufacturing component analytical record type combines fields from the transaction, transaction line, and transaction accounting line record types. This record type includes data for transactions that meet the following criteria:

<sup>1</sup> | Transaction type is one of WorkOrd, Build, Unbuild, WOComp1, WOIssue, WOClose

The manufacturing component analytical record type is included in the field list on the [Manufacturing Dataset](#). To access the record type, you must have the Assembly Items feature enabled. Some fields in the record type are also feature dependent, or display values based on custom formula definitions.

For more information about the fields included in the manufacturing component analytical record type, see the following table:

| Field          | Source Record Type | Source Field   | Custom formula or feature dependency            |
|----------------|--------------------|----------------|-------------------------------------------------|
| Build Variance | Transaction Line   | Build Variance | Displayed only with Work Orders feature enabled |
| Item           | Transaction Line   | Item           | N/A                                             |
| Line Number    | Transaction Line   | Line Number    | N/A                                             |
| Quantity       | Transaction Line   | Quantity       | abs(Quantity)//always positive                  |

| Field                  | Source Record Type | Source Field                                     | Custom formula or feature dependency                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------|--------------------|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Quantity Available     | N/A - calculated   | N/A - Calculated                                 | <pre> "CASE transaction." + TransactionTypeFragment.FieldId.TYPE + " WHEN 'WorkOrd' " +  THEN get_item_avail(componentLine." + TransactionLineCommonFragment.FieldId.ITEM + ", componentLine." + LocationSelectFragment.FieldId.LOCATION + ") " +  "ELSE NULL END" </pre>                                                                                                                                                                                                                                                                                                                                                                         |
| Quantity Back Ordered  | Transaction Line   | Back Ordered                                     | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Quantity On Hand       | N/A - calculated   | N/A - calculated                                 | <pre> "CASE WHEN transaction." + TransactionTypeFragment.FieldId.TYPE + " IN ('WorkOrd', 'Unbuild') " +  "THEN get_invitem_onhand(componentLine." + TransactionLineCommonFragment. FieldId.ITEM + ", nvl(componentLine." + LocationSelectFragment.FieldId.LOCATION + ", -1)) " +  "ELSE NULL END" </pre>                                                                                                                                                                                                                                                                                                                                          |
| Quantity Used In Build | Transaction line   | Quantity Shipped/ Received/Picked Up OR Quantity | <pre> "CASE WHEN transaction." + TransactionTypeFragment.FieldId.TYPE + " = 'WorkOrd' " +  "THEN "+  "-sign(nvl(nullif(componentLine." + TransactionLineQuantityFragment.FieldId.QUANTITY + ", 0), -1))" +  " * " +  "decode(" +  "componentLine." + TransactionLineCommonFragment.FieldId. ITEM_TYPE + ", 'Group',NULL, " +  "nvl2(" +  "accountingLine." + TransactionAccountingLineAccount Fragment.FieldId.ACCOUNT + ", " +  "decode(" + "componentLine." + TransactionLineQuantityFragment.FieldId.QUANTITY + ", NULL ", " +  "componentLine." + TransactionLine.FieldId.QUANTITY_SHIP_ RECV + ") " +  ", NULL)) " +  "ELSE NULL END" </pre> |
| Units                  | Transaction Line   | Units                                            | Displayed only with Multiple Unit of Measure feature enabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |



# Engineering Change Order

The Engineering Change Order feature is included in the Supply Chain Management SuiteApp starting version 4.0. For more information, see the help topic [Installing the Supply Chain Management SuiteApp](#).

The Engineering Change Order feature lets you create engineering change order (ECO) records. These records track changes to your Bills of Materials (BOMs) and allow you to approve those changes.

- With the ECO record, you can list the items and BOMs to update and say why you're making the change.
- With the SuiteApprovals workflow, you set up approval rules for your ECO records.
- The system evaluates each ECO record and sends it for approval based on the applicable rule.
- After approval, you can apply changes from the ECO record automatically or manually, and the system updates the affected records.
- You can track item and BOM revisions with ECO lists and implementation logs.

Make sure you're familiar with the following concepts before you turn on Engineering Change Order:

- **Advanced Bill of Materials** – List how much raw materials, assemblies, sub-components, and parts you need to make a product at one or more facilities. After your administrator enables the Advanced Bill of Materials feature, the Advanced BOM replaces the Assembly/Bill of Materials record. For more information, see [Advanced Bill of Materials](#).
- **Approval Routing** – When you use approval routing, transactions that are entered are not processed until they are approved.

Approval routing gives people in charge the ability to reject transactions if they don't want them to go through. For more information, see the help topic [Approval Routing](#).

## Availability

The Engineering Change Order feature is available in the Supply Chain Management SuiteApp. You can also contact your NetSuite account manager to get access to this SuiteApp. For information about this SuiteApp, see the help topic [Supply Chain Management SuiteApp](#).

## Engineering Change Order Considerations

Review the following points before you use the Engineering Change Order (ECO) feature:

- You can use Engineering Change Order only with inventory and assembly items.
- The feature doesn't support custom workflows for ECO approval. For more information, see the help topic [SuiteApprovals Considerations](#).
- If your NetSuite account has more than 80,000-90,000 active assembly and inventory items per subsidiary, ECO records may fail to load because of the user event script threshold.
- The feature doesn't support SOAP web services and CSV import.
- The feature works only with Advanced Bill of Materials (BOM) and it doesn't support Legacy Bill of Materials. For more information, see [Enabling Advanced BOM](#).

## Setting Up Engineering Change Order

Make sure you've completed the setup requirements before using the Engineering Change Order feature. For more information, see the following help topics:

- [Enabling Advanced BOM](#)
- [Installing the Supply Chain Management SuiteApp](#)
- [Enabling Engineering Change Order](#)
- [Customizing Roles for Engineering Change Order](#)
- [Setting the Engineering Change Order Preference](#)

## Enabling Engineering Change Order

Before you enable the Engineering Change Order feature, complete the following prerequisites:

| Task                                                         | Related Help Topic                                              |
|--------------------------------------------------------------|-----------------------------------------------------------------|
| Step 1: Enable the Advanced Bill of Materials (BOM) feature. | <a href="#">Enabling Advanced BOM</a>                           |
| Step 2: Install the Supply Chain Management SuiteApp         | <a href="#">Installing the Supply Chain Management SuiteApp</a> |



**Important:** Only users with the Administrator role can access the Supply Chain Preferences page.

### To enable Engineering Change Order:

1. Go to Transactions > Management > Supply Chain Management.
2. On the Supply Chain Management page, click the **Preferences** link.
3. On the Supply Chain Preferences page, click **Edit**.
4. On the **Features** subtab, check the **Engineering Change Order** box.
5. Click **Save**.

After you enable the feature, you'll need to set up role records for users of Engineering Change Order. For more information, see [Customizing Roles for Engineering Change Order](#).

## Customizing Roles for Engineering Change Order

The following roles can access the custom forms and records for Engineering Change Order (ECO):

- Accountant
- CEO
- CFO
- Warehouse Manager
- Administrator

Other roles for Engineering Change Order need additional permissions to use the feature. The following table shows the permissions you need to use ECO. For information about customizing roles, see the help topic [Customizing or Creating NetSuite Roles](#).

| Subtab        | Permissions                   | Level | Description                                                         |
|---------------|-------------------------------|-------|---------------------------------------------------------------------|
| Custom Record | Engineering Change Order Type | Full  | Lets you create engineering change order types.                     |
|               | Engineering Change Order Type | View  | Lets you create, view, or edit engineering change orders.           |
|               | Engineering Change Order      | Full  |                                                                     |
| List          | Custom Record Entries         | View  |                                                                     |
| Custom Record | ECO Implementation Log        | View  | Lets you access the engineering change order implementation status. |
| Setup         | SuiteScript                   | View  | Lets you implement approved engineering change orders.              |
|               | SuiteScript Scheduling        | Full  |                                                                     |

## Configuring Script Deployments for Engineering Change Order

After you customize a role to access Engineering Change Order lists and records, the Administrator needs to add the new custom roles to script deployments.

The following script deployments let you access and implement engineering change order records:

- ECO Implement SU
- ECO Implement Log SU

For more information about the custom role setup for SuiteApprovals, see the help topic [Creating or Customizing Roles for SuiteApprovals](#).

### To add custom roles to a script deployment:

1. Go to Customization > Scripting > Script Deployments.
2. Click **Edit** next to the script deployment you want to update.
3. On the **Audience** subtab, in the **Roles** field, select the roles you want to have access to the records used by this script deployment.  
You can select multiple roles by holding down the Ctrl button while selecting each role.
4. When you're done, click **Save**.

## Setting the Engineering Change Order Preference

You can manually implement approved engineering change orders, or let the system do it automatically with a preference.

### To enable the engineering change order preference:

1. Go to Transactions > Management > Supply Chain Management.

2. On the Supply Chain Management page, click the **Preferences** link.
3. On the Supply Chain Preferences page, click **Edit**.
4. On the **Engineering** subtab, check the **Implement Engineering Change Order Upon Final Approval** box.
5. Click **Save**.

 **Note:** If you don't want to use the preference, refer to the instructions in [Implementing Engineering Change Orders](#).

## Setting Up Engineering Change Order Types

With Engineering Change Order, you can manage change processes across your organization. An issue may involve a product change, component update, document revision, or changes to work instructions. You can record these issues so their change details can be approved and implemented.

Before you set up engineering change order types, identify which departments and business processes need to control change activities.

You can create as many change types as you want. The only requirement is that each type is unique.

 **Important:** Engineering Change Order supports these actions in an ECO, no matter the type: **Add**, **Replace**, and **Remove**.

### To set up an engineering change order type:

1. Go to Lists > Engineering > Engineering Change Order Type > New (Administrator).
2. On the Engineering Change Order Type page:
  - a. In the **Code** field, enter a unique code for the ECO type if needed.  
If you enter a code, it'll appear before the ECO type name.
  - b. In the **ECO Type** field, enter a unique name for the ECO type.
  - c. Optionally, enter additional information about the ECO Type in the **Description** field.
3. When you're done, click **Save**.

Alternatively, you can open the Engineering Change Order Type page by clicking **New Engineering Change Order Type**. You'll find this button on the Engineering Change Order Type List page at Lists > Engineering > Engineering Change Order Type.

### To edit an engineering change order type:

1. Go to Lists > Engineering > Engineering Change Order Type (Administrator).
2. Click the **Edit** link for the ECO Type you want to update.
3. On the Engineering Change Order Type page:
  - Enter the changes you want, then go to Step 4.
  - Check the **Inactive** box if you don't want this record to show up on lists.
  - To remove the ECO Type record, click **Delete** under **Actions**.
4. When you're done, click **Save**.

## Using Engineering Change Order

See the following help topics to learn how to use an engineering change order.

- [Creating Engineering Change Orders](#)
- [Implementing Engineering Change Orders](#)
- [Viewing Implementation Status](#)

## Creating Engineering Change Orders

Create an Engineering Change Order (ECO) to list changes in an assembly item's BOM, or to enter process changes for documentation.

When you create an ECO, NetSuite checks the ECO record for uniqueness and starts the SuiteApprovals workflow. For more information about how approval rules work, see the help topic [Approval Rule Validation](#).

## Engineering Change Order Bulk Processing

This feature supports bulk processing of ECO records. You can do any of the following actions during bulk processing:

- Multiple **Add** actions for one or more items in one ECO record.
- Multiple **Remove** actions for one or more items in one ECO record.
- Multiple **Replace** actions for one or more items in one ECO record.
- A mix of **Add**, **Remove**, and **Replace** actions for one or more items in one ECO record.
- **Add**, **Remove**, and **Replace** actions for the same BOM and BOM revision get processed together.

NetSuite emails ECO coordinators when bulk processes finish. The email includes a link to the ECO record and any error details from implementation.

 **Note:** During manual or automatic implementation, you can check the Implementation Status of any ECO and track each change action. For more information, see [Viewing Implementation Status](#).

### To create an engineering change order record.

1. Go to Lists > Engineering > Engineering Change Order > New.
2. In the Primary Information section, complete the following fields:
  - a. **ECO Type** – Select the type of change order.  
For more information about ECO types, see [Setting Up Engineering Change Order Types](#).
  - b. **Problem** – Enter the reason for the change order.
  - c. **Description** – Enter additional details for the change order.
  - d. **Date** – The ECO creation date is set to today's date by default. You can set this field to a future date.
  - e. **Effective Start Date** – Select a date of when the change order becomes effective.
  - f. **Effective End Date** – Select a date of when the change order becomes obsolete.

This field is not required.

- g. **Subsidiary** – If you have a NetSuite OneWorld account, select a subsidiary.
- h. **Approval Status** – If you are using SuiteApprovals for ECO, this field is set to Pending Approval by default. Otherwise, this field is blank.
- i. **Next Approver** – Displays the next approver when ECO enters the approval workflow.
- j. **Implementation Status** – Displays the status of the ECO when it is approved or implemented. For more information, see [Setting the Engineering Change Order Preference](#) and [Implementing Engineering Change Orders](#).

 **Note:** If you have a NetSuite Oneworld account, make sure you've selected a subsidiary before entering product change details. You can select only those items and assembly items associated with the selected subsidiary.

3. On the **Product Changes** subtab, click **Add BOM Changes**.
4. In the Add Change window, from the **Action** list, select one of the following change actions:
  - **Add** – Select this option if you want to add a component to an assembly item.
  - **Remove** – Select this option if you want to remove a component from an assembly item.
  - **Replace** – Select this option if you want to replace a component from an assembly item.
5. In the **Item** field, enter the name of the item you want to change. The field returns search results as you type.
6. When you have selected an item, click **Add BOM Change**.
7. In the sublist, provide values for each column:
  - a. **Assembly Item** – Enter the name of the assembly item affected by the change action. The field returns search results as you type.
  - b. **Quantity** – Enter the quantity of the item you want to add. The default value is 1.

 **Note:** This field is not editable for the **Remove** and **Replace** action.

- c. **Change Component Yield** – Update the percentage as necessary. This field has no value if the Use Component Yield preference is not enabled for the affected assembly item.  
For more information, see [Setting the Use Component Yield Preference](#)

 **Note:** This field is not editable for the **Remove** and **Replace** action.

- d. **BOM** – Select the BOM to which the item will be added. All BOMs associated with the affected assembly items are available for selection.
- e. **Revision** – Select the affected revision of the BOM.
- f. **New Revision** – Enter a revision name for when the change is implemented.

 **Important:** Changing the action or item field will remove all existing change line items. To add another change action, save the current one first.

8. When you are done entering values in the sublist, click **Save**.  
If you want to edit the change details again after saving, click the edit icon to open the Add Change window.
9. On the **Process Changes** subtab, enter details or note the following fields for documentation and approval:
  - a. **Process Description** – Enter the details of the process change.

- b. **Date/Time** – Displays the timestamp of when the ECO record is saved. This field cannot be edited.

If you edit the ECO record, the timestamp adjusts to the current date and time.

 **Note:** If you are using SuiteApprovals for ECO, ECO records for process changes go through approval. Ensure that you create approval rules for these change types. For more information, see [Creating Approval Rules for Engineering Change Order](#).

10. On the **Notes** subtab, in the Memo column, add additional details related to the ECO. For more information about user notes, see the help topic [Entering Communication Information to a Transaction](#).
11. On the **Files** subtab, in the **Attach File** column, select the required or relevant documents and materials about the ECO. For more information about attaching files, see the help topic [Working with Files](#).
12. When the ECO record is complete, click **Save**.

## Copying an Engineering Change Order Record

Use the following procedure to copy an engineering change order record.

### To copy an engineering change order record:

1. Go to Lists > Engineering > Engineering Change Order.

 **Important:** The Make Copy function for ECO records doesn't copy Product Changes entries.

2. On the Engineering Change Order List page, find the ECO record you want to copy.
3. Click the **View** link for the ECO record.
4. On the ECO record, from the **Actions** list, select **Make Copy**.
5. Make changes in the Primary Information fields, as needed.
6. Click **Save**.

## Implementing Engineering Change Orders

If you haven't set the **Implement Engineering Change Order Upon Final Approval** preference, an ECO administrator has to manually implement approved ECO records.

### To implement approved engineering change orders:

1. Go to Lists > Implement Engineering Change Order (Administrator).
2. Use filters to find approved ECO records:
  - Use the calendar to search by effective date range. Today's date is the default date.
  - Select a subsidiary.
3. The page lists matching records, if any.
  - To implement all records, click **Mark All**. To implement specific ECO records, check the **Select** box for each one.

- Use the pagination controls if you need to make more selections.

4. Click **Implement**.

To view the implementation status of an ECO record, see [Viewing Implementation Status](#).



**Note:** Set the **Implement Engineering Change Order Upon Final Approval** preference to have approved ECOs implemented automatically. See [Setting the Engineering Change Order Preference](#).

## Viewing Implementation Status

The Engineering Change Order Log section lets an ECO administrator see the implementation status of any approved ECO.

### To view the implementation status of engineering change orders:

1. Go to Lists > Engineering > ECO Implementation Status (Administrator).
2. The Engineering Change Order Log section lists all active ECO records.
3. Select a value in the **Implementation Status** list to filter the list:
  - **Pending** – Approved ECO records that haven't been implemented yet.
  - **In Process** – Implementation is in progress for approved ECO records.
  - **Completed** – ECO records that have been implemented.
  - **Error** – Approved ECO records with implementation errors.
4. Click **Go to Main** to go back to the full list. Click **Refresh** to show the latest data.

### To view the implementation status of action changes in approved engineering change orders:

1. Go to Lists > Engineering > Engineering Change Order (Administrator).
2. On the Engineering Change Order List page, find the ECO record you want to view using filters or sorting tools.
3. Click the **View** link of the ECO record.
4. Click the **Product Changes** subtab.
5. View the implementation status of each change line-item under the **Is Implemented** column.



**Note:** You may need to click the collapse (-) icon for a change line to see its implementation status.